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Engineering
Department of Physics



Study of photoproduction of J/ψ in p+p collisions at 510 GeV

DIPLOMA THESIS

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Studium fotoprodukce J/ψ v p+p srážkách při energii 510 GeV

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2) Rekonstrukce J/psi v datech z p+p srážek na experimentu STAR
3) Korekce hrubého výtěžku pomocí na efektivitu analýzy a detektorů
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Abstrakt: Studium centrální exkluzivní fotoprodukce J/ψ v proton-protonových srážkách při energii 510 GeV umožňuje přímé porovnání mezi experimentálními měřeními a teoretickými předpověďmi, zejména v oblasti gluonového rozdělení uvnitř protonu. V této práci podáváme zprávu o experimentálním měření fotoprodukce J/ψ v elektron-pozitronovém rozpadovém kanálu v experimentu STAR nacházejícím se v Brookhaven National Laboratory. Výsledkem měření je stanovení účinného průřezu pro dvě samostatné případy. První analýza je zaměřena pouze na fotoprodukci J/ψ v centrálním systému. Druhé měření kromě centrálního systému zahrnuje také soustavu detektorů Roman Pots, které umožňují měřit dopředné protony.

Klíčové slova: J/ψ, Pomeron, Glueball, RHIC, STAR

Title:	**Study of photoproduction of J/ψ in p+p collisions at 510 GeV**
Author:	Bc. Michal Vranovský

Abstract: The study of central exclusive J/ψ photoproduction in proton-proton collisions at $\sqrt{s} = 510$ GeV allows direct comparison between experimental measurements and theoretical prediction mainly in the area of gluon distribution inside a proton. We report on an experimental measurement of J/ψ photoproduction in the electron=positron decay channel at experiment STAR at the Relativistic Heavy Ion Collider (RHIC) at the Brookhaven National Laboratory. The results of the measurement are 2 cross sections for two separate analyses. The first analysis is focused on the J/ψ photoproduction in the central system only. In addition to the central system, the second measurement also involves the Roman Pot system of detectors, which are able to measure forward protons.

Key words: J/ψ, Pomeron, Glueball, RHIC, STAR

Contents

Introduction **1**

1 Diffractive processes **3**
1.1 Introduction to High Energy Physics . . . 3
1.2 Diffraction . . . 5
1.3 Central Exclusive J/ψ Photoproduction . . . 7
1.3.1 Kinematics . . . 8
1.3.2 Motivation . . . 9
1.3.3 Photon Flux . . . 11
1.3.4 Vector Dominance Model . . . 12
1.3.5 Color Dipole Model . . . 12
1.4 Recent Results in J/ψ Photoproduction . . . 14

2 Experimental Setup **19**
2.1 Experiment STAR . . . 19
2.2 The Time Projection Chamber . . . 21
2.3 Time of Flight Detector . . . 23
2.4 Barrel Electromagnetic Calorimeter . . . 24
2.5 The PP2PP Experiment and the Roman Pots . . . 26
2.6 Beam-Beam Counters and Zero Degree Calorimeters . . . 27
2.7 Data Acquisition and Triggers . . . 27

3 Data set and Quality Assurance **29**
3.1 Data Format . . . 29
3.2 BEMC Clustering . . . 29
3.3 Triggers . . . 31
3.4 Good Run List . . . 32

4 Event Selection **35**
4.1 List of Cuts . . . 35
4.2 Central System . . . 37
4.2.1 Primary Vertex (C2, C3) . . . 37
4.2.2 2 Good Quality BEMC Tracks (C4) . . . 39
4.2.3 Pseudorapidity (C5) . . . 40
4.2.4 Back To Back Condition (C6) . . . 40
4.2.5 Particle Identification (C7) . . . 41
4.2.6 Total Charge (C8) . . . 42

4.3 Forward System . . . 42
4.3.1 1 Track in the Roman Pots (F1.3) . . . 42
4.3.2 Roman Pot Fiducial Region (F2) . . . 43

5 Monte Carlo Simulations **45**
5.1 STARlight $J/\psi \rightarrow e^+e^-$ . . . 45
5.2 STARlight $\gamma\gamma \rightarrow e^+e^-$ . . . 47

6 Corrections and Efficiencies **49**
6.1 Trigger Topology Correction . . . 49
6.2 Trigger Veto Correction . . . 50
6.3 Pile-Up Correction . . . 53
6.4 TPC V_Z Correction . . . 54
6.5 Embedding . . . 56
6.5.1 Event Selection . . . 57
6.5.2 Reconstruction Efficiency . . . 59
6.5.3 Correction Application . . . 60
6.6 BEMC Efficiency . . . 61
6.6.1 Data-Driven Method . . . 61
6.6.2 BEMC MC Simulator . . . 63
6.6.3 BEMC Efficiency QA . . . 64
6.7 Summary of Corrections . . . 64

7 Systematic Uncertainties **67**
7.1 TPC Cuts . . . 67
7.2 Yield Extraction . . . 68
7.3 Monte Carlo BEMC Efficiency Simulator . . . 68
7.4 Topology Trigger Uncertainty . . . 69
7.5 Luminosity Error . . . 69
7.6 Summary of Systematic Uncertainties . . . 70

8 Results **71**
8.1 Yield Extraction . . . 71
8.2 Cross Section . . . 73
8.3 Discussion . . . 75

Summary **79**

References **81**

List of Figures

1.1 Elastic collision of 2 particle states to 2 particle states through the s, t, and u channel . . . 4
1.2 Pseudorapidity values according to angle θ . . . 5
1.3 Pseudorapidity and azimuthal angle distribution in a diffractive process 7
1.4 Diffractive processes: Double photon exchange, Photoproduction, and Double IPomeron Exchange . . . 8
1.5 Photoproduction of J/ψ in terms of Quantum Chromodynamics theory 9
1.6 The kinematical coverage in x-Q^2 plot of previous and current experiments . . . 10
1.7 Sketch of dipole model for quarkonium production . . . 13
1.8 Invariant mass spectrum of dielectron pairs at ALICE . . . 14
1.9 H1 Collaboration results of J/ψ photoproduction fits . . . 15
1.10 J/ψ production as a function of variable $W_{\gamma p}$ . . . 16
1.11 Measurement of central exclusive production of J/ψ and ψ(2S) at the LHCb experiment . . . 17
1.12 The photoproduction of J/ψ in Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV 18

2.1 Overview of the STAR detector and its subsystems . . . 20
2.2 Schematic view of the STAR Time Projection Chamber . . . 21
2.3 Correlation plot of energy loss per unit length and transverse momentum of a particle . . . 22
2.4 An illustrative picture of pVPD and TOFp, which determines the time of flight of a particle . . . 23
2.5 A cross section view of one of the STAR barrel TOF MRPC module . 24
2.6 The BEMC module scheme and rotation . . . 25
2.7 The BEMC position within the experiment . . . 25
2.8 An illustrative scheme of the PP2PP experiment with differences between position of detectors in Phase I and the planned positions for Phase II . . . 26
2.9 (a) is the geometrical acceptance of the Roman Pot system of detectors during Phase II. (b) shows the difference between momentum transfer acceptance between Phase I and II . . . 27
2.10 A picture of 2 Roman Pot vessels. The active area of detection is the window . . . 28

3.1 Distribution of BEMC clusters in η - ϕ plot . . . 30
3.2 Correlating distribution of cluster energy and track energy . . . 31

3.3 List of conditions for bad run list . . . 32
3.4 The averages of η and ϕ for each run . . . 33
3.5 The $\eta - \phi$ correlation plot for run 18113052 . . . 33

4.1 Summary of event selection for analysis without the Roman Pots . . . 36
4.2 Summary of event selection for analysis with the Roman Pots . . . 37
4.3 Vertex distributions for event selection . . . 38
4.4 Distribution of vertices along the z axis . . . 38
4.5 Distributions of DCA for particles and vertex . . . 39
4.6 Distributions of N_{hits}^{fit} and $N_{hits}^{dE/dx}$ for tracks . . . 39
4.7 Distribution of pseudorapidity η reconstructed in the BEMC . . . 40
4.8 Angular distribution between reconstructed tracks . . . 40
4.9 Particle identification nσ_e plots . . . 41
4.10 Correlating distributions of momentum and energy loss . . . 42
4.11 Distributions of tracks in the Roman Pots . . . 43

5.1 Distributions of rapidity and the invariant mass of the J/ψ Monte Carlo sample . . . 46
5.2 Distributions of J/ψ transverse momentum and the transverse momenta of the daughter electrons in a generated MC sample . . . 47
5.3 Distributions of pseudorapidity and invariant mass for Monte Carlo sample $\gamma\gamma \to e^+e^-$ . . . 47
5.4 Distributions of transverse momenta of daughter particles and pairs of particles for STARlight Monte Carlo sample $\gamma\gamma \to e^+e^-$ . . . 48

6.1 Selection of cuts for trigger topology efficiency . . . 50
6.2 Trigger efficiency w.r.t. the chosen invariant mass window . . . 51
6.3 Trigger efficiency distribution . . . 51
6.4 Probability of retaining an event plots . . . 52
6.5 Probability of retaining an event for trigger veto conditions separately for either BBC . . . 53
6.6 Pile-up correction plotted against instantaneous luminosity on a run-by-run basis . . . 54
6.7 Distribution of V_Z for Fill number 20562 . . . 55
6.8 Extracted parameters from V_Z correction . . . 55
6.9 The efficiency of V_Z cut w.r.t. Fill number . . . 56
6.10 The flow of cuts in embedding analysis . . . 57
6.11 Comparison plots of N_{hits}^{fit} and $N_{hits}^{dE/dx}$ between embedding and data . . . 58
6.12 Comparison plots of η and ϕ between embedding and data . . . 58
6.13 Correlating distribution of nσ_e for pairs of particles in embedding . . . 58
6.14 Comparison distributions of J/ψ transverse momentum and electrons transverse momenta . . . 59
6.15 Comparison between embedding J/ψ peak fit and the fit from data . . . 59
6.16 The reconstruction efficiency plots w.r.t. electron pseudorapidity and J/ψ rapidity . . . 60
6.17 Reconstruction efficiency w.r.t. the transverse momentum of J/ψ . . . 60
6.18 The specific number of passed events after each cut for BEMC efficiency study . . . 62

6.19 Distribution of variable δ^{dip} angle used for identification of photon conversion electrons 63
6.20 BEMC efficiency from data compared to BEMC efficiency from embedding 64
6.21 The shape of the continuum with STARsim BEMC efficiency compared to the shape from data 65
6.22 The shape of the continuum with MC BEMC efficiency simulator compared to the shape from data 65

7.1 Summary of systematic uncertainties coming from TPC cuts 68
7.2 Evaluation of the systematic uncertainty of the yield extraction . . . 69
7.3 Summary of various contributions to the systematic uncertainties . . 70

8.1 Fitted invariant mass spectrum of e^+e^- pairs without Roman Pots . . 72
8.2 Fitted invariant mass spectrum of e^+e^- pairs for analysis with Roman Pots 73
8.3 The comparison of transverse momentum distribution between the measurement and the STARlight sample 76
8.4 The comparison of J/ψ rapidity between the data and STARlight . . 77
8.5 Distribution of missing transverse momenta in the analysis with the Roman Pots 77

List of Tables

3.1 BEMC clustering parameters 30

5.1 Options used in STARlight input file for process $J/\psi \rightarrow e^+e^-$ 46
5.2 Options used in STARlight input file for process $\gamma\gamma \rightarrow e^+e^-$ 48

6.1 Summary of efficiencies 66

8.1 Table of crystal ball fit parameters for both J/ψ fits 73
8.2 Table of values used for the cross section calculation 74

Introduction

Measurement of J/ψ particle in a process central exclusive photoproduction provides a unique way of studying the inner structure of protons. The process itself has been successfully described by various models, therefore one can straightforwardly compare the experimental data to the theoretical predictions. The study of J/ψ photoproduction at energy of collision $\sqrt{s} = 510$ GeV provides insight into a specific range of Bjorken x values for studying the gluon distribution inside protons. The measured data come from experiment STAR located at the Relativistic Heavy Ion Collider (RHIC) in Brookhaven National Laboratory. Originally, STAR experiment was meant to study the quark gluon plasma (QGP). Nowadays, it is a multipurpose detector capable of measuring also diffractive processes such as photoproduction. The photoproduction of J/ψ in proton-proton collisions is characterized by 2 forward[1] protons with very small scattering angles, and a J/ψ particle. The J/ψ meson decays to daughter particles, in our case, an electron-positron pair. Due to the acceptance limits of the experiment, we measure the daughter particles in the pseudorapidity range $|\eta| < 0.9$. Furthermore, we describe the J/ψ - electron[2] system as central, and the protons scattered with very small angles as the forward system.

The study is divided into 2 separate measurements. In the first case, we only use information from the central system (meaning the electrons). In the second case, we utilize the forward system of Roman Pots to measure one of the outgoing forward protons. It is very important to note that the measurement results will be different, because the Roman Pots are capable of measuring only within certain phase space acceptance limits, and we do not attempt to correct for it nor extrapolate the results onto the entire phase space. The measurement of protons in J/ψ photoproduction is the first of its kind.

The first chapter discusses particle physics variables and observables along with a description of the process and a summary of previous similar measurements. The second chapter gives a brief overview of the experimental setup, mainly the detector systems crucial for this analysis. Further on, the dataset is described, the production of data and the very important trigger *HTTP JPsi* is described. At the end of chapter 3, the quality assurance of the dataset in the form of a good run list is described. The following chapter outlines the selected cuts to separate the signal from the background. Chapter 5 describes the STARlight Monte Carlo generator of

[1]Due to the ambiquity of forward and backward direction in a symmetrical proton-proton collision, both forward and backward protons will be referred to as forward.

[2]We use the term electron and positron interchangeably. The electron-positron pair will often be referred to as just electrons.

photoproduction, to which our results will be compared. After comes the chapter which discusses the corrections and efficiency studies necessary to obtain the cross section. The estimation of systematic uncertainties is in Chapter 6. The last chapter describes the physics results of the analysis and future improvements.

Chapter 1

Diffractive processes

The purpose of this chapter is to give a brief introduction to High energy physics (HEP), discuss several kinematic variables used in detector experiments, which will be used throughout this thesis. The chapter continues with defining particle diffraction, defining J/ψ photoproduction, and giving an overview of theoretical models which attempt to describe it. At the end, previous similar measurements are discussed.

1.1 Introduction to High Energy Physics

The area of HEP has a specific way of describing particles and their properties. Because the kinetic energies of particles are higher than their invariant mass, one uses the relativistic four-momentum description. One defines the four-momentum of a particle

$$p^{\mu} = \begin{pmatrix} E \\ p_x \\ p_y \\ p_z \end{pmatrix}, \tag{1.1}$$

where $E = \sqrt{m^2 + p^2}$ denotes the relativistic energy[1] of the particle and $\vec{p}$ is the 3D spatial momentum vector. Momentum and energy conservation from classical physics is equivalent to conservation of the four-momentum vector in relativistic physics. The conservation of four-momentum vector is especially useful when calculating the momentum and energy of outgoing particles after a collision or during transitions from one reference frame to another. The most used reference frame in HEP is the center of mass frame. The center of mass frame is defined for a system as the one, in which the sum of spatial momenta of all incoming particles is equal to 0.

[1] Throughout this thesis, natural units will be used. Constants such as the Planck constant or the speed of light in vacuum are considered to be: $\hbar = c = 1$.

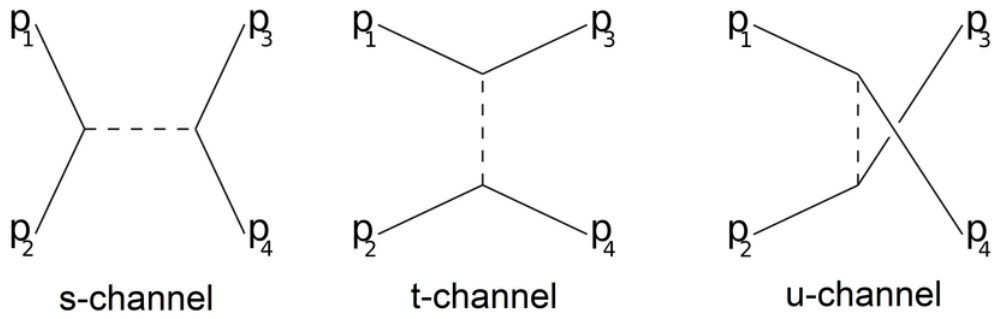


Figure 1.1: Elastic collision of 2 particle states to 2 particle states through the s, t, and u channel.

Elastic collision of 2 particle states to the same 2 particle states is the simplest interaction. There are several ways one can look at the same interaction; often referred to as the s, t, and u channels. The Feynman diagrams of the interactions can be seen in Fig. 1.1. These channels accompany the so-called Mandelstam variables

$$\begin{aligned} s &:= (p_1 + p_2)^2 = (p_3 + p_4)^2, \\ t &:= (p_1 - p_3)^2 = (p_4 - p_2)^2, \\ u &:= (p_1 - p_4)^2 = (p_3 - p_2)^2. \end{aligned} \tag{1.2}$$

The Mandelstam variables are invariant under Lorentz transformations [1]. The most used kinematic variable out of the 3 is s, or specifically the $\sqrt{s}$, which in collider physics denotes the energy of the collision. Another one fairly often used is the t variable, which represents the square of transferred energy in a collision. These 3 variables are not independent; let the invariant masses of colliding particles be denoted m_1, m_2, m_3, m_4 then the sum of s, t and u turns out to be the sum of masses squared

$$s + t + u = \sum_{i=1}^{4} m_i^2. \tag{1.3}$$

In a collider experiment, 2 bunches of particles collide along the z axis. The x and y plane is the transverse plane. A very important kinematic variable used to describe the ratio of energies in the z direction and in the transverse plane,

$$y := \frac{1}{2} \ln \left(\frac{E + p_z}{E - p_z} \right) \tag{1.4}$$

is rapidity. The particle's energy E and spatial momentum along the z axis p_z. Similar variable, which is used more often due to its dependence on 1 variable only, is pseudorapidity.

$$\eta := -\ln(\tan \frac{\theta}{2}) \tag{1.5}$$

The only independent variable in this definition is θ and represents the angle between the z axis and the direction of momentum of the outgoing particle. Visual representation of η can be seen in Fig. 1.2. Definition of pseudorapidity makes sense for outgoing particles only. In the limit of high energy, the pseudorapidity distribution in the central system continuously transitions to become rapidity-like. Pseudorapidity and rapidity play a big role in diffraction and their impact will be closely described in Sec. 1.2.

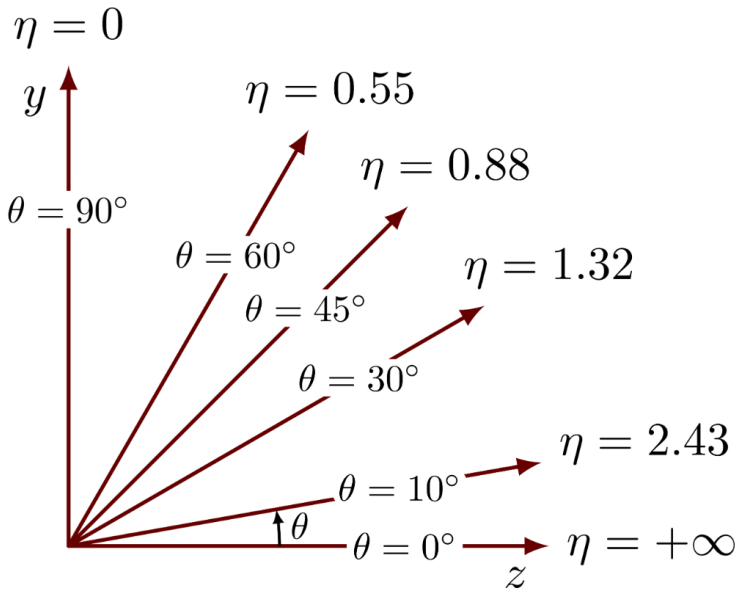


Figure 1.2: Pseudorapidity values according to angle θ between the z axis and the transverse plane.

Last variable that is important to mention is cross section. Cross section is the measure of the probability of a certain process or a type of interaction happening [2]. It is usually represented in a differential form as a function of a HEP variable such as pseudorapidity, transverse momentum or solid angle: $\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}(\theta, \phi)$. In this case it is the probability that a particle will scatter into the angle given by variables θ and ϕ. Cross section is related to the interaction rate with the equation below.

$$R_{int} := L\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} \tag{1.6}$$

Interaction rate $R_{int} = \frac{\mathrm{d}N}{\mathrm{d}t}$ is the number of events of interest per unit of time. Luminosity is represented as L, and it is independent of the process. Luminosity depends only on the properties of the collider.

1.2 Diffraction

Diffraction in particle physics has some similarities to electromagnetic diffraction, but in its essence, they are 2 separate phenomena. Typically, diffractive events in particle physics are characterized by very small scattering angles and their impact

parameters[2] are often comparable or even larger than the size of the colliding particles. Diffraction in particle physics has two definitions. The first one, formulated in 1960 by Good and Walker,

A reaction in which no quantum numbers are exchanged between colliding particles is, at high energies, a diffractive reaction. [3]

This definition accommodates 2 conditions; no quantum numbers exchanged, and that the energy of collision has to be high. The second condition is nowadays redundant due to colliders being able to accelerate particles to much higher energies than in the 20th century. The first one, though, is quite difficult to measure, especially if processes are measured inclusively[3].
The second definition was formulated because of the impracticability of the first one. It was stated by Bjorken that,

A diffractive reaction is characterized by a large, non-exponentially suppressed rapidity gap in the final state. [3]

This definition is now being used to select diffractive events. Gaps in rapidity (or pseudorapidity) can be shown in 2-dimensional distributions of pseudorapidity and azimuthal angle. An illustration of a diffractive process can be seen in Fig. 1.3. Green dots represent particles created in the central system and red dots represent the forward-scattered particles. For the purpose of this thesis, red squares are the outgoing protons, and green squares are the J/ψ decay products, electrons. Crucial for a diffractive process are two gaps in pseudorapidity denoted $\Delta\eta_1$, $\Delta\eta_2$.
Particle diffraction accommodates multiple specific processes. These processes are represented in Fig. 1.4. The first one on the left, denoted QED[4] is a process conducted solely by the electromagnetic interaction. An example of electroproduction is process $\gamma\gamma \rightarrow e^+e^-$. The last interaction in Fig. 1.4 is a process called Double $\mathbb{P}$omeron Exchange (D$\mathbb{P}$E), defined as a transfer of 2 colorless objects with vacuum quantum numbers, called $\mathbb{P}$omerons. Originally, the name $\mathbb{P}$omeron came from phenomenological Regge theory [4], a model which generalized quantum mechanical angular momentum to complex values. The theory was applied to high-energy interactions by Chew and Frautschi [5], specifically to model the strong force in low-momentum transfer collisions. The $\mathbb{P}$omeron has had multiple interpretations(for example, as DGLAP or BFKL $\mathbb{P}$omeron), but in the Quantum Chromodynamics (QCD) interpretation, it is the sum of all contributions from multiple gluon states whose quantum numbers are 0:

$$|IP >= |\mathrm{gg} > + |\mathrm{ggg} > + \ldots \tag{1.7}$$

[2]Impact parameter is usually denoted b and is the distance between the centers of the colliding particles at the time of collision.

[3]Inclusive measurements are those where one does not measure all the particles in the final state. On the other hand, exclusive processes are those when all particles coming out of a certain interaction are measured.

[4]QED stands for Quantum Electrodynamics, a quantum field theory dedicated to the description of electromagnetic processes at the quantum particle level.

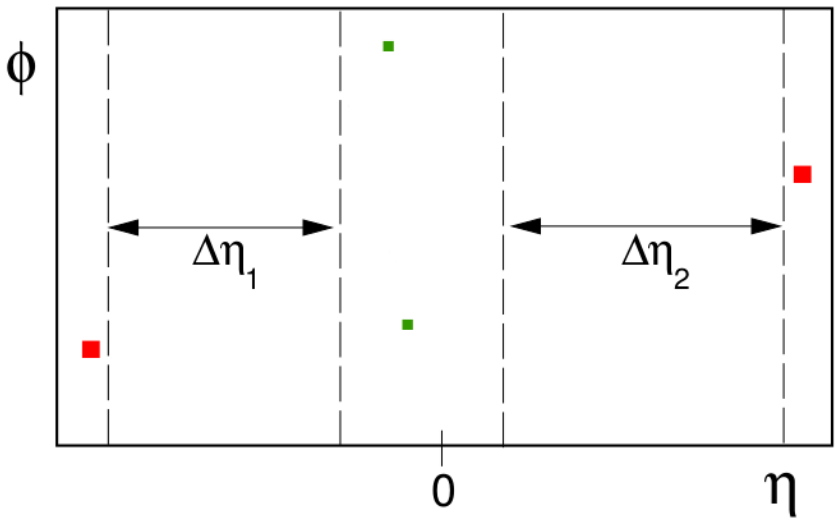


Figure 1.3: Pseudorapidity and azimuthal angle distribution in a diffractive process. Red squares represent the forward/backward particles scattered with low scattering angles. Green squares represent particles in the so-called central system. Particles in the central system are separated from the forward and backward regions of pseudorapidity.

The most dominant contribution is the lowest order contribution of 2 gluons [6]. The middle interaction shown in Fig. 1.4 is the photoproduction, the key process for this thesis. Photoproduction is a combination of the electromagnetic and the strong interaction. In terms of QCD, the process is often described with Feynman diagram shown in Fig. 1.5. The following section describes the process thoroughly and the motivation to study such processes.
One last note to end this section, the specific process which happens (out of those shown in Fig. 1.4) is determined by the restriction of the possible final states. The DPE, being a fusion of 2 objects both with 0 total angular momentum, produces systems of scalar or pseudoscalar particles. These processes involve, for example, the central exclusive production [7], [8] of h^+h^- pairs ($h = \pi,\ K,\ p$), but also other final states, which are a source of background contamination and will be mentioned further on. Photoproduction, due to the fact that a photon has spin 1, produces vector or pseudovector mesons such as ρ, ϕ or J/ψ. In the upcoming section, the focus is on J/ψ photoproduction.

1.3 Central Exclusive J/ψ Photoproduction

The attribute central exclusive refers to a case when the J/ψ particle (or its decay products) is the only particle created in the central system[5]. In the following sections, the kinematics of the process are described, then the physics motivation, and lastly, several theoretical models are mentioned and briefly illustrated.

[5]Previously, exclusive measurement has been described as the case, when all final state particles are measured. Central exclusive does not have the same meaning as exclusive.

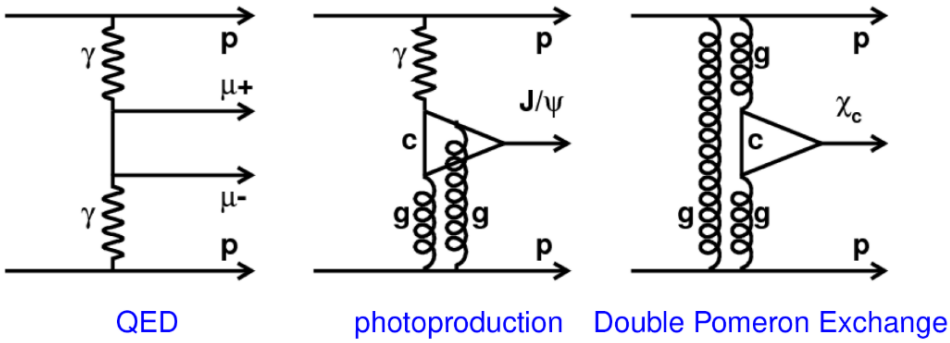


Figure 1.4: Diffractive processes: Double photon exchange, Photoproduction, and Double IPomeron Exchange. Taken from [9].

1.3.1 Kinematics

Before starting with the kinematics, one has to introduce 2 crucial variables to fully understand the picture. Both variables are used mainly for studying the inner structure of protons. The first one, called Bjorken x, represents the fraction of momentum of the proton carried by a constituent parton (quark or gluon) of the proton

$$x := \frac{p_{\text{parton}}}{p_{\text{proton}}}. \tag{1.8}$$

From the definition, the variable x reaches values from 0 to 1. The second variable is called virtuality, denoted as Q^2. It is a positive value defined as

$$Q^2 := -q^2, \tag{1.9}$$

where q is the four-vector of momentum transfer squared. Both variables are used in so-called called Deep inelastic scattering (DIS), a collision at energy high enough, after which one of the colliding particles is dissociated. DIS refers to collisions, when a highly-energetic probe (electron) is used to study the inner structure of a target (proton). More on this topic can be found in [10], [11]. Virtuality, in terms of deep inelastic scattering, can be understood as a resolution of the probe. The higher the virtuality, the bigger the resolution, and one is able to study smaller regions of the target.

In our case, the picture is less dramatic. We are mostly looking at collisions, when both protons stay intact. As can be seen in Fig. 1.5, the virtuality of the photon can be represented as the difference in momentum of one of the protons before and after the photon emission.
Another variable often used instead of the Mandelstam variable s, is the energy of the system consisting of the emitted photon and the target, denoted as $W_{\gamma p}$. The emitted photon interacts with the other proton via the strong interaction and a J/ψ is produced.
To relate the newly defined variables x and Q^2 to our process, we use relation [12]

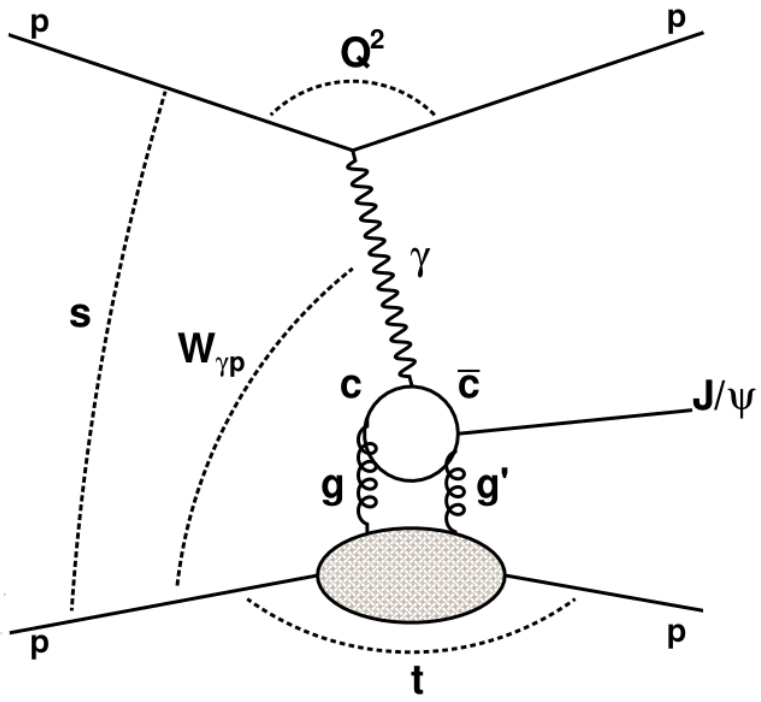


Figure 1.5: Photoproduction of J/ψ in terms of Quantum Chromodynamics theory.

$$x = \frac{M_{J/\psi}}{\sqrt{s}} e^{\pm y} \tag{1.10}$$

where $M_{J/\psi}$ is the mass of the quarkonium, s and y are the previously defined Mandelstam variable and rapidity. One often uses an equivalent relation [13]

$$y = \ln\left(\frac{2\omega}{M_{J/\psi}}\right) \tag{1.11}$$

which relates the rapidity to the energy of the photon ω. This relation will be important in defining the photon flux in Sec. 1.3.5. The scale of the process, Q^2 is approximately related to the mass of the J/ψ [12] with relation

$$Q^2 \sim \frac{M_{J/\psi}^2}{4}. \tag{1.12}$$

1.3.2 Motivation

As we said earlier, we are probing a proton target. The region we are studying is x ϵ $(0.015, 0.002)$ (by using relation Eq. 1.10 with our phase space parameters y ϵ $(-0.9, 0.9)$, $\sqrt{s} = 510$ GeV, and the mass of the J/ψ) and $Q^2 \sim 2.4$ GeV^2. The region is highlighted in red in Fig. 1.6. The blue and red lines which highlight the rapidity ranges represent values calculated for the LHC kinematics ($\sqrt{s} = 7$ TeV).

The region we are interested in, is overlapping previous data from HERA, which enables direct comparison to previous experimental results.

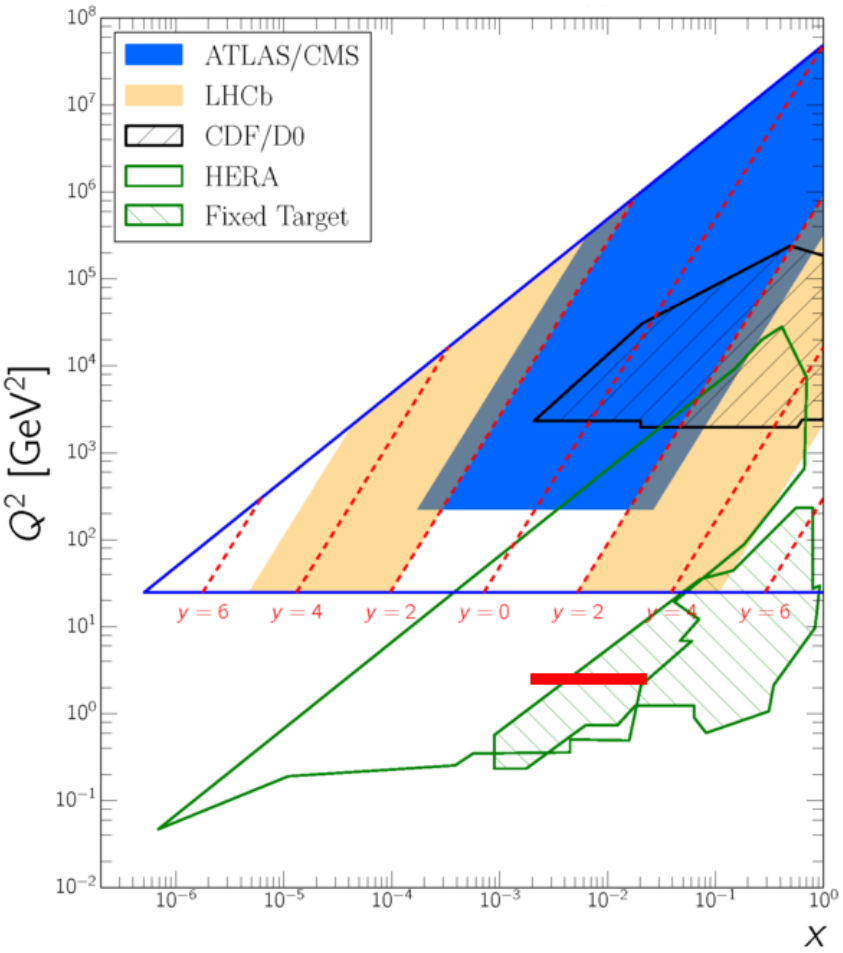


Figure 1.6: The kinematical coverage in x-Q^2 plot of previous and current experiments. The region studied with this analysis is highlighted in red.

Strong motivation comes from the fact that there exist multiple models which describe quarkonium production. This enables direct comparison between experimental data and theoretical predictions, which is rare in particle physics. Some of the models also describe proton-proton collisions such as the one implemented in the Monte Carlo generator STARlight [14]–[16] by Klein and Nystrand (KN) or the model by Goncalves and Machado [17], [18] (GM). Other models by Adeluyi and Bertulani [19] or Adeluyi and Nguyen [20], the model by Cisek, Schäfer and Szczurek [21] or by Lappi and Mäntysaari [22], [23] describe heavy ion ultraperipheral collisions only.
Both KN and GM models use photon flux as an ingredient to move from proton-proton collisions to the photon-proton system. This is described in Sec. 1.3.3. Once the virtual photon is produced, it can fluctuate into a different state limited by the Heisenberg uncertainty relations

$$|\gamma>= c_0|\gamma_0> + \sum_{V\epsilon VM} c_V|V> + \sum_q c_q|q\bar{q}> . \tag{1.13}$$

The first term is a state of a point-like photon, the first sum goes over all possible vector meson states, and the second over quarks. The first sum corresponds to the Vector Dominance Model (VDM), a phenomenological hadron model further described in Sec. 1.3.4. The second sum corresponds to the color dipole approach, further described in Sec. 1.3.5.

1.3.3 Photon Flux

The flux of photons was first mentioned by Enrico Fermi, but was fully developed independently by Weizsacker [24] and Williams [25]. It is standardly referred to as the Weizsacker Williams Approximation (see more in ref. [10]). The flux of photons produced by a fast charged particle is a fairly complicated relation (see ref. [26]). For protons though, the formula can be approximated [27]

$$N_{\gamma/p}(\omega) = \frac{\alpha_{em}}{2\pi}\left[1+\left(1-\frac{2\omega}{\sqrt{s}}\right)\right]\left[\ln(D) - \frac{11}{6} + \frac{3}{D} - \frac{3}{2D^2} + \frac{1}{3D^3}\right]. \tag{1.14}$$

The photon energy is denoted as ω as before and $D = 1 + (0.71\text{GeV}^2)/Q^2_{\text{min}}$, where Q^2_{min} is the minimal virtuality [18]. It can be simplified

$$Q^2_{\text{min}} = \frac{\omega^2}{\left(1-\frac{2\omega}{\sqrt{s}}\right)} \approx \left(\frac{\omega}{\gamma_L}\right)^2 \tag{1.15}$$

where γ_L is the Lorentz factor of the proton. There have been some modifications to this result by Kniehl [28] who included the magnetic dipole moment and the corresponding magnetic form factor. One important remark is that in proton-proton collisions, both protons can emit a photon. Therefore, the final cross section is an interference of both cases

$$\frac{\sigma_{p_1p_2\to p_1 J/\psi p_2}(y)}{\text{dy}} = N_{\gamma/p_1}(y)\sigma_{\gamma p_2\to J/\psi p_2}(y) + N_{\gamma/p_2}(y)\sigma_{\gamma p_1\to J/\psi p_1}(y). \tag{1.16}$$

The first term represents the flux of photons from the first proton, which interacts with the second proton. The other term represents the opposite meaning the second proton emits a photon which then interacts with the first proton. In addition, the cross section is written in terms of rapidity instead of photon energy ω, by using Eq. 1.11. The virtuality of the photons is, in fact, very limited, which is why they are often referred to as quasi-real photons.

1.3.4 Vector Dominance Model

The vector dominance model makes the assumption that the photon fluctuation is only restricted by its quantum numbers; therefore, it is allowed to fluctuate to a vector meson [29] ($\rho,\ \omega,\ \phi,\ J/\psi,\ \psi(2S),\ \Upsilon(1S),\ \Upsilon(2S),\ \Upsilon(3S)$). By using eikonalization and the optical theorem [12], the cross section of vector meson production is related to the cross section of elastic scattering of a vector meson on a proton

$$\left.\frac{d\sigma(\gamma+p\to V+p)}{dt}\right|_{t=0}=\frac{f_V^2}{4\pi\alpha_{\mathrm{em}}}\left.\frac{d\sigma(\gamma+p\to V+p)}{dt}\right|_{t=0}. \tag{1.17}$$

Factor α_{em} is the electromagnetic coupling constant, and f_V are the vector meson - photon coupling constants determined experimentally. The previously defined vector meson constants c_V are related to our new ones with relations

$$c_V^2=\frac{4\pi\alpha_{\mathrm{em}}}{f_V^2}. \tag{1.18}$$

The coupling constants are determined experimentally. The STARlight Monte Carlo generator uses relation [14]

$$\sigma(\gamma+p\to J/\psi+p)=\sigma_p\left[1-\frac{(m_p+m_{J/\psi})^2}{W_{\gamma p}^2}\right]^2 W_{\gamma p}^{\epsilon} \tag{1.19}$$

for J/ψ photoproduction cross section, where σ_p and ϵ are parameters restrained by a fit on HERA data shown in Fig. 1.10. Therefore, comparison to STARlight data essentially means comparison to HERA data.

1.3.5 Color Dipole Model

The dipole model is more complex; the model stands on the basis of perturbative QCD. In addition to the fluctuations to the $q\bar{q}$ states, the model also allows higher Fock states

$$|\gamma^*>=|q\bar{q}>+|q\bar{q}g>+|q\bar{q}gg>+\cdot, \tag{1.20}$$

which involve gluon emission. These states are suppressed because the more massive the state is the shorter it survives [30]. In our discussion, we will restrict ourselves to the term $q\bar{q}$ (dipole).
We are trying to obtain the amplitude of the transition from the state γ^* to the vector meson J/ψ. The relation one would use[6] [31]

$$\mathfrak{M}(s,Q^2)=\int_0^1 dz\ \int d^2r\ \Psi^{J/\psi}\sigma_{q\bar{q}}(r,s)\Psi^{\gamma^*}, \tag{1.21}$$

[6]Excluding the discussion on the spin summation.

where z is the fraction of photons momentum carried by the quark ($1-z$ is carried by the antiquark), r is the transverse size of the dipole, and $\Psi^{J/\psi}$, Ψ^{γ^*} are the wave functions of the vector meson and the photon in light cone coordinates (see more in ref. [22]), respectively. The entire scheme of vector meson production can be seen in Fig. 1.7. The wave function of the charmonium is obtained from the Schrödinger equation and various potentials between the q and the $\bar{q}$. Some of the potentials often used for modelling are standard harmonic oscillatory potential [32], powerlike potential [33], [34], logarithmic [35], or more complex, such as Buchmüller-Tye [36] or Cornell potential [37], [38]. The nonrelativistic wave functions are then boosted to the light-front frame.

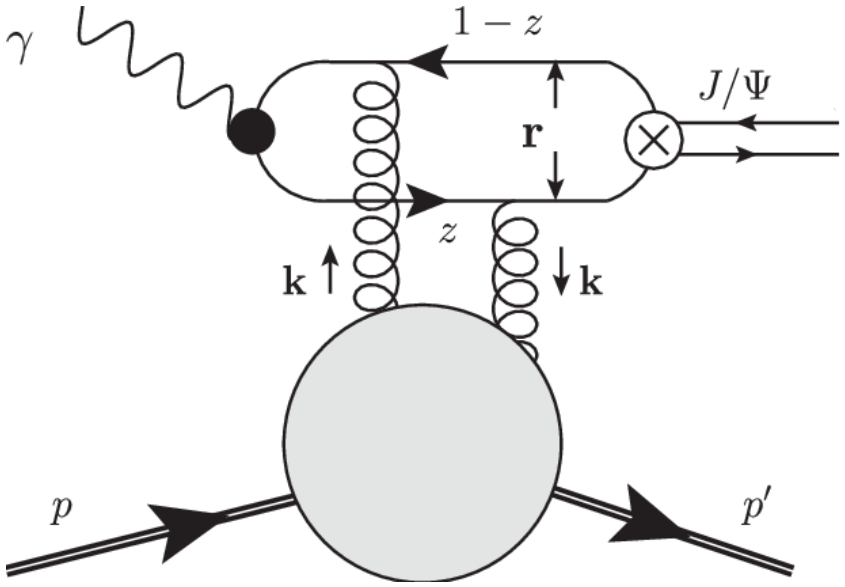


Figure 1.7: Sketch of quarkonium production using the dipole model.

One of the main ingredients for the cross section calculation is the elastic cross section of the $q\bar{q}$ dipole on the proton $\sigma_{q\bar{q}}(r,s)$. Naturally, the larger the dipole size, the larger the cross section. But the dipole size is limited by quark confinement. There are several phenomenological parametrizations available, such as Golec-Biernat-Wüsthoff [39] (GBW) or Kopeliovich-Schäfer-Tarasov [40]. In the limit of small r, both can be expressed as $\sigma_{q\bar{q}}(r,s) = C(s)r^2$.
What is more interesting in terms of QCD, is the GBW saturation model [41]. The model is improved by including DGLAP[7] evolved gluon distribution [41]. The DGLAP improved saturation model for the dipole cross section in a complicated form can be found in ref. [41], here, only the simplified version for small dipole sizes is shown

$$\sigma_{q\bar{q}}(r,x) \approx \frac{\pi^2 r^2 \alpha_s(C/r^2)}{3} x g(x, C/r^2), \tag{1.22}$$

[7]Dokshitzer, Gribov, Lipatov, Altarelli, and Parisi evolution equation [42]–[46], which is one of the most prolific equations of Quantum Chromodynamics. It describes transitions in parton distribution equations between different values of virtuality Q^2, see more in refs. [10], [11].

where $g(x, \mu^2)$ is the gluon distribution, taken at the scale $\mu^2 = C/r^2$ (see more in refs. [10], [11]). Factor C is a parameter. Again, the values were fitted to HERA data [47]. This enables direct comparison of gluon distribution functions to experimental results [48].

1.4 Recent Results in J/ψ Photoproduction

This section discusses recent results in dielectron (and dimuon) production and J/ψ photoproduction. As was mentioned before, similar measurements in the central rapidity region have been done at HERA, although the collisions were electron-proton. Proton-proton measurements have been done at the ALICE experiment and also at the LHCb in forward rapidity region.
The ALICE collaboration published results on the dielectron production in the range up to 3.3 GeV. The dielectron pairs have been produced in the central rapidity region ($|\eta_e| < 0.8$) and energy $\sqrt{s} = 7$ TeV. The distribution is shown in Fig. 1.8. Both opposite sign pairs (e^+e^-) and same sign (e^+e^+ or e^-e^-) pairs are shown. The distribution is clearly showing an exponential decrease (y axis is logarithmic) in invariant mass, which is expected. The only place where the distribution does not follow the expected shape is the region below $m_{ee} = 0.5$ GeV, where the detector efficiency effects are significant. One can also notice the peak at around 3 GeV in the red opposite-sign pairs, which corresponds to the J/ψ peak. It is the reason why there is a discrepancy between the same-sign and the opposite-sign pairs in the high m_{ee} region (> 2.5 GeV). The following results are a combination of multiple

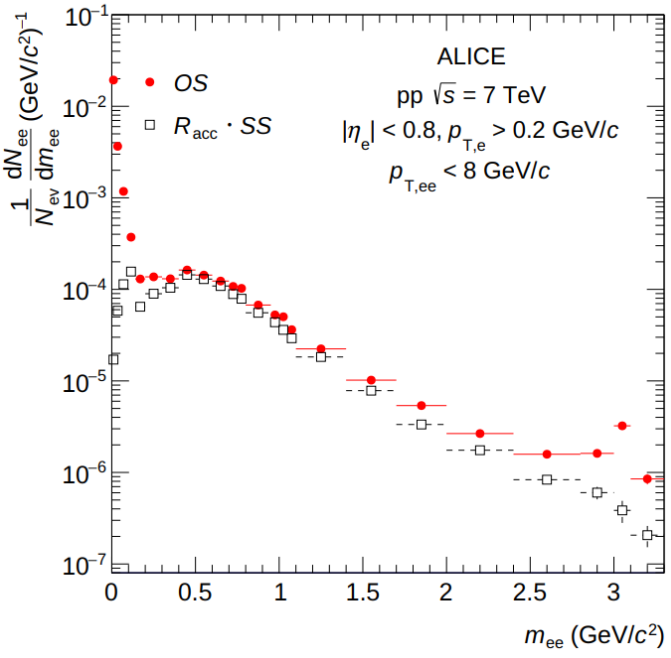


Figure 1.8: Invariant mass spectrum of dielectron pairs at ALICE. Taken from ref. [49].

experiments: HERA experiments H1 [50], [51] and ZEUS [52], LHCb [53], [54], and

previous fixed target measurements [55], [56]. The distribution shown in Fig. 1.10 is a cross section dependence on the variable $W_{\gamma p}$. One notices a clear linear dependence when both axes are logarithmic, especially in the region above 11 GeV. This distribution is used to restrict parameters in the STARlight generator [14].

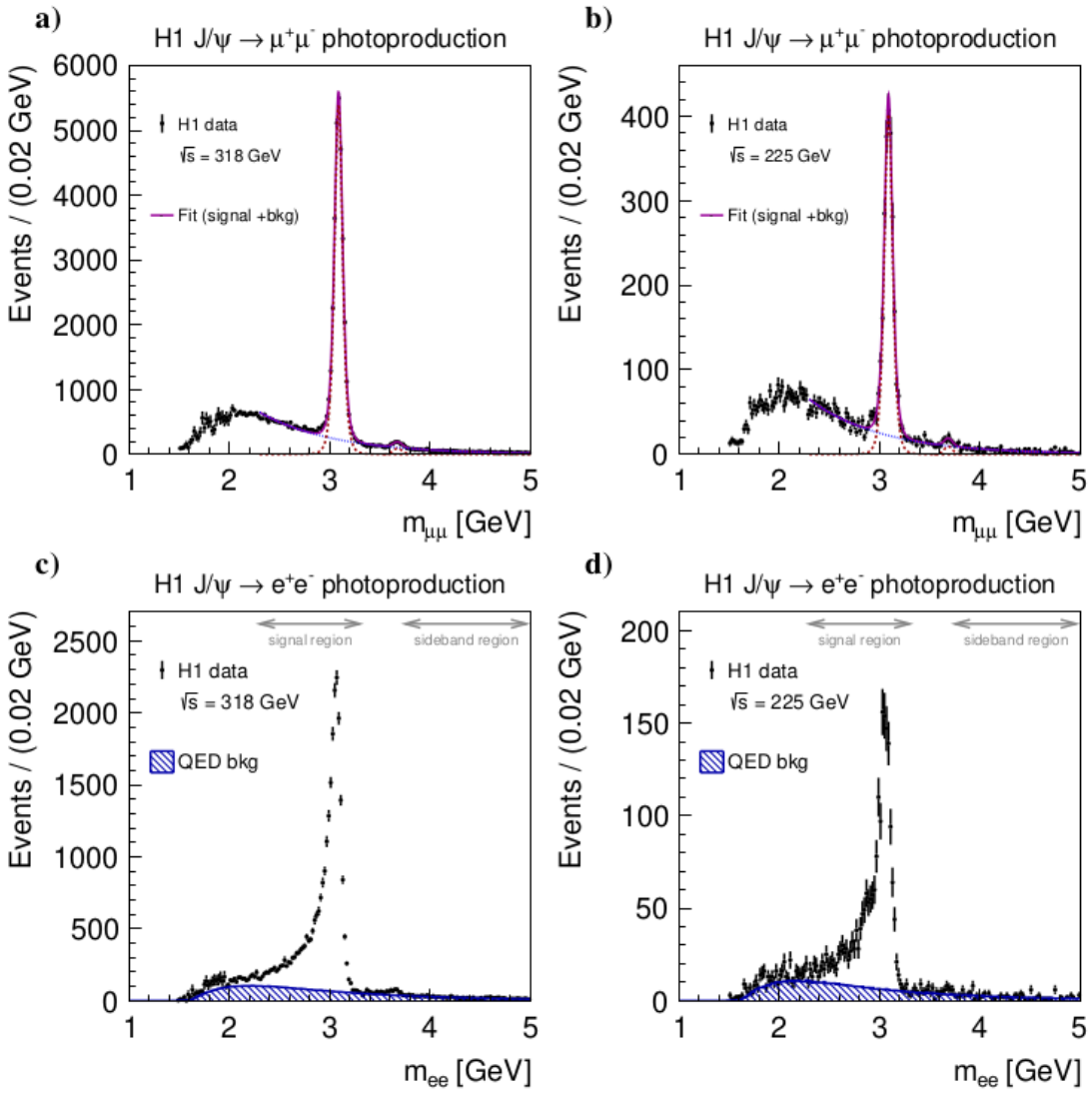


Figure 1.9: H1 Collaboration results of J/ψ photoproduction fits. Plots **a)** and **b)** show J/ψ photoproduction in the $\mu^+\mu^-$ decay channel; the plots below show the e^+e^- channel. Results are shown for 2 different values of energy of collision $\sqrt{s} = 318$ GeV and $\sqrt{s} = 225$ GeV. Taken from ref. [51].

Next are the results of the fits of invariant mass distributions from the HERA H1 collaboration, shown in Fig. 1.9. The top row shows results in $\mu^+\mu^-$ channel, bottom on the other hand e^+e^-. Both channels are relevant for this analysis, because the branching ratios for electron and muon pairs is practically the same. Left column shows results for collision energy $\sqrt{s} = 318$ GeV, right for $\sqrt{s} = 225$ GeV. In all distributions, the J/ψ peaks are clearly visible at around 3.1 GeV. In the top left corner distribution (muon channel with higher energy), one can also see a very small peak for ψ(2S) peak at around 3.7 GeV. The background process in all plots is either

$\mu^+\mu^-$, or the e^+e^- QED continuum. The continuum has a standard shape of rise and then exponential decrease, as was discussed in Fig. 1.8. The rise is due to the inefficiency of detectors for low-momentum particles and also the specific clustering parameters for track reconstruction in calorimeters.

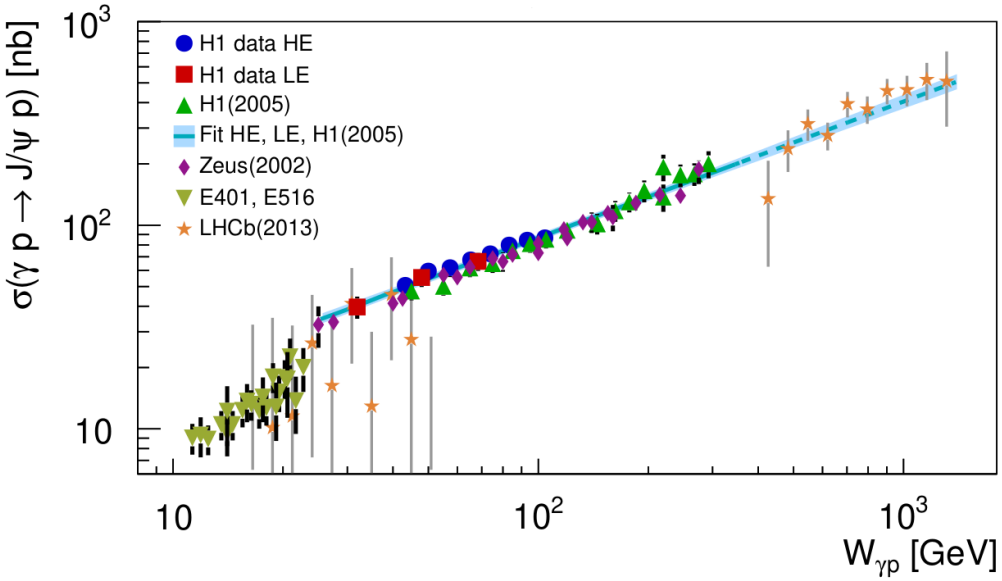


Figure 1.10: Cross section for elastic J/ψ production at multiple experiments w.r.t. energy of the photon-proton system $W_{\gamma p}$. Plot taken from ref. [51].

One more effect worth mentioning manifests in the left side of the peaks. In the muon channel, the peak looks almost symmetric, but for electrons it is not the case. This is due to the fact that sometimes a J/ψ candidate decays to 3 daughter particles $e^+e^-\gamma$ or during the propagation of electrons through the detector, they undergo bremsstrahlung process during which they emit a photon. The probability of bremsstrahlung for electrons is about 200 times higher than for muons because it depends linearly on the invariant mass of the particle, therefore, this is relevant only for electrons. The emitted photons are of quite low energy, we do not detect them or reconstruct them, their only manifestation is the asymmetric peak in the e^+e^- distribution.
Next are the results from the LHCb experiment [53], [54] of J/ψ photoproduction in the forward rapidity region. Left plot in Fig. 1.11 shows results of p+p collisions at $\sqrt{s} = 7$ TeV, right plot at $\sqrt{s} = 13$ TeV. Both distributions are in the muon decay channel. In this case, the articles also provided the measured cross sections, which in the case of lower energy are

$$\begin{aligned} \sigma_{pp\to J/\psi\to\mu^+\mu^-}(2.0 < \eta_\mu < 4.5) &= 291 \pm 7 \pm 19 \text{ pb} \\ \sigma_{pp\to\psi(2S)\to\mu^+\mu^-}(2.0 < \eta_\mu < 4.5) &= 6.5 \pm 0.9 \pm 0.4 \text{ pb} \end{aligned} \tag{1.23}$$

and in the higher collision energy case

$$\begin{aligned}\sigma_{pp\to J/\psi\to\mu^+\mu^-}(2.0<\eta_\mu<4.5) &= 435 \pm 18 \pm 11 \pm 17 \text{ pb} \\ \sigma_{pp\to\psi(2S)\to\mu^+\mu^-}(2.0<\eta_\mu<4.5) &= 11.1 \pm 1.1 \pm 0.3 \pm 0.4 \text{ pb}.\end{aligned} \tag{1.24}$$

In the first case, the errors are statistical and then systematic. In the second case, the last error is the luminosity determination uncertainty. Due to much larger collision energy, the cross section is also higher than at HERA or RHIC energies.

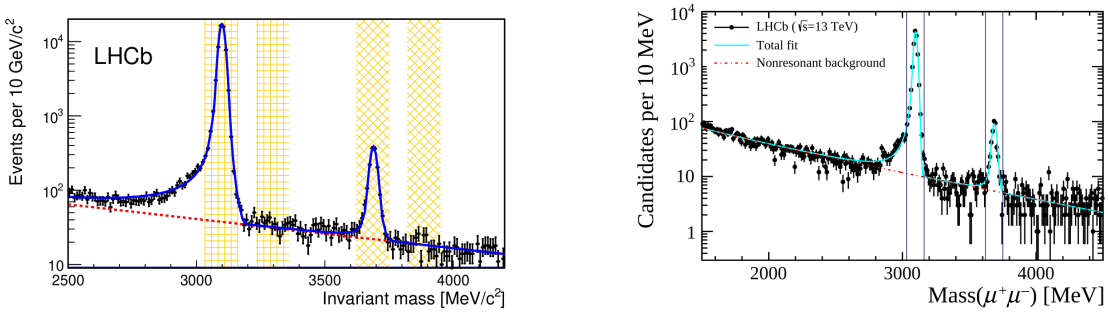


Figure 1.11: Measurement of central exclusive production of J/ψ and $\psi(2S)$ in the dimuon channel at the experiment LHCb. The left plot shows the distribution taken at energy $\sqrt{s} = 7$ TeV, the right distribution is for $\sqrt{s} = 13$ TeV. Taken from refs.[53], [54].

Finally, we mention a J/ψ measurement done at the experiment STAR [57]. The STAR experiment has not published a result on J/ψ photoproduction in proton-proton collisions, but only in Au+Au, which is slightly more complicated in terms of physics. The J/ψ photoproduction in heavy ion collisions can either be coherent, when both nuclei stay intact after the collision. However, the nuclei can be broken up by additional soft photons from Coulomb excitation. The other productions are incoherent, meaning the production is via interaction between a photon and a nucleon. There are 2 possible outcomes: either the interaction is elastic or dissociative. The dissociative production is a hard scattering when the target nucleus breaks up [57].

The various methods of production can be differentiated by studying the p_T spectra of produced J/ψ. Coherent production is characterized by very small momentum of the produced J/ψ. The majority of incoherently elastic J/ψ particles sit approximately between 0.02 to 0.5 GeV. The last category, incoherent dissociative, is characterized by higher values of transverse momenta ($p_T > 0.5$ GeV) [57]. This can be seen in Fig. 1.12. One can also study the nuclear suppression by comparing the exclusive photoproduction of J/ψ in heavy ion collisions to the same process in proton-proton collisions. Recent measurements done at STAR [58] show evidence of suppressed vector meson production in heavy ion collisions by comparing it to cross section data in proton-proton collisions at the HERA experiment.

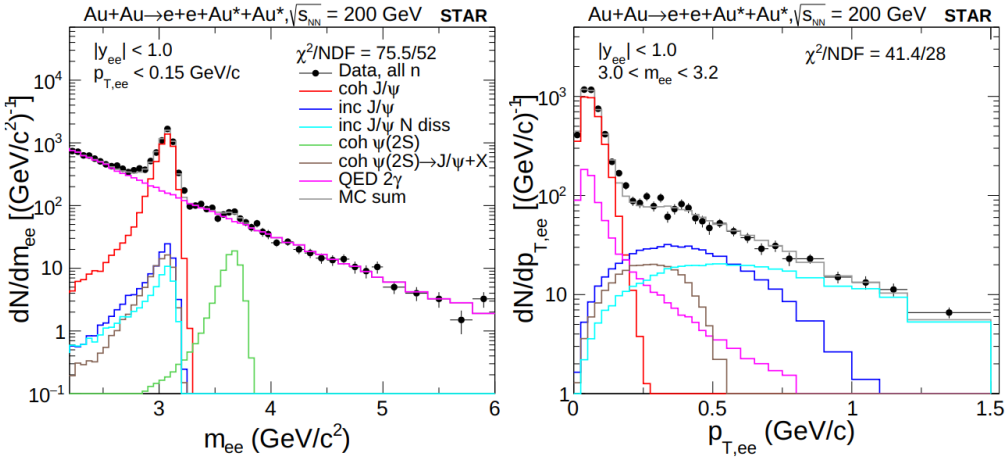


Figure 1.12: The photoproduction of J/ψ in Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. The left plot shows the distribution of the invariant mass of e^+e^- pairs. Left plot shows the transverse momentum distribution for pairs with invariant mass in the window (3.0, 3.2) GeV. Both plots include samples from Pythia Monte Carlo generator, which are scaled to the data and highlight the differences mainly in transverse momentum. Taken from ref. [57].

Chapter 2

Experimental Setup

Experiment STAR is located at the Relativistic Heavy Ion Collider (RHIC), currently the second-largest particle collider and accelerator. It is located in the United States, in the Brookhaven National Laboratory in the state of New York. The RHIC is remarkable in its ability to collide a wide range of collision systems. Whether it is a broad range of nuclei from hydrogen (protons) and deuteron up to gold and uranium nuclei, or by varying the center-of-mass energy of collision from several GeV up to 510 GeV. At the forefront of RHIC is the Solenoidal Tracker At RHIC (STAR). In the following sections, the experiment is described with its detector and subdetector systems which are used in the analysis. Detectors such as the Time Projection Chamber (Sec. 2.2), the Barrel Electromagnetic Calorimeter (Sec. 2.4), or the Roman Pot system (Sec. 2.5) are described, which are crucial for the analysis, but so are the detectors used in the triggering, such as the Time of Flight (Sec. 2.3) or the Beam Beam Counters (Sec. 2.6). At the end, Sec. 2.7 discusses the data acquisition process and triggering crucial for particle physics experiments.

2.1 Experiment STAR

Experiment STAR is the only experiment located at the Relativistic Heavy Ion Collider (RHIC) that has been taking data throughout RHIC operations. Originally, the purpose of the detector was to study Quark Gluon Plasma (QGP), its properties, the phase diagram of Quantum Chromodynamics, and the spin of protons. Over the years, the detector subsystems have been upgraded and it has become a multipurpose detector system. This system consists of many different detectors where each one has its own purpose, efficiency, phase space range, and acceptance.
The coordinate system at STAR is identical to all other HEP experiments. Bunches of particles moving through the beam pipe collide at the interaction points at which detector systems are located. Particles are moving along the z axis, also called the beamline. The transverse plane is the x-y plane. Most of the detectors are in a solenoidal magnetic field created by large coils. A cross section of the STAR detector can be seen in Fig. 2.1. Starting from the point exactly in the middle, the interaction point, the light green cylinder around is the Heavy Flavor Tracker (HFT), orange is the Time Projection Chamber. The following pink layer is the

Time of Flight detector, followed by the green Barrel Electromagnetic Calorimeter. STAR has never had a hadron calorimeter, which would be positioned furthest from the interaction point. Purple detectors close to the beamline are the Beam Beam Counters. The Roman Pot system of detectors and Zero Degree Calorimeters are positioned further along the beamline, thus they are not in the picture. This is an older picture which corresponds to the experimental setup at the time of data taking for this thesis except for HFT. Since then, STAR has been upgraded several times and detectors such as the forward calorimetry system and event plane detector have been added.
The main detector to track the produced particles in the central system is the Time Projection Chamber, which is discussed in Sec. 2.2. Time of Flight detector and its properties are discussed in Sec. 2.3 followed by the description of the 5. experiment at RHIC- the PP2PP. This experiment and the Roman Pot system of detectors are discussed in Sec. 2.5. The last section covers some of detectors, which are not directly used in this analysis, nonetheless are important for many other measurements done at STAR.

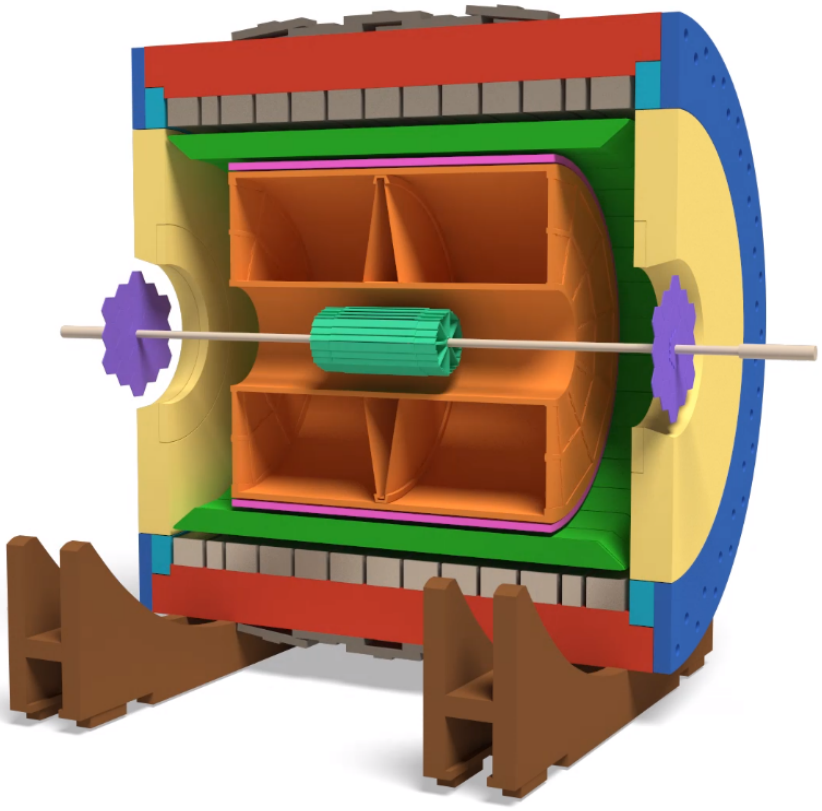


Figure 2.1: Overview of the STAR detector and its subsystems. The green detector in the middle is a Heavy Flavor Tracker, orange is the Time Projection Chamber, pink is the Time of Flight detector followed by magnetic coils and calorimeters around the cylinder as well as at the endcaps. Taken from ref. [59].

2.2 The Time Projection Chamber

The Time Projection Chamber (TPC) is the heart of STAR. Its purpose is to track and reconstruct charged particles created in collisions. It was created as a high multiplicity detector with relatively low readout times compared to, for example, the Time of Flight detector. It is of cylindrical shape with a high-voltage membrane in the middle. It is 420 cm long, its inner diameter is 100 cm, and its outer 400 cm [60].

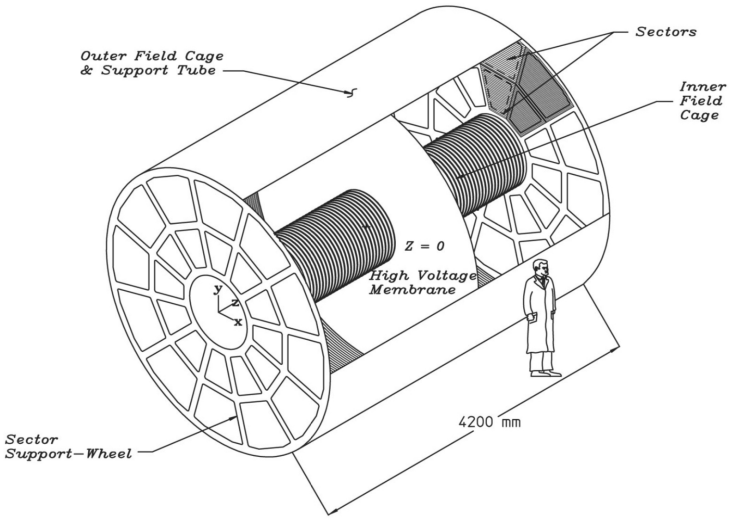


Figure 2.2: Schematic view of the STAR Time Projection Chamber. Taken from ref. [60].

As a charged particle is created in a collision, it is scattered at some angle and moves along a curved trajectory due to the magnetic field. To measure a particle that enters the TPC, it needs to be moving with transverse momentum larger than a minimal value of 150 MeV, otherwise it will spiral inside the TPC and never reach the outer radius. The magnetic field, which causes this curvature, can operate at values $\pm$ 0.25 or $\pm$ 0.5 T [60].
The Time Projection Chamber is filled with a P-10 mixture of ionizable gas consisting of 90 % of Ar, 10 % of CH_4, and a small amount of isobutate [61]. The particle moves through the medium and creates electron-ion pairs, which drift in the opposite directions due to the electric field in the longitudinal direction. The electric field between the membrane and both endcaps was constructed for values of 140 V/cm and the central membrane is operated at values of -28 kV [60]. Electrons, which are moving much faster than ions, drift towards the endcaps of the detector, where the signal is amplified and read out by a combination of Multi Wire Proportional Chambers (MWPC) and cathode pads. The average drift velocity of electrons is 5.45 cm/μs [60].
Cathode pads at the endcaps are organized into an inner and an outer sector. The inner sector contains 1,750 pads in 13 rows, and the outer sector contains 3,942

pads organized in 32 rows [60]. Part of the Beam Energy Scan (BES) II[1] inner TPC (iTPC) upgrade was to increase the number of rows in the inner sector to 40 [62]. Part of the iTPC upgrade was the increase of the pseudorapidity range from $|\eta| < 1$ to $|\eta| < 1.5$ [62]. Thanks to the cylindrical shape, TPC covers the full azimuth.
A point on the trajectory of a particle in TPC is reconstructed from a cluster of multiple electron hits. Based on multiple reconstructed points of a particle, one can determine its trajectory in the transverse x-y plane. The longitudinal z axis position is determined by the drift time of electrons from the point of ionization to the readout [60].

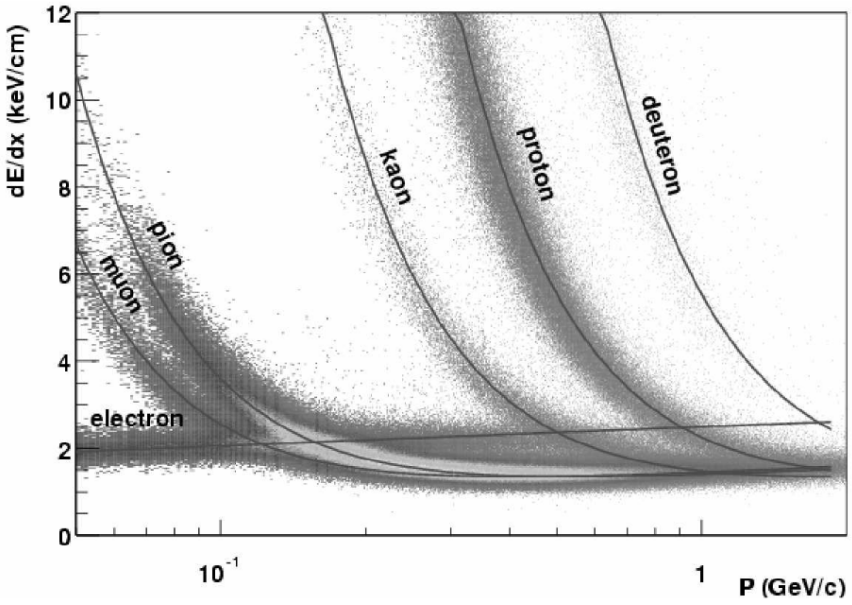


Figure 2.3: Correlation plot of energy loss per unit length and transverse momentum of a particle. For a specific momentum, one can identify the particle. Taken from ref. [60].

Thanks to the proportionality of MWPC, based on the amplitude of the signal, one can differentiate among particles with respect to their energy loss per unit of length [60], as one can see in Fig. 2.3. To eliminate the momentum and energy loss dE/dx from the measurement, one can use the following formula,

$$n\sigma = \frac{ln(dE/dx_{mea}) - ln(dE/dx_{th})}{\sigma(ln(dE/dx))} \tag{2.1}$$

where dE/dx_{mea} is the measured energy loss, dE/dx_{th} is the theoretical prediction for a certain particle, and $\sigma(ln(dE/dx))$ is the resolution [63]. The theoretical prediction is based on Bichsel formalism [64], which is a modified version for STAR of the famous Bethe-Bloch formula for particle energy loss traversing a medium.

[1]The data for the analysis are from Run17, which took place in 2017, so the upgrades for BES II have not been installed at that time yet.

2.3 Time of Flight Detector

Time of Flight (TOF) is the next layer of detectors after the Time Projection Chamber. It covers the full azimuthal angle and pseudorapidity range $|\eta| < 1$. As was mentioned before, it is a fast detector used to reduce pile-up[2] and background. The quick timing also plays a role in triggering of interesting events. As the name suggests, it is used to measure the time of flight of a particle from the moment of collision until it reaches the barrel ToF [65].
The position of the vertex is determined by a pair of Vertex Position Detectors (VPD), but what is more interesting for timing is the Pseudo Vertex Position Detector (pVPD). The pVPD is an integrated system in each VPD that determines the "start" time of collision. It works closely with the Time Of Flight patch (TOFp) that is the "stop" detector integrated within TOF [66]. Both of the systems are illustrated in Fig. 2.4. The time of flight of a particle from the point of collision until the barrel TOF determines the velocity of a particle, and if momentum is known, one is able to determine the invariant mass of the particle and identify it [67].

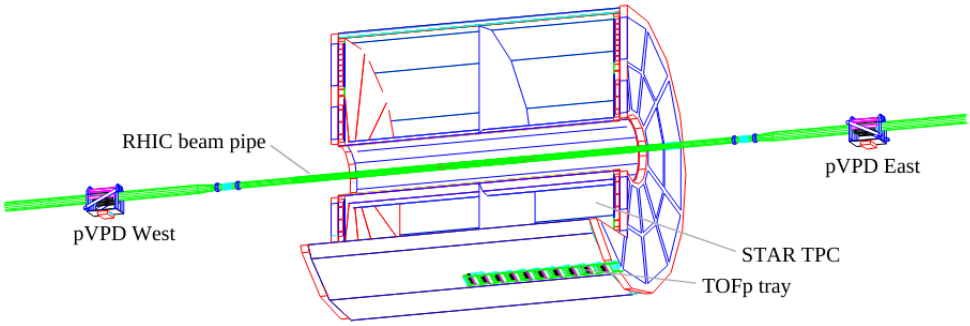


Figure 2.4: An illustrative picture of pVPD and TOFp, which determine the time of flight of a particle. Taken from ref. [66].

The barrel Time of Flight detector consists of 120 trays, each tray contains 32 Multi-Gap Resistive Plate Chamber (MRPC) modules [68]. An illustrative picture of a MRPC can be seen in Fig. 2.5. Each module has 7 layers of glass with a resistivity of 10^{12} Ω cm with 6 gaps of thickness around 0.22 mm. The whole structure is in a gas mixture 95 % $C_2H_2F_4$ and 5 % i-C_4H_{10} [68]. Signal from a particle moving through a module is amplified in an electric field of size 106 kV/cm [68]. The amplification works in avalanche mode, which is fast but lacks proportionality.
The original efficiency of a single tray for R&D in 2003 was tested in CERN and determined to be 98 % with a time resolution of 50 ps [69]. After installing trays at STAR, the time resolution was measured again and determined to be approximately 80 ps. Obviously, with damage to the material over more than 15 years of data taking, the efficiency and time resolution have degraded.

[2]Pile-up in terms of HEP and detectors happens when there are multiple independent collisions in a bunch crossing. This causes difficulty for detectors to differentiate which track belongs to which event/collision.

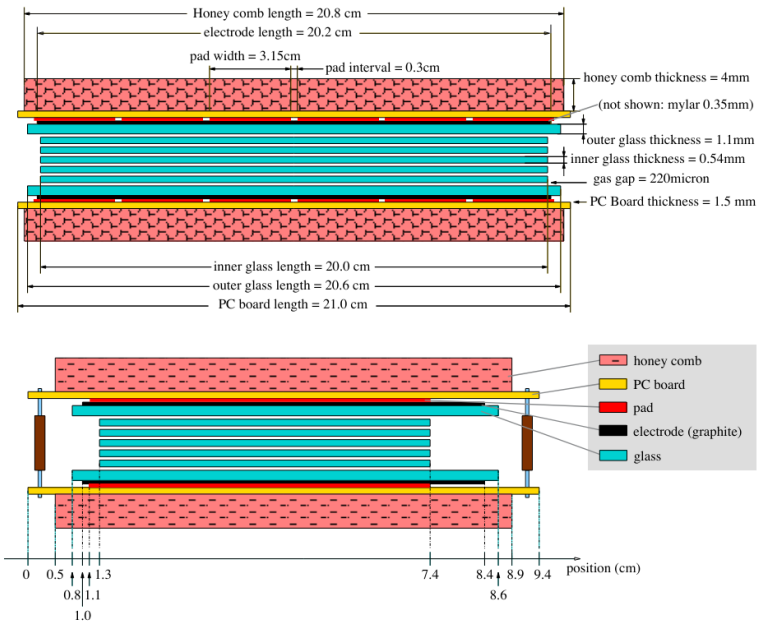


Figure 2.5: A cross section view of one of the STAR barrel TOF MRPC module. Taken from ref. [70].

2.4 Barrel Electromagnetic Calorimeter

The Barrel Electromagnetic Calorimeter (BEMC) matches the TPC's pseudorapidity and the azimuthal angle range, therefore, it covers $|\eta| < 1$ and full azimuth [71]. The detector itself plays a major role in the offline analysis as well as in triggering. The calorimeter was created to measure electrons and photons, which are used as probes of the quark gluon plasma (QGP). The goal was to use the BEMC for reconstruction of W, Z leptonic decays, direct photons from the QGP, Drell-Yan processes, and for studying jets and jet substructure. This means the BEMC is required to measure a wide energetic range of particles with high spatial resolution. The maximum depth of the detector was chosen to be approximately 20 radiation lengths [71] at a pseudorapidity value $\eta \approx 0$.

The calorimeter is located behind the TOF and the solenoidal magnet at a radius of about 220 cm from the z axis. The detector consists of 120 modules, in which each module is made up of 40 separate towers. The position with respect to TPC and TOF can be seen in Fig. 2.7. Each tower is a separate sampling calorimeter composed of lead and plastic scintillator plates, which are pressed together to minimize the distances between them. The entire scheme of a module can be seen in Fig. 2.6 on the right. The signal is read out by Photomultiplier tubes (PMT). Each scintillator is connected to a wave length shifting (WLS) fiber, which leads light from the plate to a PMT box on the back side of the calorimeter. The PMT boxes are located away from the magnetic field due to their high sensitivity. All 21 fibers that lead light from 21 scintillating plates are connected to a single PMT where they uniformly illuminate the photo-cathode and transform light into electrical signal [71].

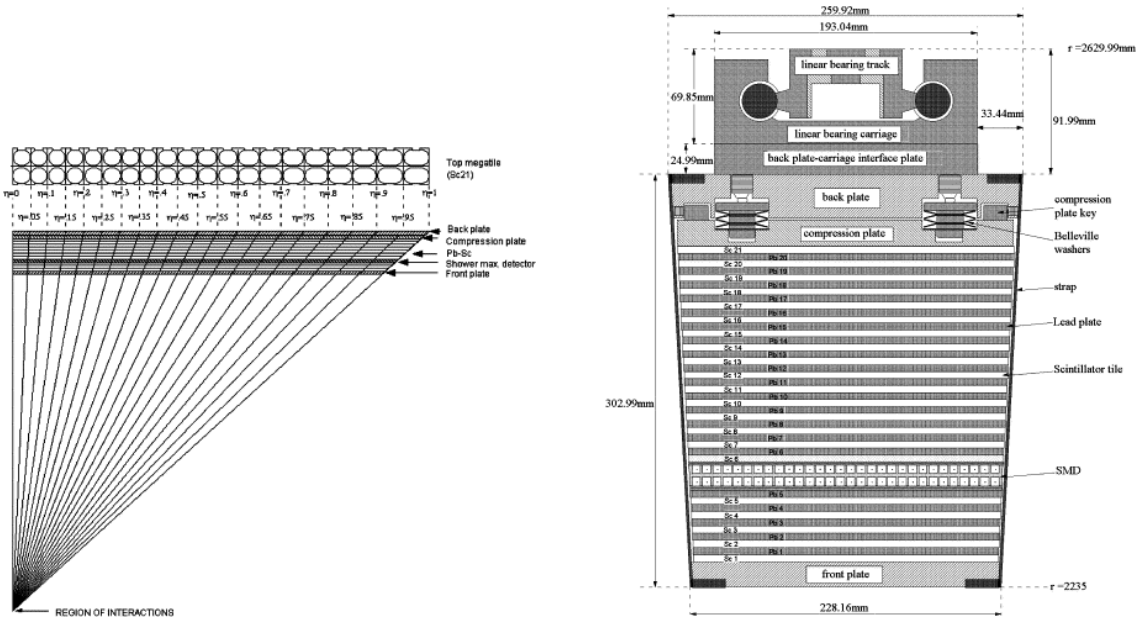


Figure 2.6: Right: The BEMC rotation w.r.t. the interaction point to maximize the depth of the detector and minimize cross-talk between towers. Right: The BEMC module scheme with specific lead and scintillator plates.

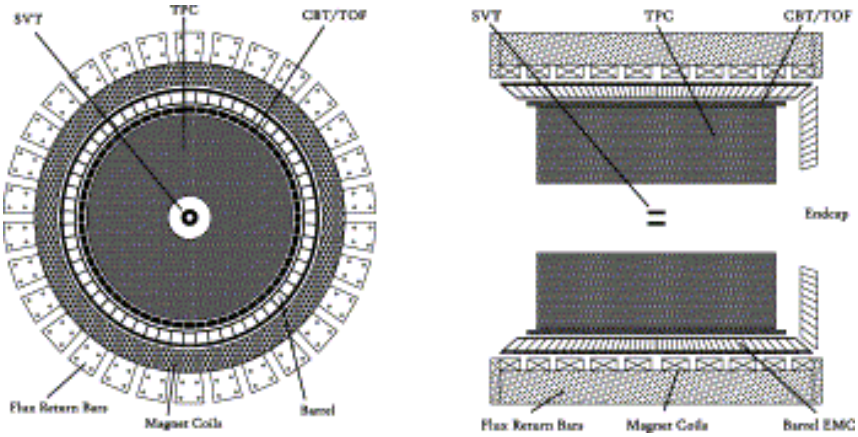


Figure 2.7: The position of the BEMC with respect to the TPC, TOF, and the solenoidal magnet. The left picture is the view of the beam, the right is from the perpendicular plane to the direction of the beam. Taken from ref. [71].

BEMC is considered to be a fast detector, with readout times below 700 ns. Therefore, information from the BEMC is used in the STAR level-0 trigger information, which is more thoroughly described in Sec. 2.7. For this purpose, only the BEMC primitives are used and not the entire reconstructed data. This analysis relies on the BEMC triggered events as well as uses the fully reconstructed information in the offline analysis.

2.5 The PP2PP Experiment and the Roman Pots

The PP2PP experiment was at the beginning considered to be the 5. experiment at the Relativistic Heavy Ion Collider with a separate goal from all the other experiments. As STAR is a heavy ion collider created to measure Quark Gluon Plasma, PP2PP was designed to measure elastic proton collisions with small momentum transfer. The experiment used a detection system called Roman Pots. This system was first used at ISR experiment at CERN [72].

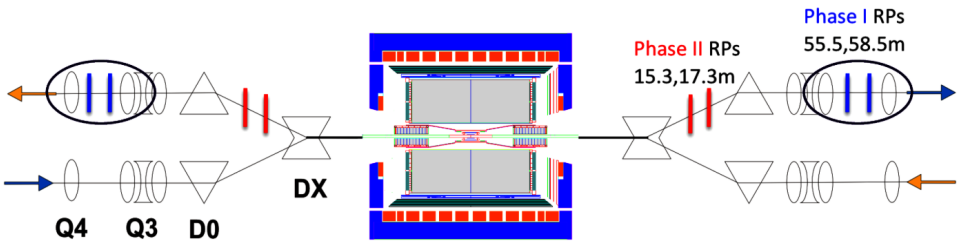
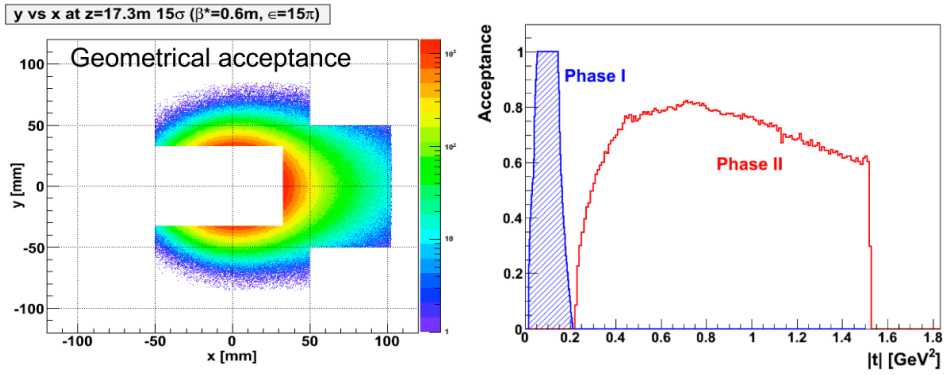


Figure 2.8: An illustrative scheme of the PP2PP experiment with differences between position of detectors in Phase I and the planned positions for Phase II. Taken from ref. [72].

The operation of the experiment was divided into 2 phases. Phase I setup officially started data taking in May 2003. The Roman Pot system of detectors was installed in 2 stations on each side of the interaction region. Each station hosts 2 Roman Pot vessels positioned above or below the beamline, so altogether, the system is made up of 8 vessels [7]. These vessels are not set in place; their vertical position with respect to the beamline can be changed.
The position of detectors during Phases I and II can be seen in Fig. 2.8. Stations during Phase I were positioned 55.5 and 58.5 m from the interaction region after 2 dipole magnets D0, DX, and 3 quadrupole magnets. The goal during Phase I was to measure events with very low four-momentum transfer [72]. During Phase II, which began in 2010, PP2PP experiment merged with the experiment STAR. The positions of stations changed to be between the 2 dipole magnets, 15.8 and 17.6 m from the interaction point [7]. Geometrical acceptance during Phase II setup can be seen in Fig. 2.9.
A Roman Pot vessel can be seen in Fig. 2.10. The active area, which detects incoming particles, is the translucent window of size 79×49 mm [7]. The detector itself consists of 4 silicon strip detectors (SSD) with a thickness of 400 μm [72] and one scintillator used for triggering. Silicon strips measure the position of a traversing particle; 2 are set to measure in the x direction with horizontal orientation of strips,

2 in the y direction with vertical orientation [7]. From the measured positions, one can determine their momenta. The vessel also includes 2 lightguides that lead light from the scintillator to photomultiplier tubes, which convert light to signal.
The Roman Pot system of detectors is used to efficiently trigger and measure outgoing forward scattered particles, mostly protons. The maximum readout rate of a vessel measured during Run3 was set at approximately 500 events/s. In the first years during Phase I, average reconstruction efficiency was set at 99.98 % [72].



(a) (b)

Figure 2.9: (a) is the geometrical acceptance of the Roman Pot system of detectors during Phase II. (b) shows the difference between momentum transfer acceptance between Phase I and II. Taken from ref.[73].

2.6 Beam-Beam Counters and Zero Degree Calorimeters

The last detectors, which will be mentioned in this section, are the Zero Degree Calorimeters (ZDC) and the Beam Beam Counters (BBC). In diffractive events, they are only used for triggering events at level 0. ZDCs [74] are located on each side of the interaction region at positions (x, y)=(0,0) behind the first dipole magnet, which curves all charged particles. ZDCs measure neutral particles, mostly neutrons, created in a collision. BBCs are symmetrically positioned 3.75 m from the interaction point and cover pseudorapidity region of $2.1 \leq |\eta| \leq 5.0$ [75]. The lack of any particles triggered in these detectors is used to confirm Large gaps in rapidity, which are typical for diffractive events.

2.7 Data Acquisition and Triggers

A crucial part of STAR for successful and fast operation is its triggering system. The number of all collisions in any collider is enormous compared to the events of interest. It is necessary to exclude the events that are already well described to measure the underlying events. To maximize the statistics of physics of interest, several rules in the online data processing are used.
STAR uses a trigger system, Data Acquisition system (DAQ)[76], [77], to select

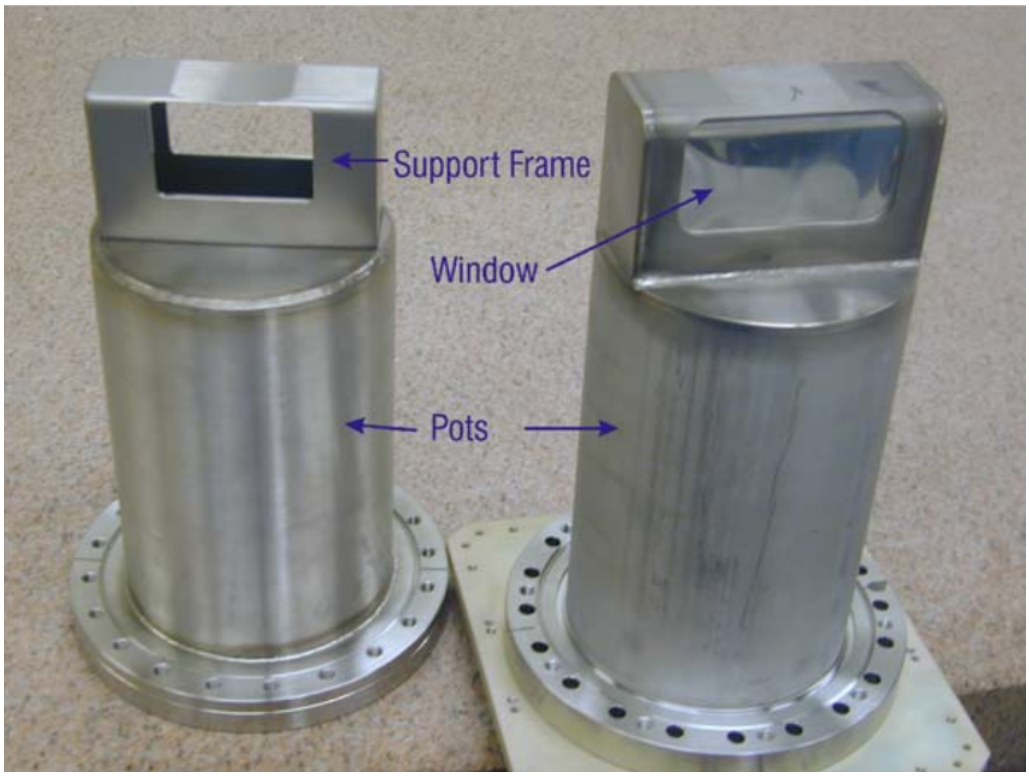


Figure 2.10: A picture of 2 Roman Pot detectors. The active area of detection is the window. Taken from ref. [72].

events with physics motivation. Over the years of data taking and changes in detectors, the triggering system has changed as well. The decision-making process is more complex, but in essence, it works on 4 levels. The first level, level 0 (L0), is based on hit/no hit triggers in specific detectors and works at the rate of bunch crossing. The level 0 trigger was developed to make quick decisions without creating any downtime[3]. Level 1 (L1) is made to analyze gross features and protect against pile-up. Level 2 (L2) is used the least; it analyzes data from calorimeters in more depth. Level 3 (L3) reconstructs data from TPC and makes the final decision whether to keep or discard an event.

[3]The time when data is being processed for higher level triggers, all the detectors are being readout. During this time, STAR is inactive.

Chapter 3

Data set and Quality Assurance

The chapter begins with the production of data, the software used for reconstruction, and, very importantly for this analysis, the BEMC clustering. The following sections discuss the trigger and its conditions, and the conditions required for data to be considered.

3.1 Data Format

The entire analysis has been performed in the *star-upcDst* data format. Originally, it was created by Jaroslav Adam [78] to facilitate analyses on Run14 Au-Au ultraperipheral collisions (UPC), it was modified by Tomas Truhlar to include information from the forward system of Roman Pots.
The data format is created from the official STAR *MuDst* format. The format was made to be light, therefore it only includes events with a primary track[1] that is matched to BEMC or TOF. The data format includes only the information necessary for low-multiplicity ultraperipheral heavy-ion and proton-proton collisions. The original production was done on rp_stream data with the addition of J/ψ trigger events.
The data have been produced with STAR libraries version SL20c with the following Big Full Chain (BFC) options: *DbV20200225, pp2017a, StiCA, btof, mtd, mtdCalib, pp2pp, PicoVtxDefault, UseBTOFmatchOnly, VFStoreX, fmsDat, fmsPoint, fpsDat, BEmcChkStat, OSpaceZ2, OGridLeakFull, -evout, -hitfilt.*
The main information for each event, such as the event id, the run number, and the number of vertices, is stored in class StUPCEvent. Each track and each vertex is saved separately in classes StUPCTrack and StUPCVertex, respectively. The tracks from the Roman Pot system are saved in StUPCRpsTrack.

3.2 BEMC Clustering

The clustering algorithm used for the BEMC in Run17 is done during the production of the *star-upcDst* data format. Format *star-upcDst* uses an algorithm implemented

[1] A primary track is a track matched to a primary vertex.

in class `StEmcOldFinder` which takes 4 different parameters. The parameters and the values used are listed in Tab. 3.1.

Parameter	Value
bemc_size_max	4
bemc_energy_seed	0.3
bemc_energy_add	0.001
bemc_energy_threshold_all	0.1

Table 3.1: Table of BEMC clustering parameters used in `StEmcOldFinder` class.

The first parameter represents the maximum number of hits a cluster can have. The energy seed is the minimum energy of a hit to start looking for a cluster. The third parameter is the minimum energy a hit must have to be added to the cluster, and the last parameter is the minimum energy of a reconstructed cluster to continue to the next stage.
Once the clusters are found, they are matched to tracks from the TPC. First, the TPC track trajectories are projected to the BEMC. This is done by using official the STAR class `StEmcPosition`. After, the TPC track positions are compared to existing clusters with STAR class `StEmcGeom`, specifically in η - ϕ variables. This entire process is done during the production of the data format *star-upcDst*.

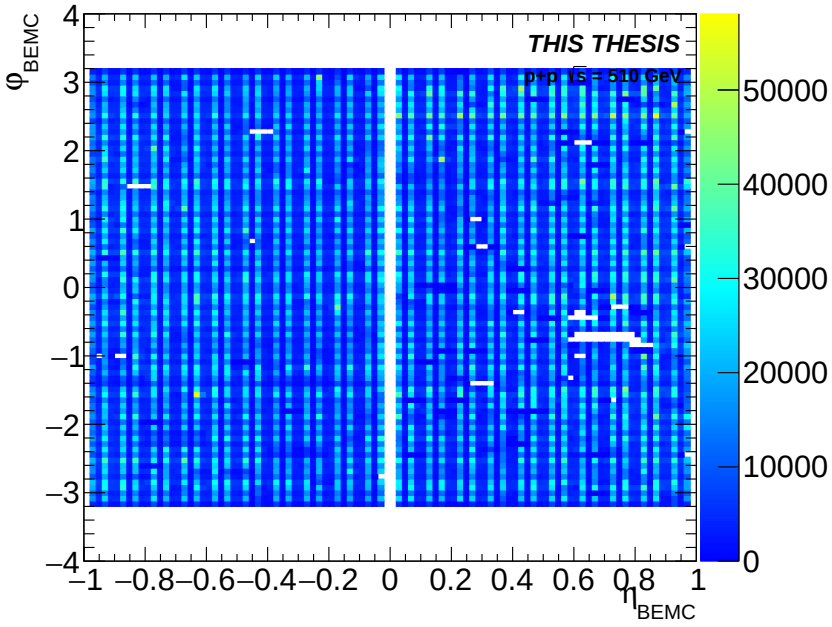


Figure 3.1: Distribution of BEMC clusters in η - ϕ plot.

The distribution of clusters in η - ϕ correlating distribution can be seen in Fig. 3.1. One can see the straight lines, which comes from how the BEMC hardware was designed, previously described in Sec. 2.4. The correlating distribution of track energy against the cluster energy is shown in Fig. 3.2. Here, one sees a hard cutoff at 0.3 GeV, which exactly corresponds to the lower limit for the BEMC clustering. In addition, it is obvious that the track energy is always smaller than the energy of a cluster.

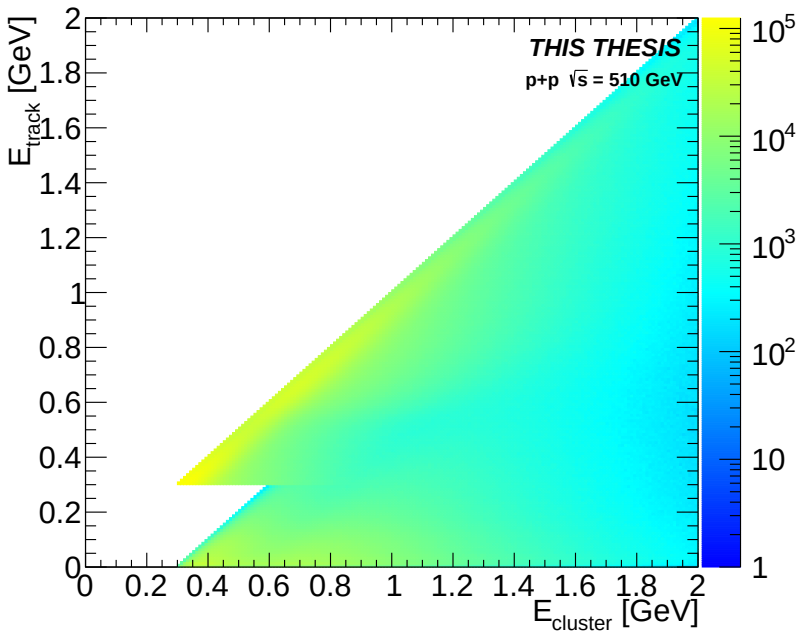


Figure 3.2: Correlating distribution of cluster energy and track energy.

3.3 Triggers

The analysis utilizes one specific trigger called *HTTP JPsi* and has 3 different ids: 570209, 570219, 570229. The trigger is a combination of conditions to ensure the selection of J/ψ photoproduction events and, at the same time, to reject contamination from processes such as ρ meson photoproduction. The trigger itself consists of the following conditions:

- T1 Hits in opposite sectors of the BEMC detector. The BEMC is divided into 6 sectors, each of them covering azimuthal angle of $\pi/3$.
- T2 Number of TOF hits below 4, ensuring low multiplicity events.
- T3 No hits in the BBC detector on both sides.
- T4 Laser protection against laser calibration runs.

The first condition is a positive condition, when the trigger requires signal in a detector. The BEMC is divided into 6 equally sized sectors in azimuthal angle. The other 3 conditions are so-called veto conditions, meaning that if any of them is not satisfied, the event is discarded. The reason for the first condition is that diffractive events such as this one belong to soft physics. Soft physics is characterized by fairly low momentum transfers, up to units of GeV/c. J/ψ is a heavy particle itself, therefore its transverse momentum is expected to be very small, close to 0. From simple kinematics and the law of momentum conservation, the daughter particles are expected to be back-to-back.
The first 2 veto conditions are to ensure low-multiplicity diffractive events. The second one (T3) ensures no particles are produced in the ranges $3.4 < |\eta| < 5.0$.

Diffractive events are defined as having large rapidity gaps between produced particles [3]. Laser is used for TPC calibration, by requiring protection, we discard all events which could have additional tracks in the TPC not related to J/ψ photoproduction.

3.4 Good Run List

This section discusses the minimal requirements for each run to be considered. At STAR, each Run (a large period of data taking denoted with capital R) is divided during data-taking into a number of small periods called runs. A run can consist from tens of events up to millions, depending on problems that happened during the data-taking period.

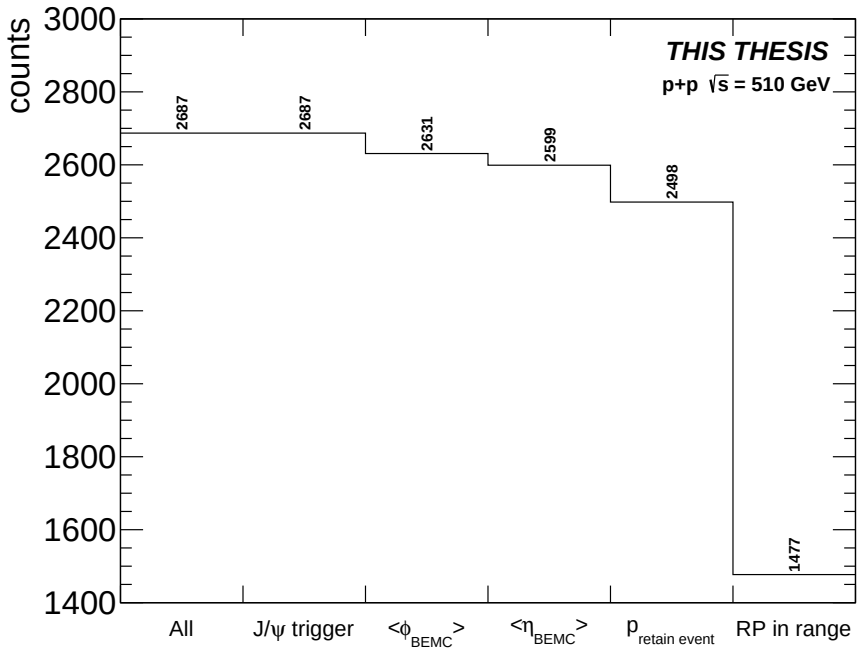


Figure 3.3: List of conditions and the respective number of passed runs for a good run list.

Out of all possible runs from data production (2800), several have been discarded for not satisfying the minimal conditions. The conditions and their respective numbers of passed runs are shown in Fig. 3.3. The first condition is to have at least one event triggered by the *HTTP JPsi* trigger. The next 2 conditions are imposed to make sure all independent parts of BEMC are working. The averages of pseudorapidity η and the azimuthal angle ϕ of all tracks in a run are plotted in Fig. 3.4. The distributions of averages for both variables are fitted with a normal distribution. The runs that are outside the $\pm$ 9σ range are investigated separately for signs of BEMC not working properly. One of such runs is shown in Fig. 3.5.

The next condition is connected to a correction on luminosity called the probability of retaining an event. The correction is thoroughly described in Sec. 6.2. In this

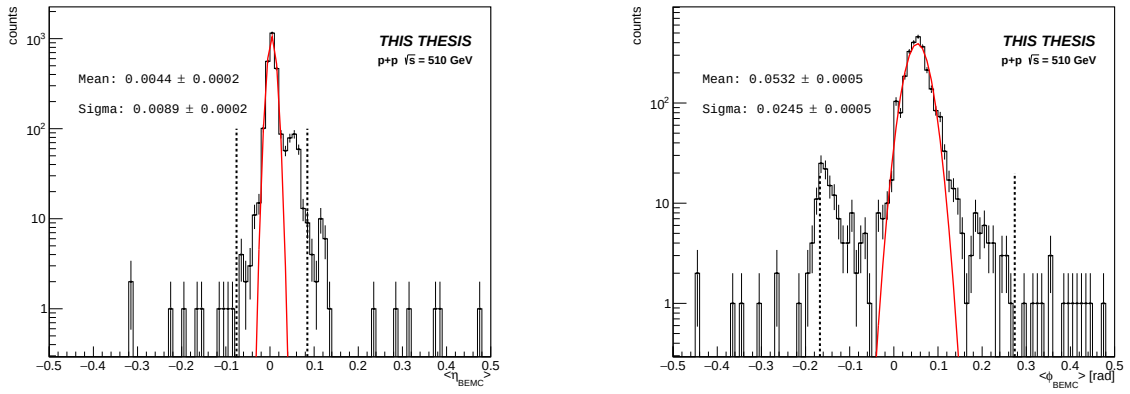


Figure 3.4: The averages of η (left) and ϕ (right) for all tracks in each run are computed, plotted, and fitted with a Gaussian distribution. The runs that are outside the $\pm$ 9σ range of the fit are each investigated for possible signs of BEMC not working properly.

case, we exclude runs, which are outside the $\pm$ 9σ range of the fit, which can be found in Fig. 6.1.
The last condition imposed on the runs concerns only the analysis when the Roman Pot system of detectors is used. For each fill[2] the first and the last run, the Roman Pots are switched off to keep them safe from being burnt by the beam. The switching off is done by electronically moving them away from the beamline. The condition was set that if a single Roman Pot vessel is more than 4 cm away from the beamline, the run is discarded.

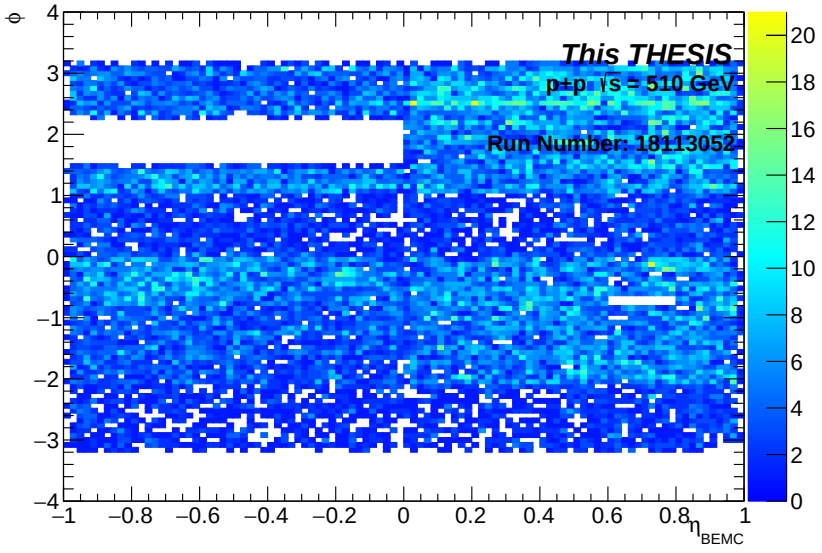


Figure 3.5: The $\eta - \phi$ correlation plot for run 18113052. An empty space with no particles indicates that the BEMC is not working properly.

[2]A fill means a new set of bunches of particles filled into the collider. During one fill, there are usually several runs.

Chapter 4

Event Selection

This chapter provides a description of the various cuts used in the analysis. First, the list of cuts is given. Following are descriptions of the cuts divided into 2 separate parts. The first section describes the central system, the second the forward system. The second part is used only in the case when the Roman Pots are used.

4.1 List of Cuts

The list of conditions imposed on the events goes as follows:

- C1 *HTTP JPsi* trigger
- C2 Exactly 1 vertex with at least 1 TPC track matched to signal in the TOF or BEMC
- C3 Vertex position in the centre of the TPC: $|V_Z| < 100$ cm
- C4 Exactly 2 good quality BEMC tracks
  - C4.1 Each TPC track must have good quality reconstruction: $N_{hits}^{fit} > 15$
  - C4.2 Each track must have good quality energy measurement: $N_{hits}^{dE/dx} > 15$
  - C4.3 Each TPC track must satisfy: $|DCA_Z| < 1$ cm
  - C4.4 Each TPC track must satisfy: $DCA_{XY} < 1.5$ cm
  - C4.5 Each TPC track must be matched to a hit in BEMC
  - C4.6 There is exactly 2 TPC tracks that satisfy the conditions above
- C5 Condition on the pseudorapidity of daughter particles $|\eta| < 0.9$
- C6 Back to back signals in the BEMC
- C7 Particle identification
  - C7.1 Electron pairs satisfy $\chi^2 < 9$

C7.2 Each TPC track must satisfy the condition given by the equation below

$$\begin{aligned} |n\sigma_K| &> 1.3, \\ |n\sigma_p| &> 1.3. \end{aligned} \tag{4.1}$$

C8 Total electric charge of pair: $Q_{tot} = 0$

F1 Exactly 1 Roman Pot track

F1.1 Each track is reconstructed in 2 RP modules

F1.2 Each module detects the track with at least 3 out of 4 SSDs

F1.3 Exactly 1 track satisfying the conditions above

F2 The track detected in the Roman Pots is inside the fiducial region:

$$\begin{aligned} (p_x + 0.6 \text{ GeV})^2 + p_y^2 &< 1.25 \text{ GeV}^2/\text{c}^2, \\ 0.4 \text{ GeV/c} &< |p_y| < 0.8 \text{ GeV/c}, \\ p_x &> -0.27 \text{ GeV/c}. \end{aligned} \tag{4.2}$$

The first conditions are imposed on the central pseudorapidity system, denoted with C. The last 2 conditions are on the forward and backward system, denoted with F. The number of events after each imposed cut are shown in Fig. 4.1 and Fig. 4.2 for analysis without and with the Roman Pots, respectively.

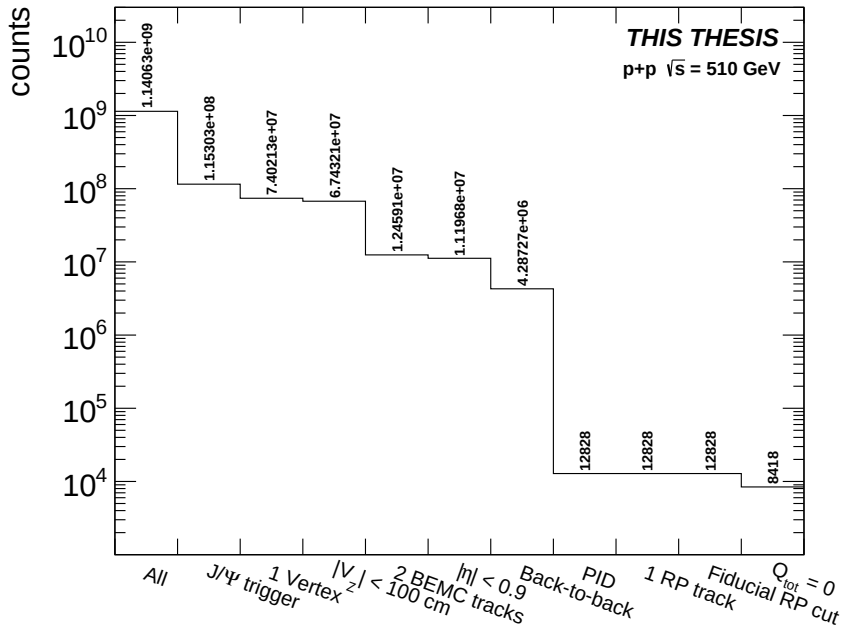


Figure 4.1: The summary of event selection for analysis without the Roman Pots. The number on top of each bin represents the number of events after each imposed cut.

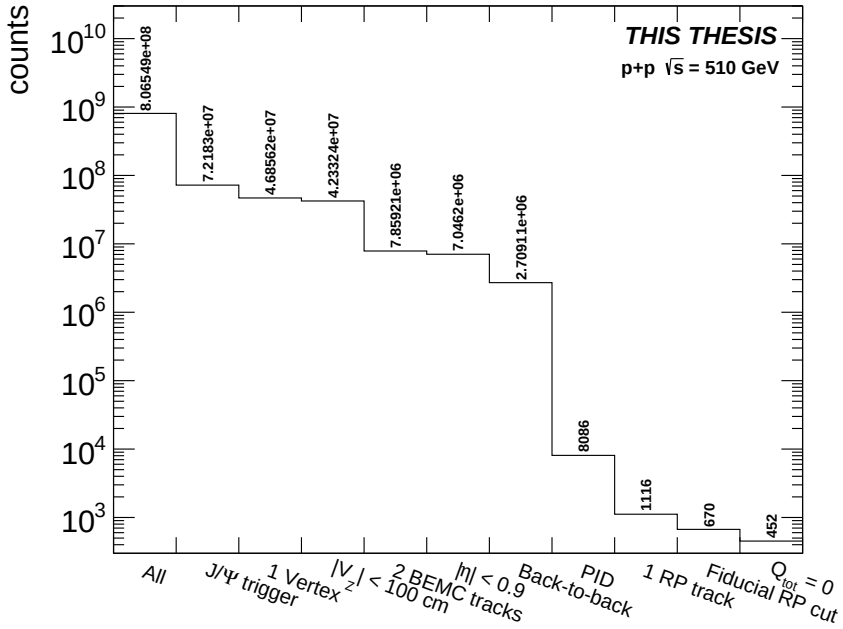


Figure 4.2: The summary of event selection for analysis with the Roman Pots. The number on top of each bin represents the number of events after each imposed cut.

4.2 Central System

The cuts on the central system are imposed on the data, which have been triggered by the HTTP J/ψ trigger described in Sec. 3.3. The cuts are in the order as they were used in the analysis.

4.2.1 Primary Vertex (C2, C3)

The first condition in the central system is exactly 1 reconstructed primary vertex. With this cut, the goal is to minimize pile-up events and contamination. In addition, if there were multiple collisions/primary vertices reconstructed in the central system, when matching the information from the forward system to the central system, it is impossible to match them to each other.
In fact, the vertex which is reconstructed is truly the secondary vertex, meaning the point at which the J/ψ decays to daughter electrons. That is because the vertex is reconstructed only from the reconstructed tracks. The STAR vertex reconstruction algorithm still considers this secondary vertex to be the primary one. The original primary vertex is not being reconstructed, because only the neutral J/ψ is it's product in the central system[1]. The other condition on the primary vertex (C3), is on its reconstructed z position. We limit ourselves to events with vertices in the middle of the TPC because on the sides, acceptance effects are more pronounced. The distribution of vertices along the z axis can be seen in Fig. 4.4.

[1]The information from the Roman Pots is not used for vertex reconstruction, therefore, I neglect the protons in this discussion.

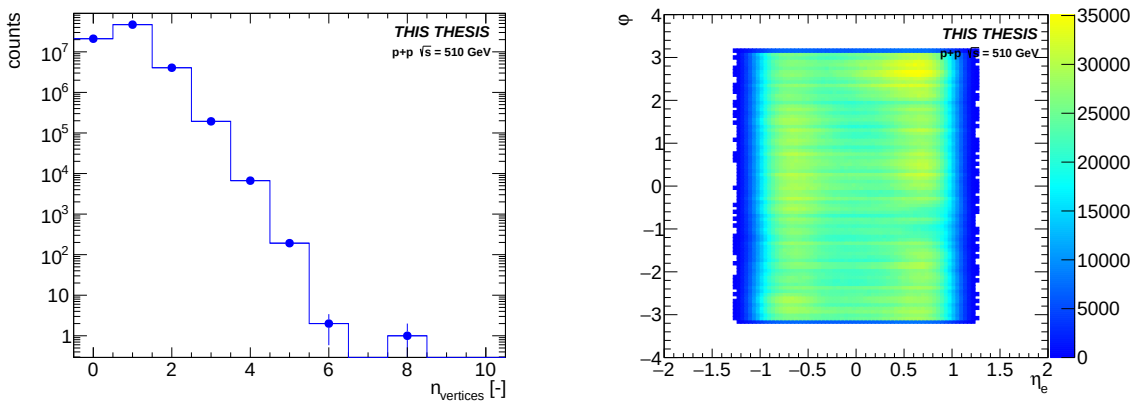


Figure 4.3: Left: Distribution of the number of vertices per event. The majority of events contain 1 event. Right: Correlating distribution of pseudorapidity η and azimuthal angle ϕ.

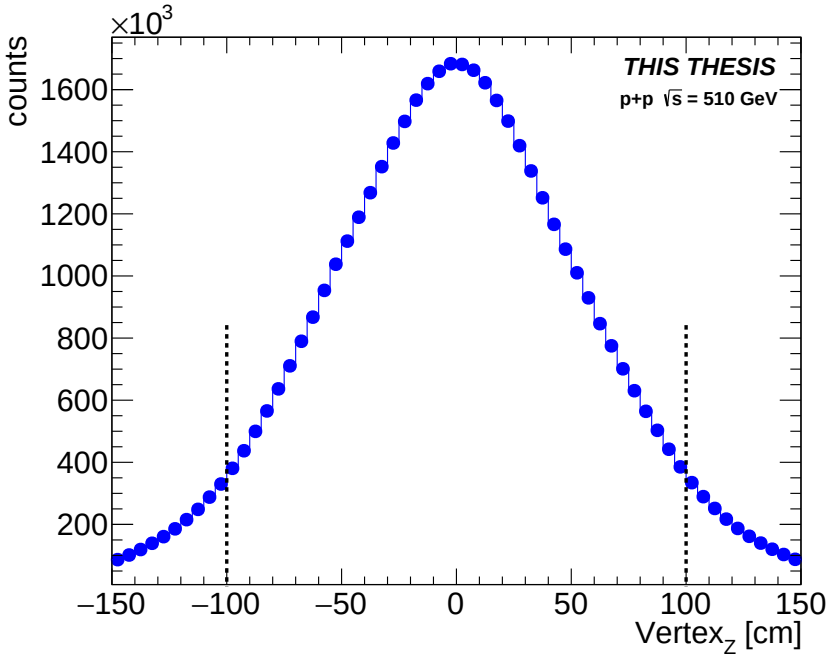


Figure 4.4: Distribution of vertices along the z axis. Black dashed lines represent the cut $|V_Z| < 100$ cm.

4.2.2 2 Good Quality BEMC Tracks (C4)

As described in the list, this condition is actually a set of multiple conditions imposed on the data. The first 2 conditions are related to the primary vertex. The helix of each TPC track is extrapolated towards the beamline and the distance of closest approach is the vector between the vertex and track trajectory. The z component is handled separately from the transverse component. The distributions of DCA variables with the respective cuts are shown in Fig. 4.5.

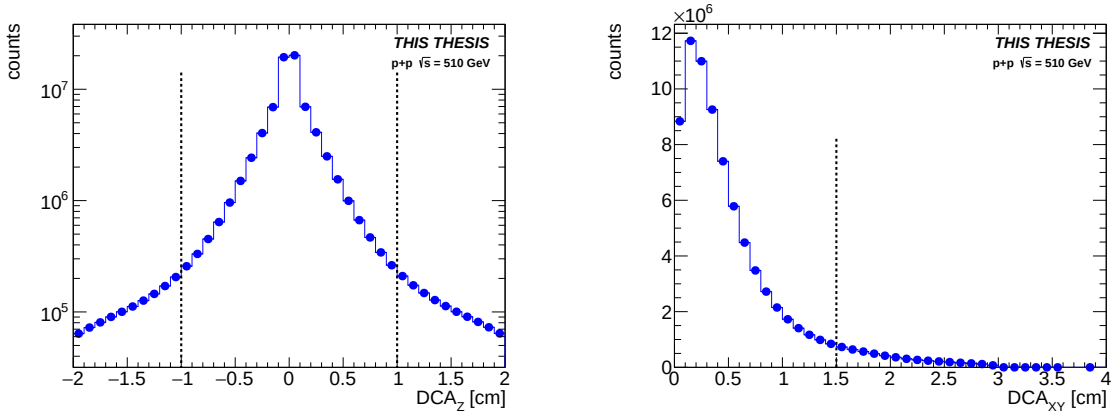


Figure 4.5: Left: Distribution of DCA_Z for particles w.r.t. the reconstructed vertex. Right: Distribution of DCA_{XY} w.r.t. the reconstructed vertex. The black dashed line represents the cuts.

Each particle needs to be reconstructed in the TPC with a certain level of quality. This is done by selecting tracks with minimum reconstruction hits $\mathbf{N^{fit}_{hits} > 15}$. For high quality particle identification, the chosen cut is $\mathbf{N^{fit}_{dE/dx} > 15}$. Each particle that passes the conditions above has to be matched to a reconstructed hit in the BEMC. Exactly 2 tracks per event will pass these conditions.

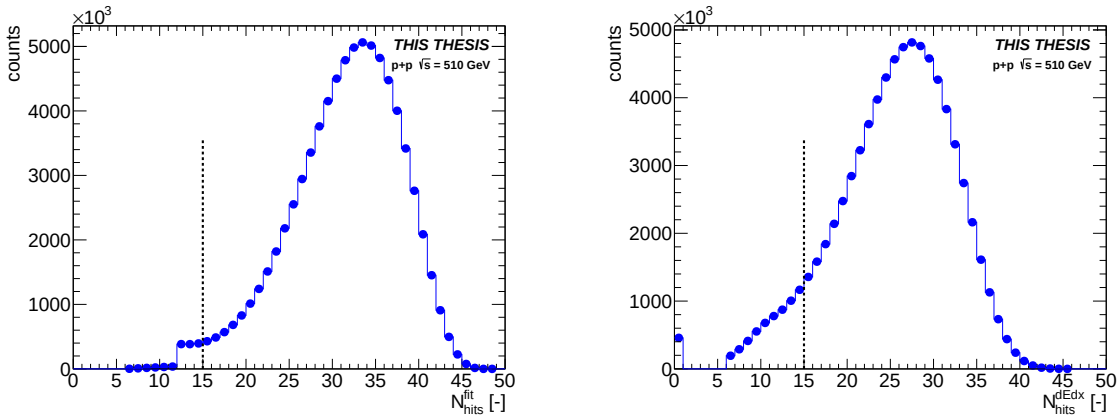


Figure 4.6: Left: Distribution of hits for track reconstruction N^{fit}_{hits}. Right: Distribution of the number of hits for energy loss measurement $N^{dE/dx}_{hits}$. Black lines represent the respective cuts.

4.2.3 Pseudorapidity (C5)

The daughter tracks in the central system have to pass the condition on the pseudorapidity $|\eta| < 0.9$. To ensure a low level of contamination, the cut is chosen as it is even though the acceptance of the TPC and the BEMC is $|\eta| < 1$. The pseudorapidity of the tracks reconstructed in the BEMC can be see in Fig. 4.7.

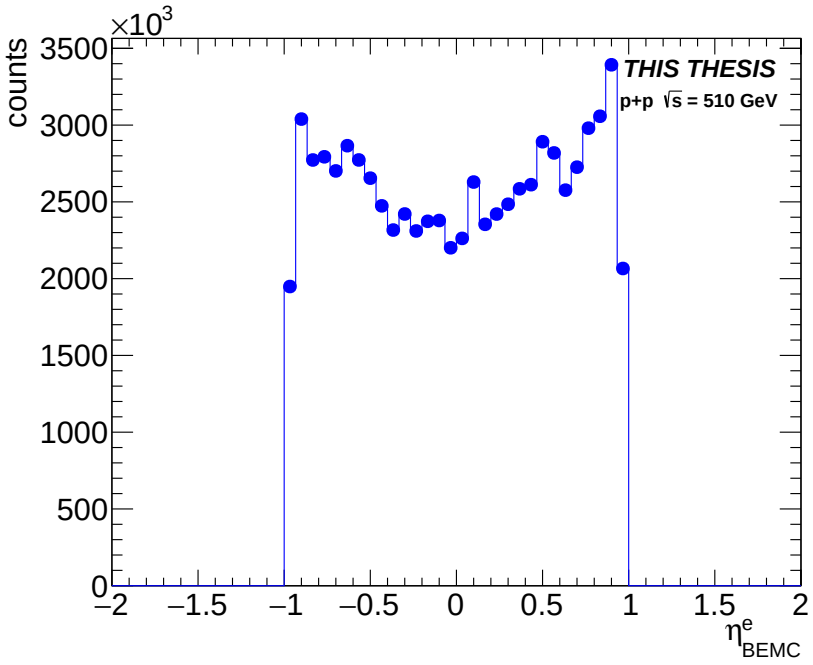


Figure 4.7: Distribution of pseudorapidity η reconstructed in the BEMC.

4.2.4 Back To Back Condition (C6)

Once the 2 good quality BEMC tracks are chosen, we impose a condition on their azimuthal angle. This cut is applied to match the condition requirement in the trigger described in Sec. 3.3. We aim to analyze events that we trigger.

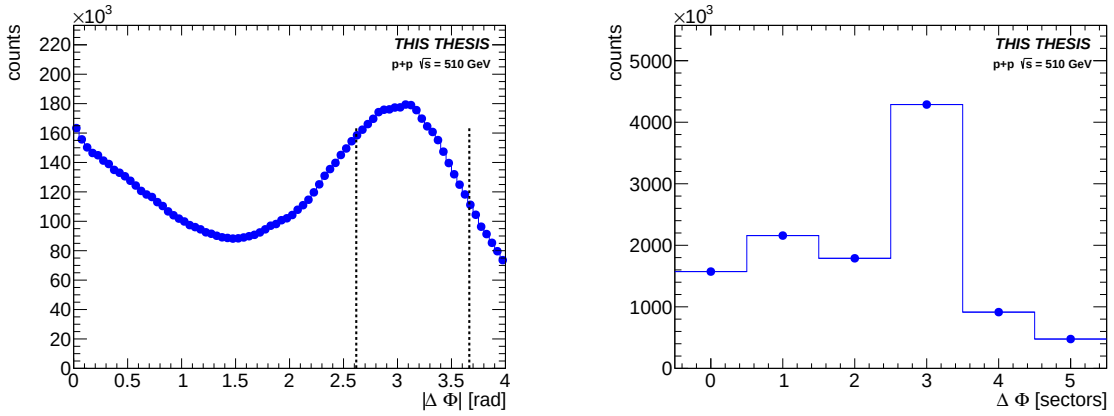


Figure 4.8: Angular distribution between reconstructed tracks. The black line represents the condition. The left plot shows the difference in angle, right plot shows the difference in sectors on which we apply the condition.

The distribution of the difference in azimuthal angle $\Delta\phi$ can be seen on the left in Fig. 4.8. On the right, that is the distribution of sector difference. Back-to-back tracks mean their sector difference is 3. In the online triggering, only a primitive signal is processed, and tracks are not fully reconstructed. Therefore, the condition is only on signals from opposite sectors of the BEMC.

4.2.5 Particle Identification (C7)

Particle identification (PID) is based solely on the energy loss measurement from the TPC. The signal-to-noise ratio under the J/ψpeak is very significant even without additional conditions on the p/E ratio in the BEMC.
The standard energy loss PID was described in Sec. 2.2, specifically Eq. 2.1. In our case, we identify particles at a point after the previously described conditions. Hence, we already have 2 good quality back-to-back BEMC tracks. We use condition

$$\chi^2 := (n\sigma_e^{(1)})^2 + (n\sigma_e^{(2)})^2 < 3^2 \tag{4.3}$$

also called χ^2 method. Instead of applying the $n\sigma$ condition to both particles, we create a circle in the $n\sigma_e^{(1)} - n\sigma_e^{(2)}$ plane with radius 3 and apply the condition there. This plot can be seen in Fig. 4.9 on the left side. We can see a large amount of pairs in the bottom left corner; those are pion pairs. In Fig. 4.9 on the right side is the χ^2 distribution. Again, the pions can be seen there as well, but they are excluded by the cut.

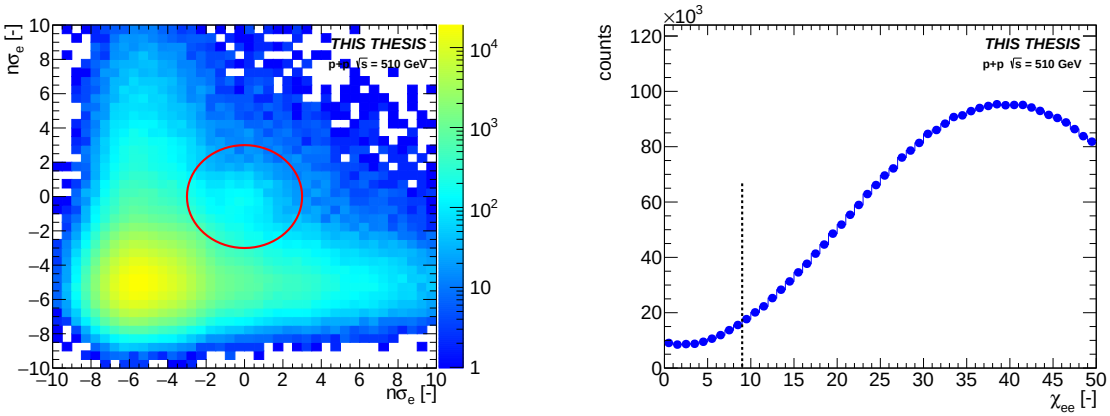


Figure 4.9: Left: Correlating distribution of $n\sigma_e^{(1)}$ and $n\sigma_e^{(2)}$ of the pair of tracks. The red circle represents the χ^2 cut. Right: Distribution of variable χ^2 defined in Eq. 4.3 for electron pairs with a cut at a value of 9.

Not necessarily for J/ψ signal extraction, but important for uncovering the $\gamma + \gamma \to e^+ + e^-$ spectrum, are conditions for p, K exclusion. The spectrum is then used in the BEMC efficiency study as a quality assurance, further described in Sec. 6.6. With this condition, we cut out the overlapping regions in the correlation plots of momentum and energy loss. The plots can be seen in Fig. 4.10. The state before the cut is shown on the left, after the cut on the right.

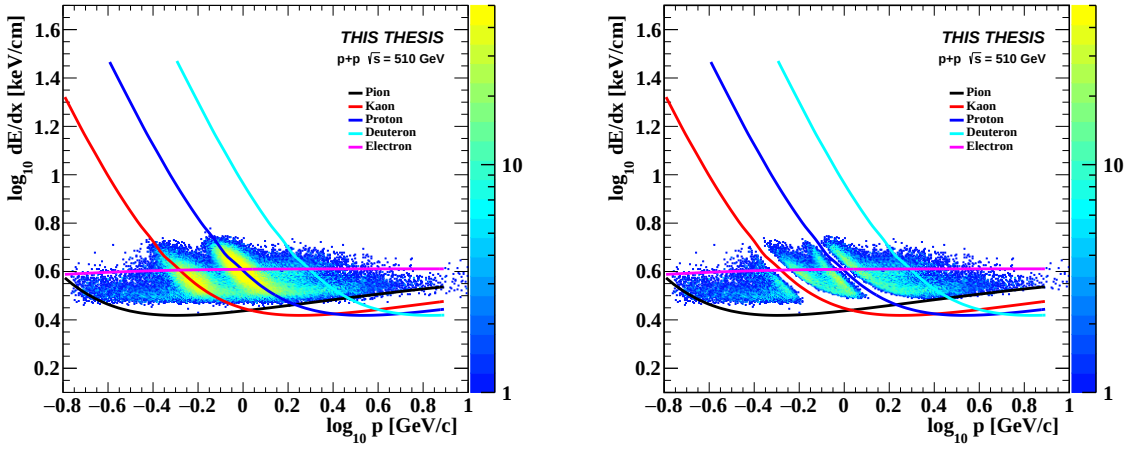


Figure 4.10: Correlating distributions of track momenta and the measured energy losses. Left distribution shows all the tracks after the χ^2 condition and before the specific condition on $n\sigma_K$, and $n\sigma_p$ is applied. The right distribution shows after both conditions.

4.2.6 Total Charge (C8)

The last condition in the central system is the total charge of the pair. So far, we have been dealing with 2 good quality tracks reconstructed in the BEMC, which pass the above mentioned electron PID conditions. Now, we separate the signal (e^+e^-) pairs from the background. In the end, there is 8418 events that pass as a signal. This is the end of event selection for the case when we do not use the Roman Pots. In the analysis with the Roman Pots, 2 additional conditions are applied. They are described in the following subsection.

4.3 Forward System

As mentioned before, the following conditions are used only in the analysis with the Roman Pot system. Altogether, there are only 2 conditions, the first one being on a single reconstructed track and the second one on its momentum.

4.3.1 1 Track in the Roman Pots (F1.3)

A track is reconstructed in the Roman Pots if there are coincidence hits in 2 successive RP modules, also called RP branch[2]. A hit in a module means at least 3 of 4 SSD strips detect the particle and reconstruct its position. If the requirement were looser, meaning 2 SSD hits are enough, some tracks could be reconstructed without the x or y position.
We require exactly 1 hit in the Roman Pots. That means detection of a track only on one side of the interaction point and a veto on the other. The assumption behind this requirement is that only the proton that interacts via the strong interaction has enough transverse momentum to reach the acceptance of the Roman Pots. Such

[2]An RP branch means 2 Roman Pot modules on the same side of the interaction point, both above or below the beamline.

strict conditions are necessary because there is no reasonably accurate way to match the central system information to the forward system, it is possible only if there is 1 separate event in each.

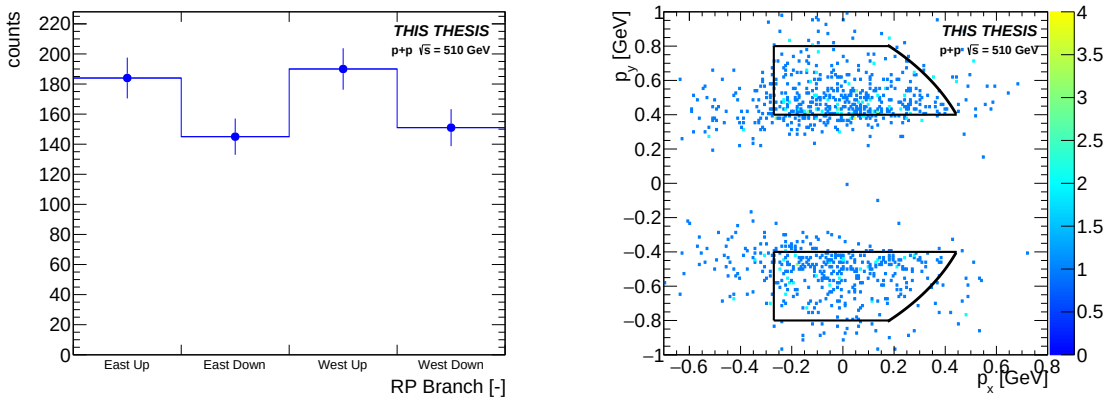


Figure 4.11: Distributions of tracks in the Roman Pot detector system. The left plot shows the number of events with signal in a specific branch after the central system conditions. The right plot shows the correlating distribution of transverse momenta p_X, p_Y of tracks in the Roman Pots.

4.3.2 Roman Pot Fiducial Region (F2)

The Roman Pot fiducial area is chosen not according to the measured positions, but according to the reconstructed momenta. The fiducial volume is shown on the right in Fig. 4.11. It is a carefully chosen area in the $p_X - p_Y$ plane. The chosen area is selected so to maximize statistics at the same time as minimize contamination from the beam halo. This condition has been applied in 2 previous STAR papers [7], [8].

Chapter 5

Monte Carlo Simulations

This chapter discusses the use of Monte Carlo (MC) simulations in the analysis. We use the STARlight event generator [14], a generator focused on simulating ultraperipheral heavy-ion collisions. The first section discusses the generated MC sample of $J/\psi \rightarrow e^+e^-$ used in the embedding as well as for comparison to the final cross section. The following section discusses the MC sample of $\gamma + \gamma \rightarrow e^+ + e^-$, a background process to J/ψ photoproduction used mainly as a quality assurance in simulating BEMC efficiency.
The generated MC samples are used for evaluation of reconstruction efficiency described in the following chapter and for a direct comparison between our experimental result and theory. This is further described in the last chapter, chapter 8.

5.1 STARlight $J/\psi \rightarrow e^+e^-$

In addition to heavy-ion collisions, STARlight can simulate diffractive events such as vector meson photoproduction in proton-proton collisions. We generated a sample using the options described in Tab. 5.1. The first 4 settings are the proton and nucleon numbers of the 2 colliding bunches, the energy of collision is not defined as usual with $\sqrt{s}$, but with a relativistic speed γ factor. The W value is the energy of the $\gamma + p$ system. When set to -1, the program uses the default values. The condition on the transverse momentum (CUT_PT) is turned off. Important is also the ETA_CUT, which turns on the condition on the pseudorapidity of daughter particles. Due to the acceptance of STAR, we are only interested in $|\eta| < 1$. We are generating a sample of 100 thousand events in production mode 2, which is narrow coherent vector meson production[1]. Lastly, we are interested in the decay mode $J/\psi \rightarrow e^+e^-$, we use the PROD_PID channel 443011. The rest of the conditions are described in more detail in ref. [14].

Once the sample is generated, the distributions are saved to plots in Fig. 5.1 and Fig. 5.2. In Fig. 5.1, the plot on the left shows the rapidity distribution of generated J/ψ particles. The right plot shows the invariant mass spectrum of J/ψ. The peak

[1] In proton-proton collisions, wide and narrow coherent vector meson production mean the same thing.

OPTION	Value
BEAM_1_Z	1
BEAM_1_A	1
BEAM_2_Z	1
BEAM_2_A	1
BEAM_1_GAMMA	271.635
BEAM_2_GAMMA	271.635
W_MIN	-1.
W_MAX	-1.
W_N_BINS	35
RAP_MAX	9.
RAP_N_BINS	180
CUT_PT	0
CUT_ETA	1
ETA_MIN	-1.
ETA_MAX	1.
PROD_MODE	2
N_EVENTS	100 000
PROD_PID	443011
RND_SEED	7540069
OUTPUT_FORMAT	2
INTERFERENCE	0

Table 5.1: Options used in the STARlight input file for process $J/\psi \rightarrow e^+e^-$. More information on the specific options can be found in Ref. [14].

is very narrow, the width we see in the data (for example, Fig. 8.1) comes from the resolution of the detector.

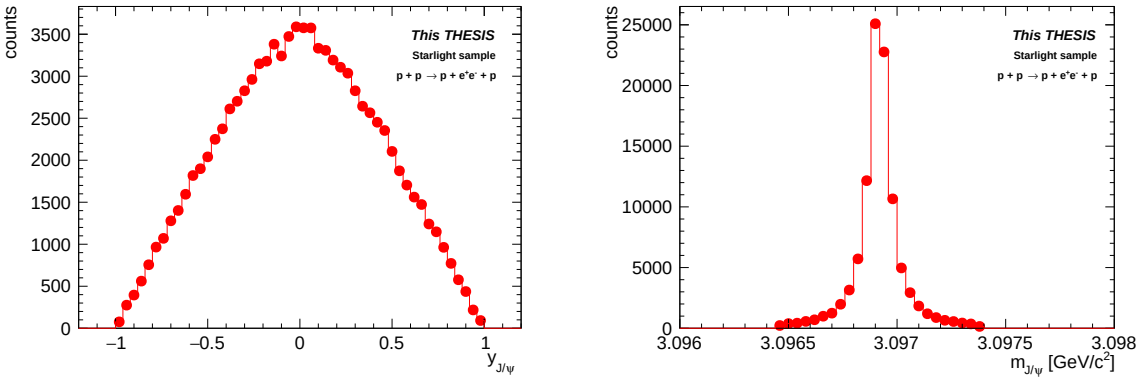


Figure 5.1: Distributions of rapidity and the invariant mass of the J/ψ Monte Carlo sample generated in STARlight.

The next 2 plots show the transverse momentum distributions. The left histogram shows the p_T of daughter electrons, the right plot shows the transverse momentum of the J/ψ. As was mentioned before, the majority of the produced J/ψ are of low momentum up to 1 GeV.
In addition to generating events, STARlight also computes the cross section of the

process. The entire cross section of J/ψ photoproduction in all channels without a condition on the pseudorapidity is 6.2 nb. The branching ratio for the e^+e^- channel is (5.971 $\pm$ 0.032) % [79], therefore, the cross section in the e^+e^- channel is 370 pb. In our case, we restrict ourselves to the central pseudorapidity region of $|\eta_e| < 0.9$, which corresponds to STARlight cross section $\sigma(p + p \to J/\psi \to e^+ + e^-) = 40$ pb.

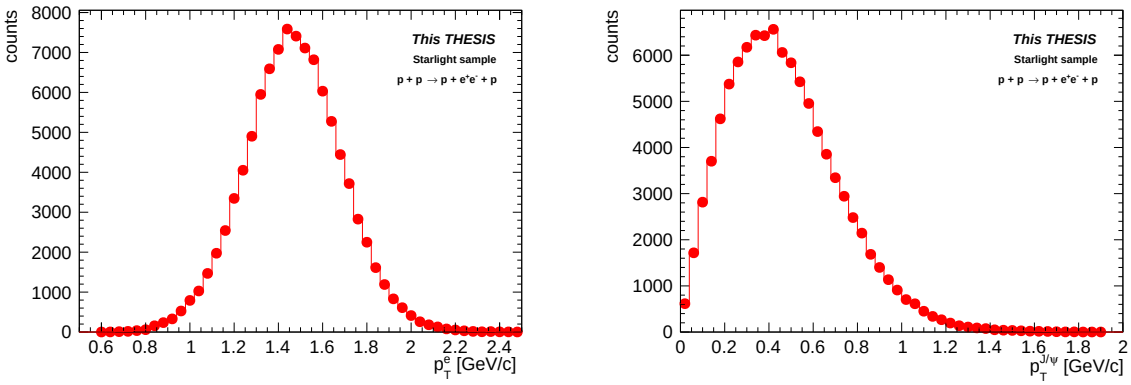


Figure 5.2: Distributions of J/ψ transverse momentum and the transverse momenta of the daughter electrons in a generated STARlight MC sample.

5.2 STARlight $\gamma\gamma \to e^+e^-$

In this case, we generated a sample of a background process to J/ψ photoproduction. The process is very similar to the case in the previous section, only the options which are used are slightly different. The specific options used are in Tab. 5.2.

The main difference is the chosen range of W values: the $\gamma\gamma \to e^+e^-$ has a typical exponential decrease along the entire invariant mass spectrum. We have chosen the minimum value at 1.0 GeV, which translates to electron transverse momentum well below the limit for a reasonable BEMC range. The production mode is now channel 1, meaning 2-photon mode instead of 2, which was photon-IPomeron mode. The last difference is the change in PROD_PID value to 11, with which we specify the electron-positron pair production.

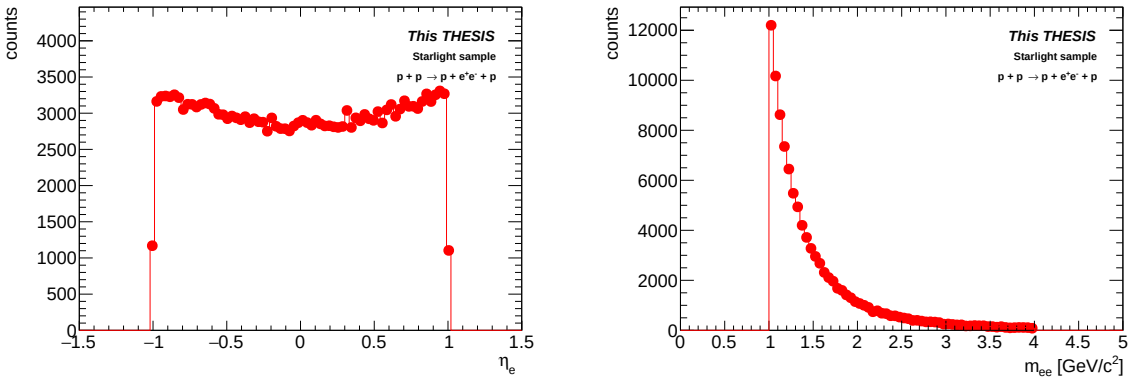


Figure 5.3: Distributions of pseudorapidity of daughter particles and invariant mass of the pair for the STARlight Monte Carlo sample $\gamma\gamma \to e^+e^-$.

OPTION	Value
BEAM_1_Z	1
BEAM_1_A	1
BEAM_2_Z	1
BEAM_2_A	1
BEAM_1_GAMMA	271.635
BEAM_2_GAMMA	271.635
W_MIN	0.2
W_MAX	5.0
W_N_BINS	35
RAP_MAX	9.
RAP_N_BINS	180
CUT_PT	0
CUT_ETA	1
ETA_MIN	-1.
ETA_MAX	1.
PROD_MODE	1
N_EVENTS	100 000
PROD_PID	11
RND_SEED	7540069
OUTPUT_FORMAT	2
INTERFERENCE	0

Table 5.2: Options used in STARlight input file for process $\gamma\gamma \rightarrow e^+e^-$. More information on the specific options can be found in Ref. [14].

The generated sample consists of 100 thousand events, whose distributions can be seen in Fig. 5.3 and Fig. 5.4. The first 2 distributions are the distribution of pseudorapidity of daughter electrons and positrons and the invariant mass of the pair. In the pseudorapidity distribution, one can see clearly the condition used on minimal and maximal values. In the invariant mass, one can see the exponential decrease. The other 2 plots show distributions of transverse momentum, for the pair on the right and for daughter particles which enter the BEMC is the left.

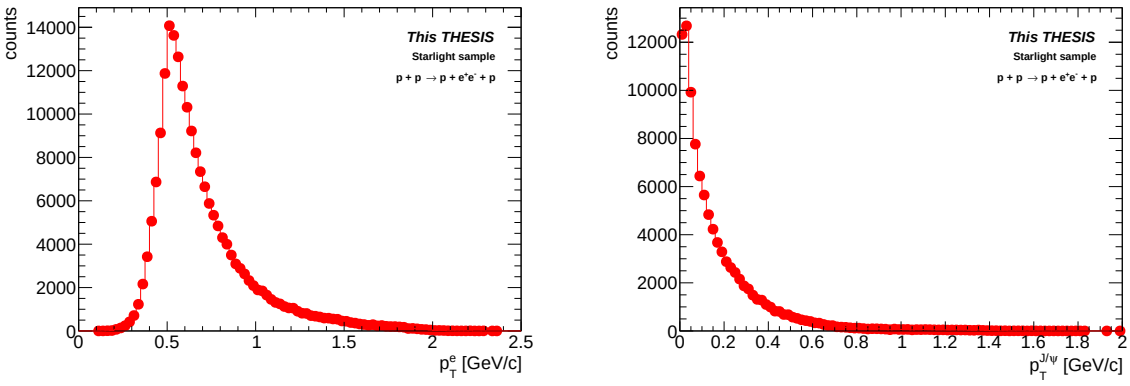


Figure 5.4: Distributions of transverse momenta of daughter particles and pairs of particles for STARlight Monte Carlo sample $\gamma\gamma \rightarrow e^+e^-$.

Chapter 6

Corrections and Efficiencies

In this chapter, the methodology of various corrections and efficiency studies are described. First, the trigger efficiency is studied in the first 2 sections; the first one concerns the positive condition in the trigger, and the second one concerns the veto conditions. Next is the correction to pile-up. Following is the correction to the z position of the primary vertex. Subsequently, the central embedding, namely the reconstruction efficiency is described, and the corrected BEMC efficiency is discussed. In the final cross section, the vertex reconstruction efficiency and the Roman Pot efficiency are also used. These efficiencies have been studied previously by Tomas Truhlar for the Central Exclusive Production study [7], we only mention the final values in the last section.

6.1 Trigger Topology Correction

Trigger topology efficiency represents the probability that an event will be processed and saved (trigger is triggered). This efficiency is studied on the dataset rp_zerobias stream. Zerobias data has only 1 condition and that is that at least 1 primary interaction per bunch crossing happens. The trigger efficiency would generally be calculated by reconstructing J/ψ candidates in the dataset, and then, out of the found candidates, one would be looking at how many of those events satisfy the *HTTP JPsi* conditions mentioned in Sec. 3.3.
In this analysis, it is done differently because the zerobias dataset is statistically not large enough to contain a reasonable amount of J/ψ candidates. Instead, we look at pairs of electrons with invariant mass in a certain mass window and check how many of them satisfy the *HTTP JPsi* condition. The invariant mass window is chosen to be above 2 GeV, to minimize contamination from ρ mesons. From performed studies, it turns out the final efficiency depends strongly on the chosen invariant mass window, as can be seen in Fig. 6.2.
The selection of cuts is similar to the event selection for J/ψ events. We analyze zerobias trigger events, therefore, we use 570704 trigger in the offline selection. The condition C4 is applied, but with the exception of C4.6 cut. The tracks are paired in the following back-to-back (C6) condition, when we loop over all BEMC tracks and look for a pair which produces back-to-back signals. Next, we identify electrons

with the χ^2 condition (C7.1). Last condition is a combination of V_Z of the primary vertex (C3) and the pseudorapidity of tracks η (C5).

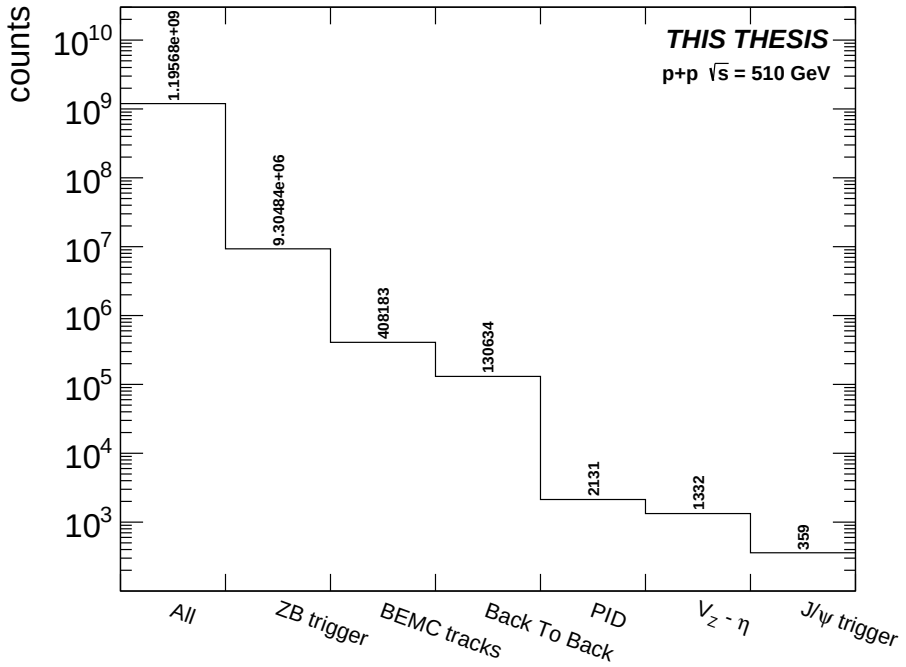


Figure 6.1: Selection of cuts applied to rp_zerobias stream to create a sample of pairs for studying the trigger condition T1.

To account for the dependence on the range of the invariant mass window, instead of choosing one value, we performed a thorough study by varying the lower and upper limits. They are chosen systematically with a step size 10 MeV in ranges (2.0, 2.8) and (3.3, 4.1) GeV, respectively. Altogether, there are 80 lower limits and 80 upper limits, therefore, we systematically look at $80 \times 80 = 6400$ mass windows. Efficiency from each window is then filled into the histogram shown in Fig. 6.3. This distribution leads to a normal distribution, which is fitted with a Gaussian distribution in red. The resulting mean value is the efficiency used in the cross section measurement, and the standard deviation is the systematic error.

6.2 Trigger Veto Correction

The next correction is called the probability of retaining an event. In simple terms, it represents that a J/ψ event of interest happens, but it is vetoed by noise in the detector. This noise can come from various sources, but it is the noise which happens in either one of BBCs (T3) or it increases the TOF multiplicity beyond the required limit (T2). The so-called probability of retaining an event is calculated on a run-by-run basis on the zerobias data

$$P_{\text{veto}} = \frac{\begin{array}{c}\#\text{ zerobias events which satisfy veto trigger conditions} \\ \text{with 0 reconstructed vertices}\end{array}}{\#\text{ zerobias events with 0 reconstructed vertices}}. \tag{6.1}$$

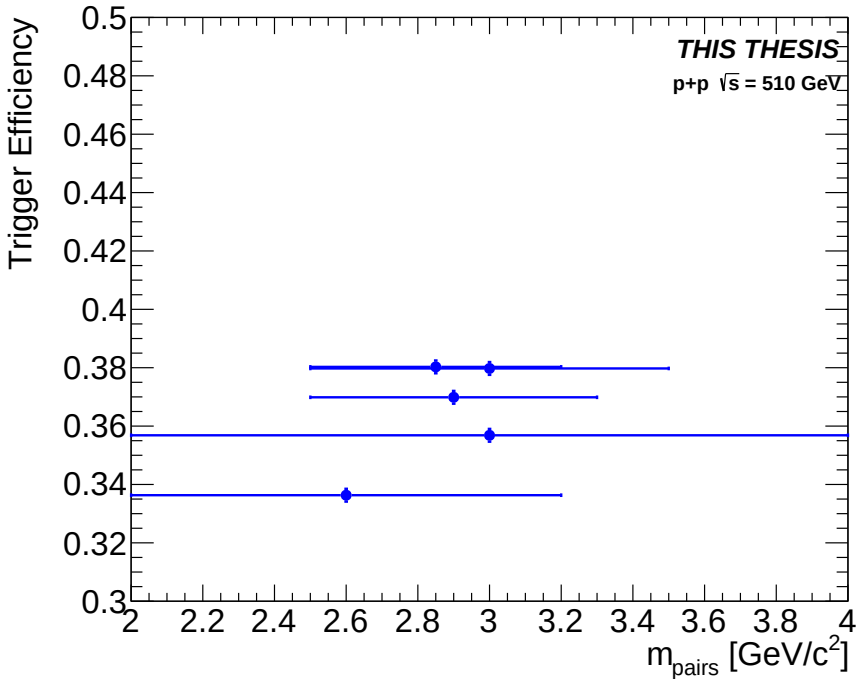


Figure 6.2: Values for trigger efficiency for certain specific invariant mass windows. The x error bars denote the invariant mass range used for the calculation of the trigger topology efficiency.

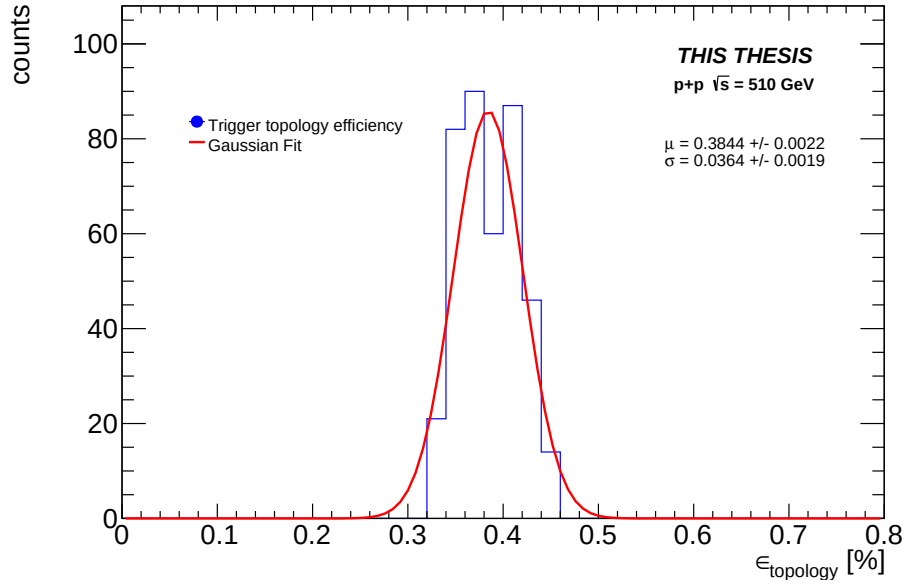


Figure 6.3: Distribution of trigger efficiency for various invariant mass windows of e^+e^- pairs. The distribution is fitted with a Gaussian function. The resulting mean value and standard deviation are printed on the plot.

We are only looking at events with 0 reconstructed vertices, because we analyze events with exactly 1 vertex (C2) and we are only interested in correcting the noise produced by 0 primary interactions. The probability of retaining an event is calculated on a run-by-run basis, because it strongly depends on the instantaneous luminosity Λ aside from the real time luminosity delivered by the collider. Purposefully, it is denoted as Λ to not cause confusion with luminosity of *HTTP JPsi* trigger. The instantaneous luminosity is the integrated luminosity for each run, which can take from several seconds up to 30 minutes of data taking. The distribution of probability of retaining an event w.r.t. instantaneous luminosity is shown in Fig. 6.4, on the left.
In Fig. 6.5, there are similar distributions to the probability of retaining an event, only in this case, we use only the veto condition in BBC East (left) and BBC West (right). One can see that the majority of the rejected events come from vetoes in these 2 detectors. The TOF multiplicity condition is almost always satisfied for bunch crossings without any events.
The distribution was fitted with a linear function with only 1 parameter. The constant parameter was fixed to value 1, and only the parameter describing the slope of the function was left loose for fitting. The distance to the fit along the y axis was plotted on the right. This distribution was then fitted with a normal distribution, and the $\pm 9\sigma$ range has been highlighted in both plots. If the value is outside the range, the entire run is discarded, as was described in Sec. 3.4.

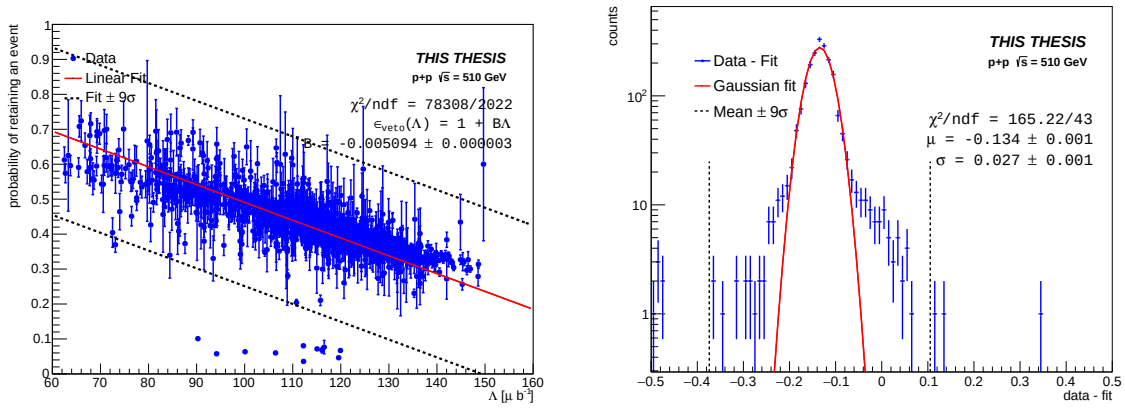


Figure 6.4: Left: the probability of retaining an event, also called the veto efficiency. The x axis represents the instantaneous luminosity, and the y axis is the probability of retaining an event. The distribution was fitted with an linear function. Right: the fit value for each run's luminosity subtracted from the probability of retaining an event value. The distribution was fitted with a normal distribution. The black lines in both plots represent the $\pm$ 9σ range of the Gaussian fit.

Integrated luminosity for each run is multiplied by the function value depending on the instantaneous luminosity obtained from a linear fit function with the fit parameters. The final effective luminosity is given by relation in Eq. 6.3 with the pile-up correction described in the following section.

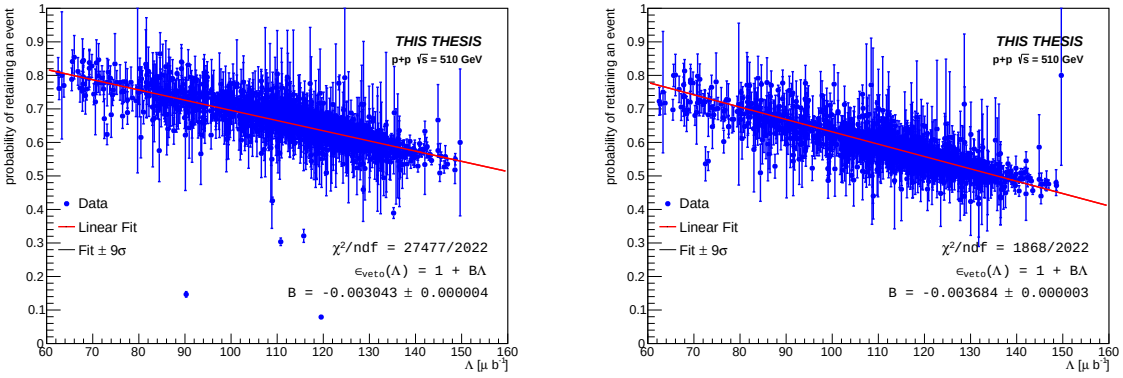


Figure 6.5: Number of zerobias events which satisfy the BBC East (left) and the BBC West (right) condition in the trigger divided by the number of zerobias events with 0 vertices in each run. Fitted with linear functions. Results of the fits are printed on the plots.

6.3 Pile-Up Correction

The Run17 dataset is considerably marked by pile-up, meaning there are multiple independent primary collisions per bunch crossing. In the data, we select events with exactly 1 reconstructed primary vertex (C2), therefore, we have to correct for this. This is done by analyzing zerobias data. For each run, we divide the number of events with exactly 1 reconstructed vertex by the number of events with at least 1 reconstructed vertex

$$P_{\text{pile-up}} = \frac{\#\text{ zerobias events with exactly 1 reconstructed vertex}}{\#\text{ zerobias events with at least 1 reconstructed vertex}}. \tag{6.2}$$

The distribution is plotted in Fig. 6.6 again, against instantaneous luminosity. The distribution, which falls slowly, is fitted with a linear function. In this case, both parameters were left loose.

The effective integrated luminosity L_{int}^{eff} is obtained for each analysis separately with

$$L_{int}^{eff} = \sum_{run} L_{int}^{run} \epsilon^{veto}(\Lambda^{run}) \epsilon^{pile-up}(\Lambda^{run}), \tag{6.3}$$

where the L_{int}^{run} is the integrated (not corrected) luminosity for a specific run, and Λ^{run} is the instantaneous luminosity for the corresponding run. 2 separate corrections are applied ϵ^{veto}, $\epsilon^{pile-up}$. The first one comes from veto conditions described in the previous section, $\epsilon^{pile-up}$ is the pile-up correction defined as

$$\epsilon^{pile-up}(\Lambda^{run}) = A_p + B_p \Lambda^{run}, \tag{6.4}$$

where B_p is the parameter from the fit shown in Fig. 6.6.

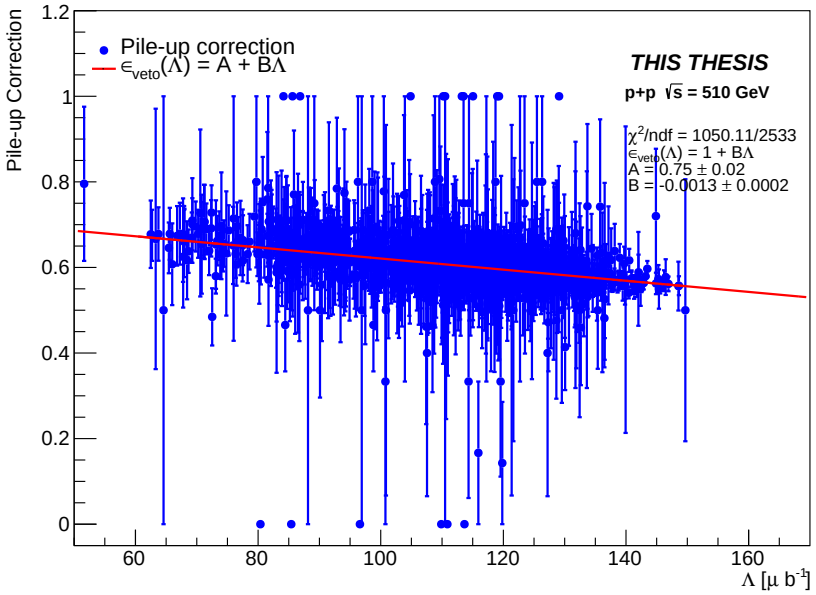


Figure 6.6: Pile-up correction plotted against instantaneous luminosity on a run-by-run basis. Fitted with a linear function with parameter results printed on the canvas.

6.4 TPC V_Z Correction

The luminosity of the collider is lowered by the application of the V_Z cut (C3) on the data. Generally, this would be part of the embedding. Unfortunately, the production of embedding data for proton-proton collisions is not capable of this as will be described in Sec. 6.5. Therefore, vertex reconstruction and correction to V_Z cut have to be done separately.

The efficiency is measured by looking at distributions of z position of primary vertices on a Fill-by-Fill basis and extracting the percentage of vertices that are discarded by the cut. This is done with 2 independent methods; the first one is fitting the distribution with a Gaussian function, and then integrating the function fit

$$\epsilon_{V_Z}^{\text{fit}} = \int_{V_Z^{\text{Min}}}^{V_Z^{\text{Max}}} \mathcal{N}(V_Z;\mu,\sigma)\, dV_Z = \frac{1}{2}\left[\text{Erf}\left(\frac{V_Z^{\text{Max}}-\mu}{\sqrt{2}\sigma}\right) - \text{Erf}\left(\frac{V_Z^{\text{Min}}-\mu}{\sqrt{2}\sigma}\right)\right], \quad (6.5)$$

where the $V_Z^{max/min}$ are the integration limits, in our case (-100,100) cm. The error function is defined as

$$\text{Erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}\, dt. \quad (6.6)$$

The second method is simply addition of the bins in the (-100,100) range, while assuming that 100 % of the vertices are in range (-200,200) cm. Then, the efficiency is defined as

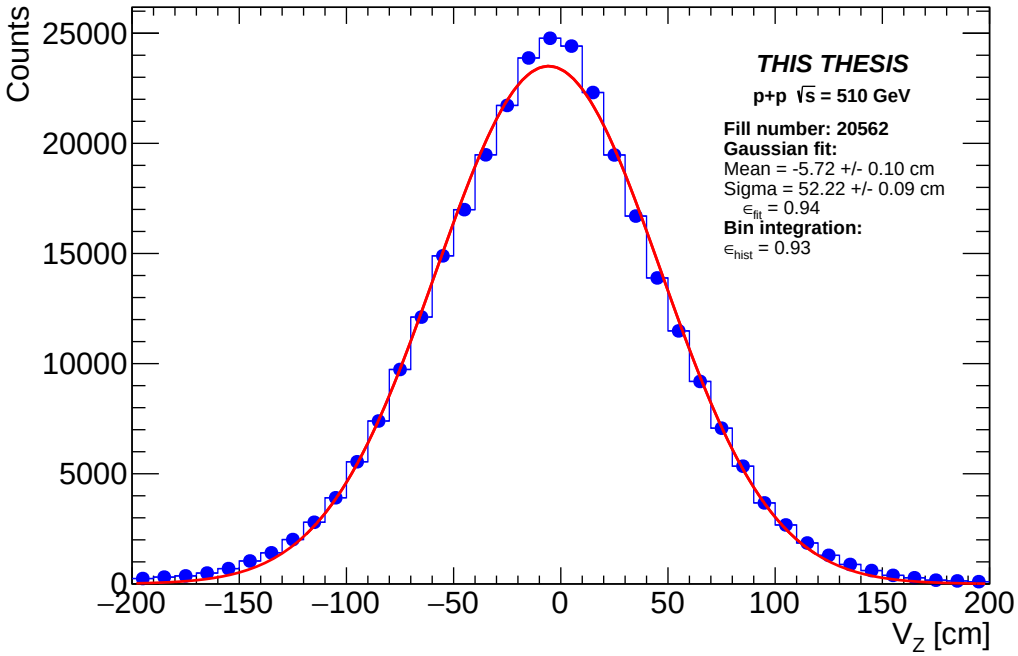


Figure 6.7: Distribution of z position of primary vertices for Fill number 20562. Fitted with a Gaussian distribution, the fit results of the parameters μ and σ are printed on the canvas. The final efficiency for both methods is printed on the canvas as well.

$$\epsilon_{V_Z}^{\text{int}} = \frac{\sum_{i=i_{min}}^{i_{max}} n_i}{\sum_{i=1}^{N} n_i} \tag{6.7}$$

where N is the number of all bins, n_i is the number of entries in the i-th bin, and i_{min}, i_{max} are the first and last bin of the range in which we look for the efficiency.

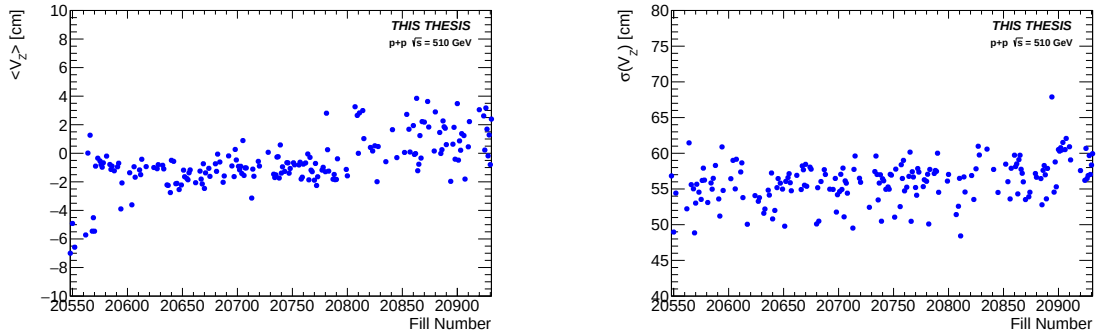


Figure 6.8: Extracted parameters from fits of V_Z distribution along the z axis. The left plot shows the expected mean values w.r.t. Fill numbers, right plot shows the standard deviations.

An example of a fit for fill number is shown in Fig. 6.7, with the results of the fit and the efficiency results of both methods. Distribution of mean expectation values and the standard deviations of the fits are shown in Fig. 6.8. Last of all,

the efficiency results w.r.t. fill numbers are shown in Fig. 6.9. Blue circles are the efficiency values obtained from the fit, and the red squares are the bin integration values. For each distribution, the mean value is calculated and plotted as a straight line. The resulting correction value used in the cross section is calculated as the average value between the 2 averages of the methods. The difference between the 2 methods divided by 2 is the systematic error of the combination of the 2 methods.

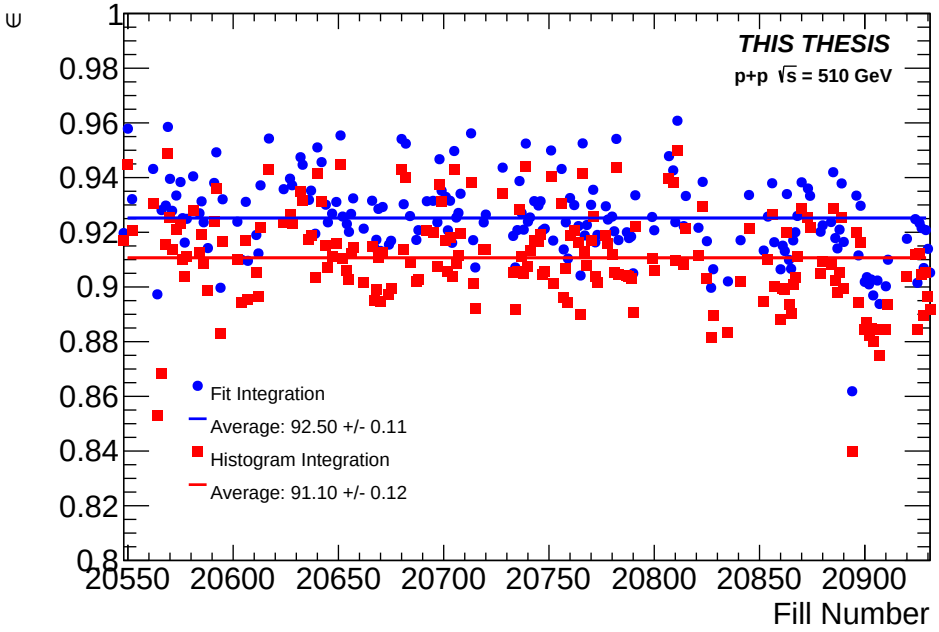


Figure 6.9: The efficiency of cut on V_Z w.r.t. Fill number. Blue points are from the fit method, red from the bin integration method and straight lines are the mean values of each method.

6.5 Embedding

The central embedding simulates the detector output, the data reconstruction, and the data production. The procedure of the entire process is described below.

E1 The first part is the generation of a sample of events with a Monte Carlo (MC) generator. For our purpose of generating a coherent production of J/ψ events, we decided to use STARlight [14]. The STARlight MC sample and the generation has been previously described in chapter 5.

E2 We used STARsim to simulate the crossing of particles through the different detectors and materials. The output from STARsim is an *.fzd* file.

E3 The data production is done by using the so-called Big Full Chain (BFC), which takes the *.fzd* file from the simulation, and the *.daq* files of rp_zerobias data stream.

E4 Once the BFC is finished, the output then needs to be converted from the standard *MuDst* format to *star-upcDst* format.

6.5.1 Event Selection

Ideally, once the produced sample is converted into *star-upcDst*, we use the same conditions as were used for the data sample. Unfortunately, this is not possible. The vertex reconstruction does not work properly in embedded low-mulitplicity pp collisions. Only a small fraction of the embedded events are reconstructed. That is why the vertex reconstruction efficiency is done separately. Without the reconstructed

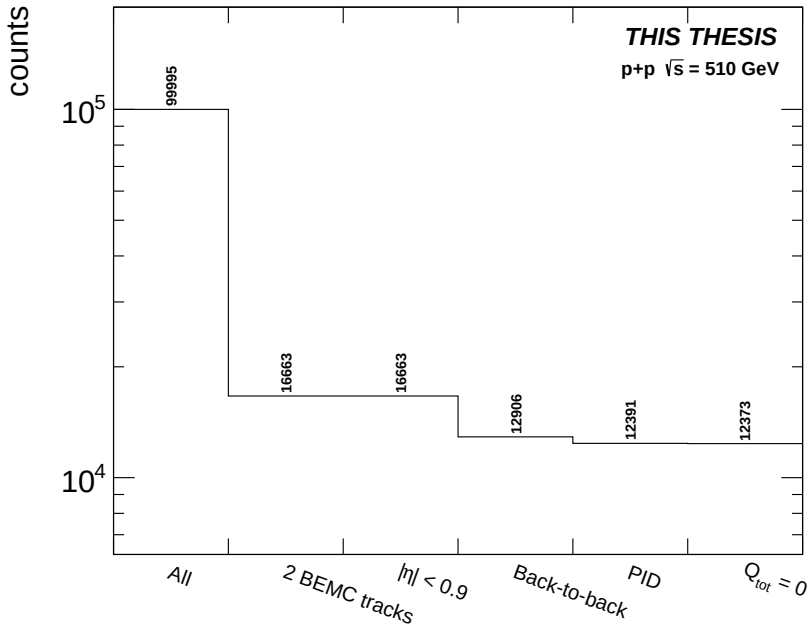


Figure 6.10: The flow of cuts in embedding analysis.

primary vertex, one has to use the information *idTruth* of daughter particles. First, the program loops over all tracks and selects those that come from the embedded MC sample. This is possible because we are only interested in the final yield of reconstructed J/ψ candidates.
The cuts that were not used in the embedding analysis are the 1 primary vertex (C2), the z position of vertex (C3), and the distance of closest approach cuts (C4.3, C4.4). The cut on the exactly 2 BEMC tracks has also been removed. The number of events after each condition is shown in Fig. 6.10.

After all the cuts, the distributions of variables $N_{\text{hits}}^{\text{fit}}$ and $N_{\text{hits}}^{\langle dE/dx \rangle}$ are shown in Fig. 6.11 with the distributions that come from the data. The number of entries of embedding is scaled to match the number of entries in data. The distributions look almost identical. The distributions of pseudorapidity η and azimuthal angle ϕ are shown in Fig. 6.12. The embedding sample is scaled to match the statistics of the data. The particle identification cut on the χ^2 of electrons (C7.1) is shown in Fig. 6.13. The limit is the red circle.

Distributions shown in Fig. 6.14 are the transverse momentum comparisons. The left plot shows the reconstructed transverse momentum of J/ψ candidates, the right shows the transverse momenta of daughter particles.

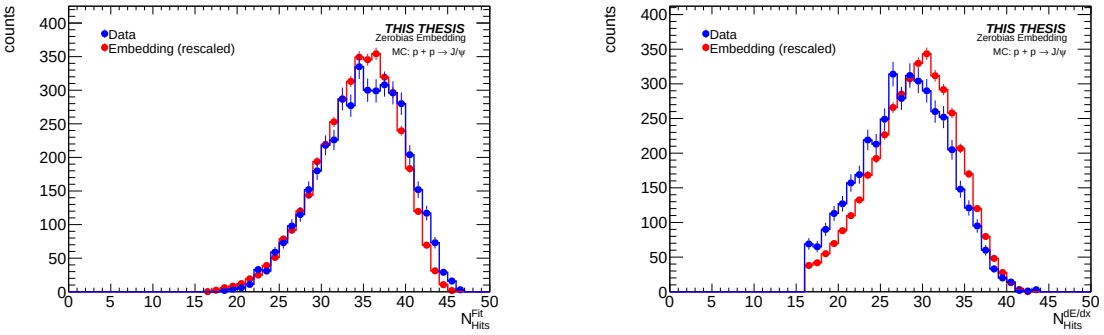


Figure 6.11: Comparison plots of N_{hits}^{fit} (left), and $N_{hits}^{dE/dx}$ (left) between embedding (red) and data (blue).

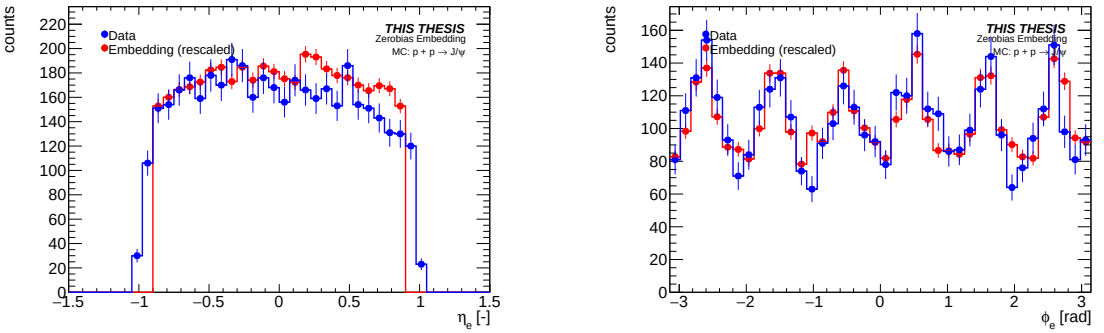


Figure 6.12: Comparison plots of η (left), and ϕ (right) between embedding (red) and data (blue).

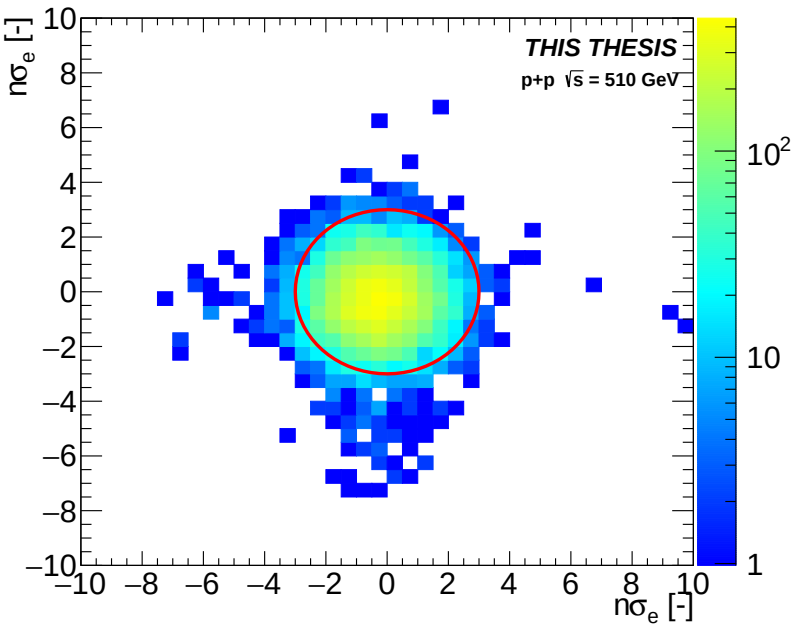


Figure 6.13: Correlating distribution of nσ_e for pairs of particles in embedding. Red circle represents the χ^2 cut (C7.1).

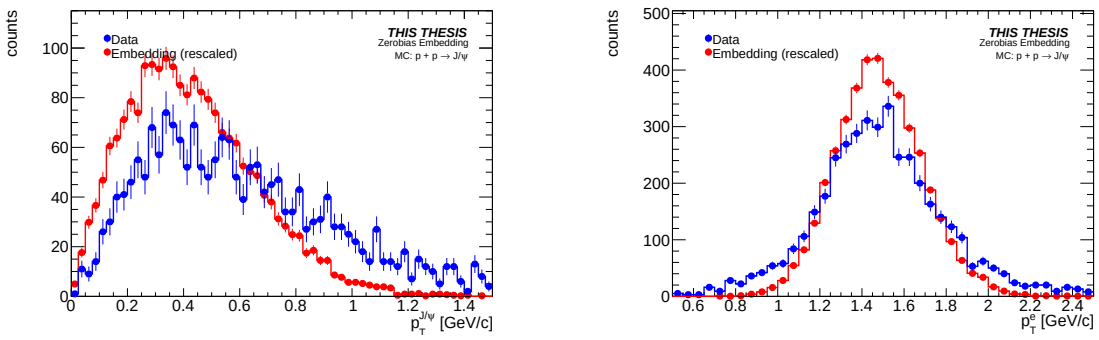


Figure 6.14: Left: distribution of reconstructed J/ψ particle transverse momentum in data (blue) and embedding (red). Right: distributions of reconstructed electrons' transverse momenta in data (blue) and embedding (red).

6.5.2 Reconstruction Efficiency

The final reconstruction efficiency is the number of J/ψ candidates that were reconstructed, passed the data production, and the reconstructed tracks passed the conditions described in the previous section divided by the number of J/ψ candidates generated originally. The final number of reconstructed J/ψ candidates is shown in Fig. 6.15. The embedding data are fitted with a crystal ball function (Eq. 8.1) from which the yield is extracted. The red peak is a crystal ball function with parameters from the data, scaled (by yield) to match the embedding. The point of this is to display that the embedding peak is wider than the peak from the data. In the opposite case, further smearing of the reconstructed tracks would be necessary. In our case, the smearing done by STARsim and reconstruction is enough.

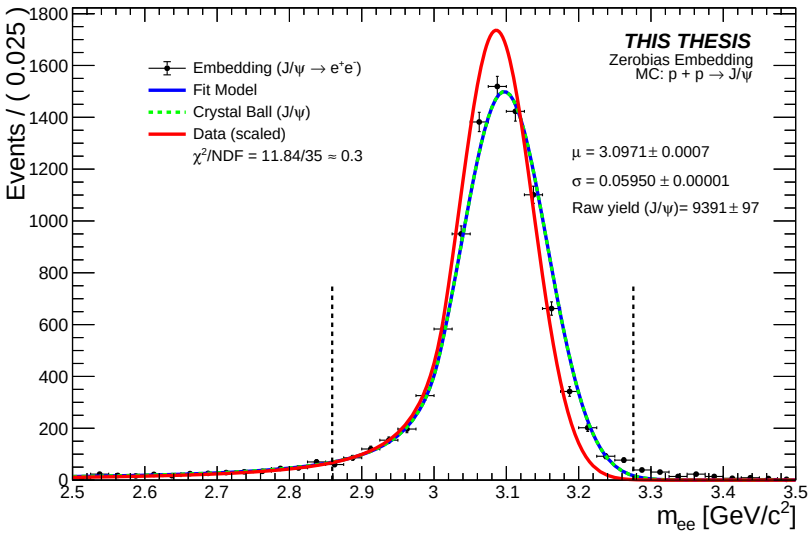


Figure 6.15: Comparison between embedding J/ψ peak fit and the fit from data. The fit from data is shown in red, the embedded data are shown and fitted with a crystal ball distribution, and integrated in the range of $(\mu - 4\sigma,\ \mu + 3\sigma)$.

Reconstruction efficiency is also shown as a function of pseudorapidity of the daughter particles (Fig. 6.16 left) and the rapidity of the J/ψ (Fig. 6.16 right). Both are

constant in the central region, and are only lower on the sides, where acceptance issues play a role.

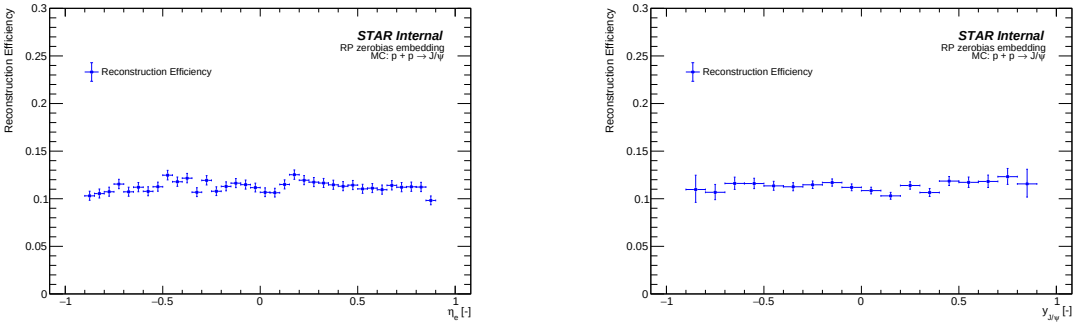


Figure 6.16: The reconstruction efficiency plots w.r.t. electron pseudorapidity η and J/ψ rapidity y.

The last distribution in this subsection is the reconstruction efficiency w.r.t. the transverse momentum of the J/ψ candidate. Throughout the range, the efficiency is monotonously decreasing, almost in a linear fashion. The correction of reconstruction efficiency to the final cross section is applied using this distribution. The method of application is further described in the following subsection.

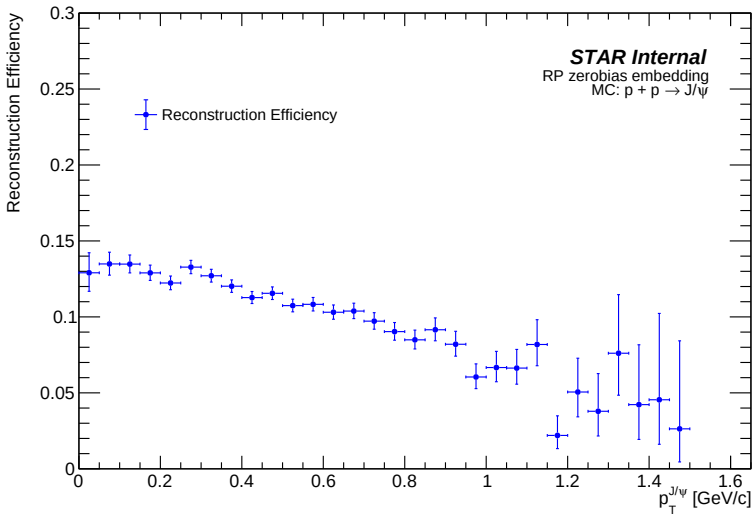


Figure 6.17: Reconstruction efficiency w.r.t. the transverse momentum of J/ψ. The reconstruction efficiency was applied to calculate the cross section using this distribution.

6.5.3 Correction Application

The correction to the reconstruction efficiency is applied by using the efficiency w.r.t. the transverse momentum of the J/ψ, shown in Fig. 6.17. The method of application goes as follows: after fitting the peak (shown in Fig. 8.1), the transverse momentum of all candidates in the fitting range $(\mu - 4\sigma,\ \mu + 3\sigma)$ is plotted. Based on the fit

result, we determine how many of the events in the range are the J/ψ events. We determine the fraction of the signal by using relation

$$f = \frac{Y_s}{Y_s + Y_b}, \tag{6.8}$$

where Y_s is the yield of the signal, and Y_b is the background yield. We are not able to separate the signal from the background, so we will assume that the background is distributed the same way as the J/ψ candidates along the entire p_T spectrum. Then, the correction is applied by applying relation

$$Y_c = \sum_{i=1}^{n_{bins}} \frac{b_i(p_T) * f}{\epsilon(p_T)}, \tag{6.9}$$

where Y_c is the corrected yield, the sum goes over all bins of the p_T spectrum, $b_i(p_T)$ is the bin content for the i-th bin (which translates to a specific p_T), and the $\epsilon(p_T)$ is the reconstruction efficiency shown in Fig. 6.17.

6.6 BEMC Efficiency

As was mentioned before, the simulation of the BEMC does not match the BEMC efficiency obtained from the data. To correct this, we have created our own method of simulating the BEMC efficiency and applied it. The data driven method used to obtain the BEMC efficiency is described in Sec. 6.6.1. The method of application of the obtained efficiency in the embedding is described in Sec. 6.6.2. The quality assurance of the method using the $\gamma+\gamma \rightarrow e^+ + e^-$ spectrum is described in Sec. 6.6.3.

6.6.1 Data-Driven Method

The list of conditions below is used to select events with photon conversions to electron pairs ($\gamma \rightarrow e^+e^-$). The event selection does have some similarities to event selection for J/ψ events, although with some differences. The different cuts and the number of passed events can be seen in Fig. 6.18.

B1 Exactly 1 primary vertex

B2 Condition on the z position of vertex: $|V_Z| < 100$ cm

B3 Exactly 2 good quality TOF tracks

- B3.1 Each TPC track must have good quality reconstruction: $N_{hits}^{fit} > 15$
- B3.2 Each track must have good quality energy measurement: $N_{hits}^{dE/dx} > 15$
- B3.3 Each TPC track must satisfy: $|DCA_Z| < 1$ cm
- B3.4 Each TPC track must satisfy: $DCA_{XY} < 1.5$ cm
- B3.5 Each TPC track must be matched to a hit in TOF
- B3.6 There is exactly 2 TPC tracks that satisfy the conditions above

B4 Both tracks are projected to BEMC acceptance

B5 Pair satisfies $\delta^{dip} < 0.03$

B6 Condition on the pseudorapidity of the daughter particles $|\eta| < 0.9$

B7 Particle identification

B7.1 Electron pairs satisfy $\chi^2 < 9$

B8 Invariant mass condition: $m_{ee} < 0.1$ GeV

B9 Total electric charge of pair: $Q_{tot} = 0$

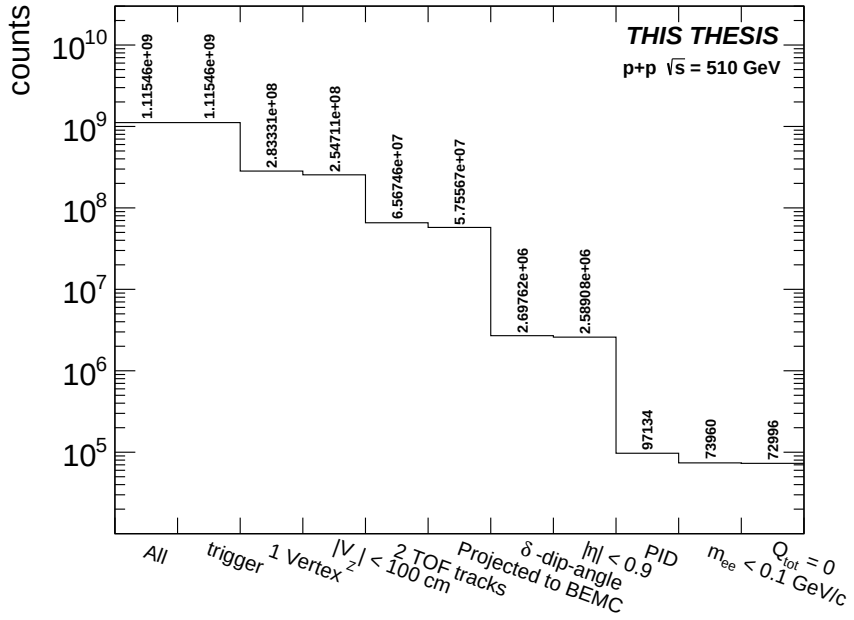


Figure 6.18: The specific number of passed events after each cut for BEMC efficiency study.

Instead of the BEMC match, we select pairs of electrons matched to hits in TOF. The TPC tracking conditions and the $V_Z - \eta$ cuts are the same. The particle identification condition is looser, only the χ^2 cut is used. Instead of looking at back-to-back tracks, we use a new condition called δ-dip angle. The δ-dip angle is a combined function of daughter particles' momenta $p^{(1)}$, $p^{(2)}$, $p_T^{(1)}$, $p_T^{(2)}$, $p_Z^{(1)}$, $p_Z^{(2)}$ and is defined as

$$\delta = \cos^{-1}\left(\frac{p_T^{(1)}p_T^{(2)} + p_Z^{(1)}p_Z^{(2)}}{p^{(1)}p^{(2)}}\right). \tag{6.10}$$

The variable represents the opening angle at the origin (the point at which the 2 tracks have the smallest distance of closest approach, which represents the decay vertex). Electron pairs coming from photon conversions usually are parallel to each other, therefore have a small δ-dip value [80]. The majority of photon conversion electron pairs are well below the chosen cut B5, as can be seen in Fig. 6.19.

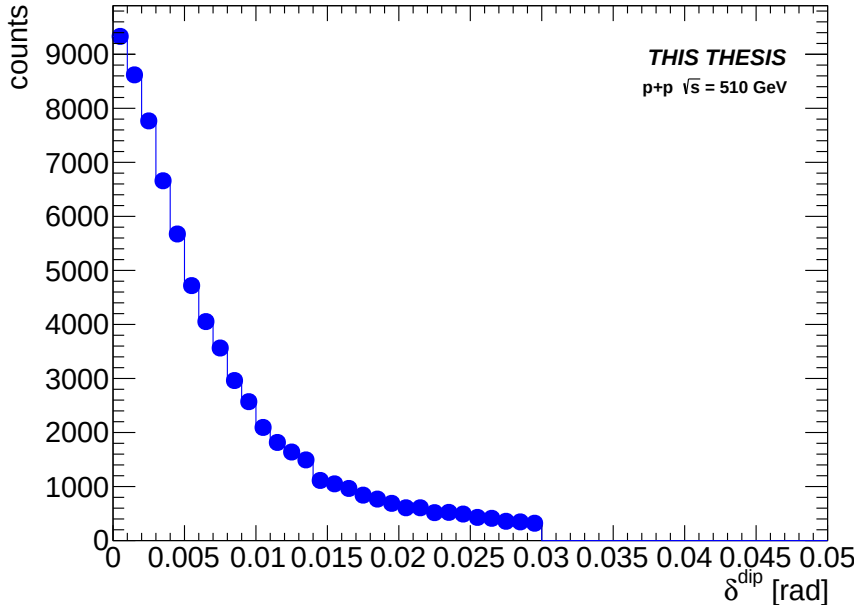


Figure 6.19: Distribution of variable δ^{dip} angle used for identification of photon conversion electrons with a cut applied at $\delta^{dip} < 0.03$.

To ensure the electron pairs are coming from a single massless photon, we cut on the reconstructed invariant mass of the pair (B8). Last of all, the pair has to be of the same charge.
The distribution of efficiency w.r.t. the transverse momentum of the traversing particle is shown in Fig. 6.20, along with efficiency which comes from the embedding simulation. Clearly, the efficiency rises with the transverse momentum up to a value around 1.5 GeV, when it becomes constant at around 85 %. The distribution is fitted with the function defined in Eq. 6.11. The function has 3 independent parameters, the results of the fit are printed in Fig. 6.12. The fit is later used for the simulation of BEMC efficiency, as is described in the following section.

6.6.2 BEMC MC Simulator

To match the efficiency of BEMC from data, we introduce a Monte Carlo BEMC simulator. Instead of looking at which track has a match with the BEMC, we use the transverse momentum of the track and the fit function with the parameters determined from data.

$$f(p_T) = n\left[1 + \mathrm{Erf}\left(\frac{p_T - p_T^{thr}}{\sqrt{2}\sigma}\right)\right]. \tag{6.11}$$

For each particle, we generate a random number in the range (0,1) with `TRandom3` class. At the same time, we calculate the BEMC efficiency value for the specific transverse momentum of the track with Eq. 6.11. If the generated random number is smaller than the efficiency value, the track is said to have a BEMC hit. If the random value is larger, then no hit in the BEMC. The difference between the BEMC STARsim (embedding) efficiency, and the BEMC efficiency from data can be seen

in Fig. 6.20. The red points represent the STARsim BEMC efficiency. The blue points in the graph represent the BEMC efficiency, which is fitted with the function defined in Eq. 6.11. Although the difference is not large in the region above 1 GeV, the difference is significant below 1 GeV, which is why it was necessary to create our own BEMC simulation.

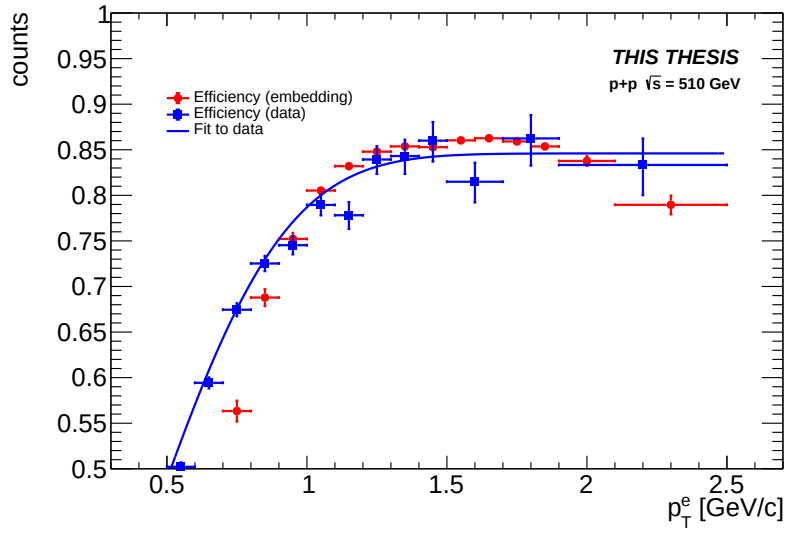


Figure 6.20: BEMC efficiency from the data (blue) compared to the BEMC efficiency from embedding (red). The efficiency obtained from the data are fitted with function Eq. 6.11.

6.6.3 BEMC Efficiency QA

To ensure the BEMC efficiency is applied correctly, we look at spectrum of electron-positron pairs in the process $\gamma + \gamma \rightarrow e^+ + e^-$. As shown in Fig. 8.1, we have fitted the peak of J/ψ along with the spectrum. The shape of the spectrum is then compared to the shape of the embedded Starlight MC sample of the same process.

The shape of the spectrum is fitted with a function defined as

$$f_{bkg}(m_{ee}) = (m_{ee} - c_1)\exp\left(c_2(m_{ee} - c_1)^2 + c_3 x^3\right) \tag{6.12}$$

where c_1, c_2, and c_3 are the parameters. The fit of the background, when we used the BEMC embedding simulation, is shown in Fig. 6.21. The magenta line is the shape taken from data. When we used our own MC simulator of BEMC efficiency, the spectrum changed its shape to match the data much more closely. What is especially important is the exponential decrease of the spectrum from around 1.5 GeV, because that is the area of interest for us.

6.7 Summary of Corrections

This section summarizes the efficiency studies and corrections. Some corrections are applied globally, meaning they are applied to the final integrated cross section.

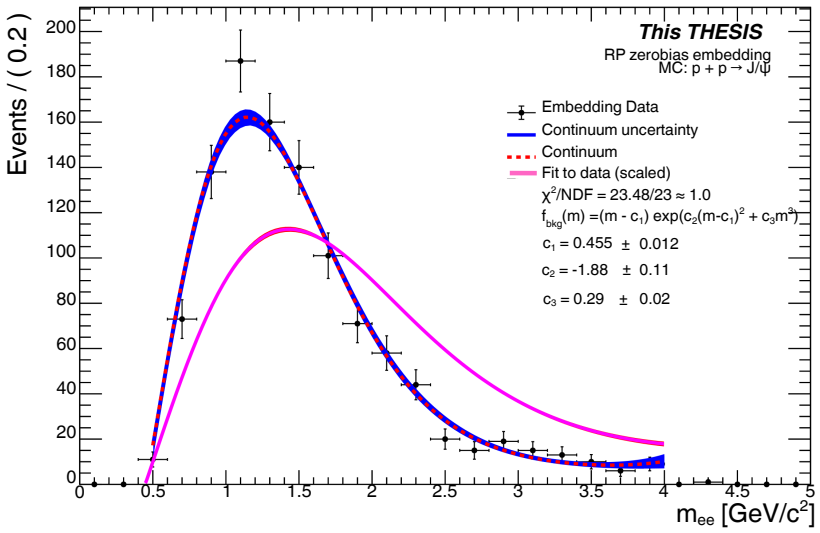


Figure 6.21: The shape of the continuum using the STARsim BEMC efficiency compared to the shape from data. Shape fitted with function Eq. 6.12.

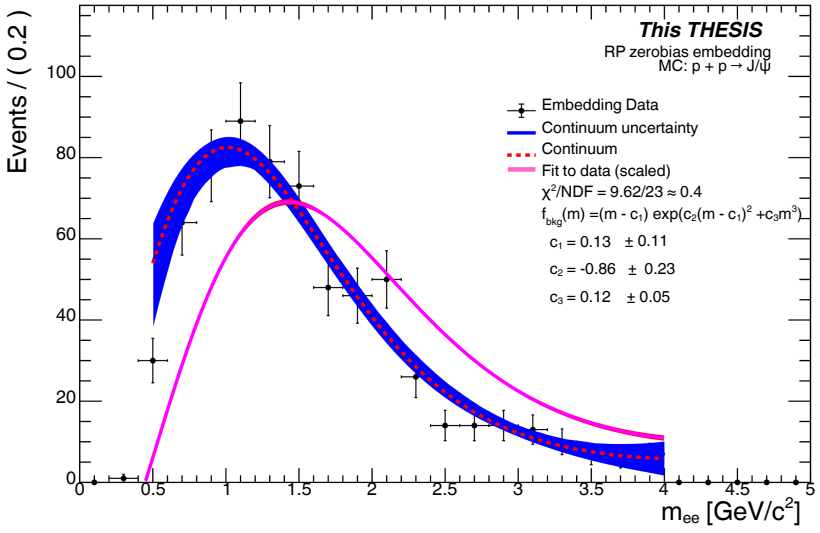


Figure 6.22: The shape of the continuum using the MC BEMC efficiency simulator compared to the shape from data. Shape fitted with function Eq. 6.12.

These corrections are summarized in Tab. 6.1. The second type of corrections is applied either depending on the instantaneous luminosity (trigger veto correction and pile-up correction) or based on the transverse momentum of the J/ψ candidate (reconstruction efficiency). The BEMC efficiency described in the previous section is part of the reconstruction efficiency.
The vertex reconstruction efficiency and the Roman Pot detector efficiency was done previously by Tomas Truhlar for Central Exclusive Production; the methodology is described in ref. [7].

	ϵ [%]
Trigger Topology	39.0
Vertex Reconstruction	89.6
TPC V_Z Correction	92.5
Roman Pots	88.9

Table 6.1: Summary of efficiencies which will be applied globally on the integrated cross section.

Chapter 7

Systematic Uncertainties

This chapter discusses the systematic uncertainties, their sources, and the various methods for their evaluation. The first section, Sec. 7.1, discusses the systematic uncertainties coming from multiple cuts in the track and event selection process. Following is the uncertainty coming from yield extraction, namely the fit of the peak. The following section discusses the systematic uncertainty stemming from the implementation of our own Monte Carlo BEMC efficiency simulator. Next is the discussion of the luminosity systematic error combined with the trigger and pile-up uncertainties. The systematic errors for vertex reconstruction and the Roman Pot detector system efficiency have been studied by Tomas Truhlar for Central Exclusive Production analysis [7]. The same values will be used; they are shown in Sec. 7.6.

7.1 TPC Cuts

The Time Projection Chamber is used for various aspects in the offline analysis, therefore, its combined systematic uncertainty is expected to be large. Both analyses use the TPC for track reconstruction, vertex reconstruction, and particle identification. In addition, we create conditions in the offline analysis to create a clean signal of J/ψ. The systematic uncertainty coming from the event selection cuts is evaluated by varying the values at which we cut. The nominal values, described in Sec. 4.1, are compared to so-called loose and tight conditions. The TPC cuts which are evaluated are the following: Vertex z position cut (C3) N_{hits}^{fit} (C4.1), $N_{hits}^{dE/dx}$ (C4.2), distance of closest approach to vertex cuts DCA_Z, DCA_{XY} (C4.3, C4.4), pseudorapidity η (C5), particle identification conditions χ_e^2 (C7.1), and the exclusion of kaons and protons with nσ cut (C7.2).
Obviously, just using the loose conditions would lead to a higher cross section and with the tighter cut, the opposite. When combined with a change in the reconstruction efficiency (we apply the same loose/tight condition on the embedding sample), the final value should theoretically lead to the same cross section as with the nominal values. The difference in the cross sections between the nominal and the tight/loose conditions will be used as the systematic uncertainties of the various cuts.
Because we are not able to reconstruct vertices in the embedding, we were forced to evaluate the systematic uncertainty of the V_Z cut (C3) in a different manner. This is

done by varying the limits in the efficiency of the z position of the vertex, described in Sec. 6.4. In addition to this, the efficiency estimation with the 2 methods brings its own uncertainty, evaluated as 1/2 of the difference between the 2 values. For DCA cuts (C4.3, C4.4) we make an assumption, that the reconstruction efficiency does not depend on the chosen cuts and only vary the tight/loose conditions in the data.

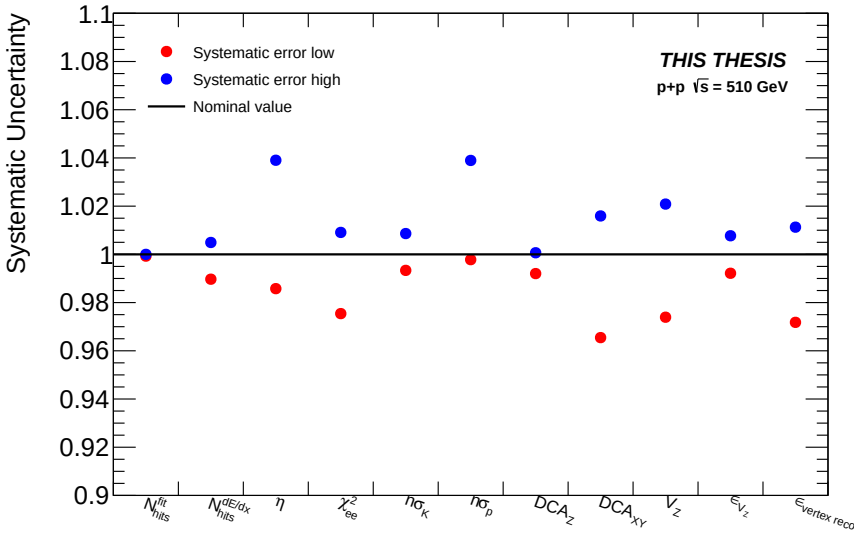


Figure 7.1: Summary of systematic uncertainties coming from TPC cuts. The blue values come from the looser cuts, the blue ones from the tighter cuts.

7.2 Yield Extraction

The yield extraction uncertainty comes from the uncertainty of the fit. The error is evaluated by varying the number of bins, the lower and the top limit of the invariant mass histogram in a reasonable range of values. All the yields are then filled into a histogram. The distribution of the yields (shown in Fig. 7.2) is then fitted with a Gaussian distribution. The error is the σ parameter of the Gaussian printed in the plot. The error is then propagated to the resulting cross section. It is evaluated at 0.03 %, which is negligible compared to other systematic uncertainties.

7.3 Monte Carlo BEMC Efficiency Simulator

The systematic uncertainty stemming from the Monte Carlo BEMC efficiency simulator is evaluated by varying the parameters n, p_T^{thr}, and σ of the fit shown in Fig. 6.20 within the errors of the fit. The uncertainty is then propagated to the resulting cross section and compared to the nominal value. The error has been evaluated at $^{+6.1}_{-3.4}$ %.

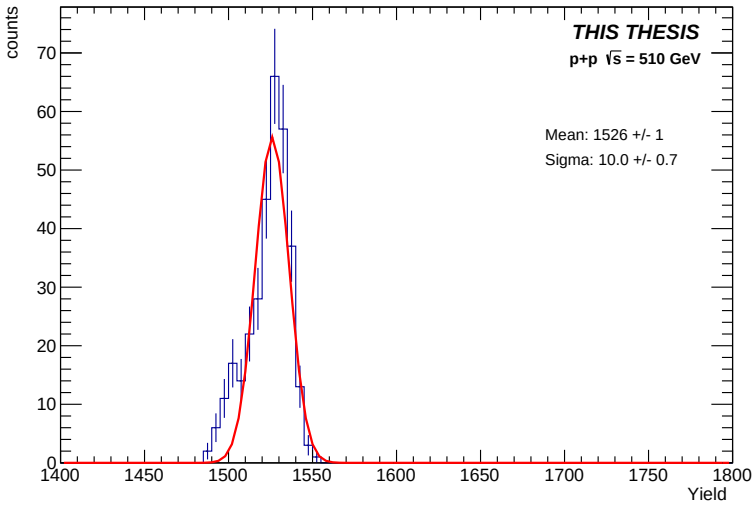


Figure 7.2: Evaluation of the systematic uncertainty of the yield extraction. The values are obtained by varying the number of bins and the limits of the invariant mass histogram. The extracted yields are then filled into this plot, and the distribution is fitted with a Gaussian function. The resulting parameters are printed on the canvas.

7.4 Topology Trigger Uncertainty

The topology trigger efficiency is described previously in Sec. 6.1. The error comes from varying the lower and upper limit of the invariant mass range, in which we are measuring the efficiency. Again, the efficiency has a certain normal distribution, which is fitted with a Gaussian function and the error is propagated to the cross section. The fit can be seen in Fig. 6.3. The uncertainty from the trigger efficiency estimation is evaluated at $^{+4.9}_{-4.5}$ %.

7.5 Luminosity Error

The luminosity error stems from 3 separate sources. The first comes from the measurement of the luminosity using Van der Meer scans, evaluated at 5 % [81]. It is one of the largest contributions to the entire uncertainty. The other contributions to the luminosity determination comes from the probability of retaining an event, described in Sec. 6.2, and the pile-up correction, described in Sec. 6.3. Both errors are evaluated similarly to the previously described MC BEMC efficiency simulator uncertainty. The parameters of the fits are varied in the range of the fit errors. The errors are then propagated to the effective integrated luminosity and, subsequently, to the final cross section. Compared to the nominal value, the uncertainty for veto correction (probability of retaining an event) has been evaluated at $^{+0.1}_{-0.1}$ %, and the pile-up correction $^{+2.4}_{-2.3}$ %.

7.6 Summary of Systematic Uncertainties

All the described systematic uncertainties are combined into 2 values: one for the upper limit uncertainty estimation, the other for the lower limit. They are combined as the square root of quadratures

$$\sigma = \sqrt{\sum_{i=1}^{n} \sigma_i^2}. \tag{7.1}$$

The resulting estimates for the upper and lower systematic uncertainties differ for the 2 analyses because of the Roman Pots. For the analysis without the Roman Pots, the combined systematic uncertainty has been evaluated at $^{+11.6}_{-9.9}$ %. In the case with the Roman Pots, the uncertainty is $^{+11.8}_{-10.1}$ %. A summary of all sources and their contributions can be seen in Fig. 7.3. The largest contribution comes from the uncertainty from the TPC combination, the luminosity measurement, specifically the Van der Meer scans, and the uncertainty stemming from the BEMC efficiency estimation.

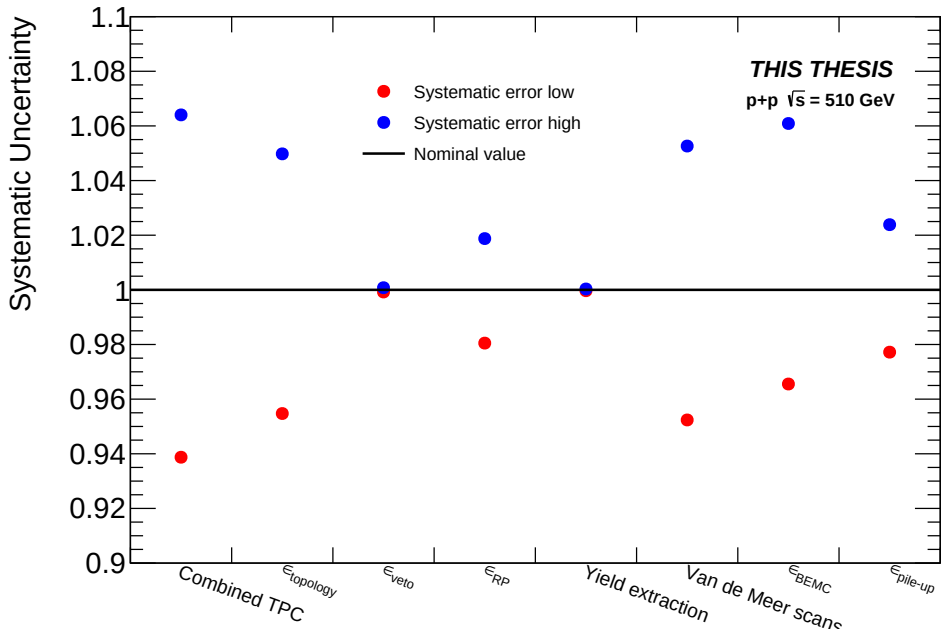


Figure 7.3: Summary of various contributions to the systematic uncertainties. Blue values represent the upper estimation, red values represent the lower estimation. All values are shown w.r.t. the nominal value.

Chapter 8

Results

This last chapter discusses the obtained results of the analysis. The first section shows the resulting plots of the invariant mass spectra, fits of the peaks, and the extracted yields. Following is a section which discusses the main result of the thesis; cross section of J/ψ photoproduction. In addition, we show the rapidity distribution of the measured cross section. Finally, we discuss the obtained results, compare them to similar measurements or theoretical predictions, and examine the potential sources of the differences. In addition, future improvements of the analysis are presented.

8.1 Yield Extraction

The extraction of the raw yield of J/ψ candidates is done by fitting the invariant mass distribution in `RooFit` package, a toolkit integrated in the ROOT [82] library. The signal is fitted with a crystal ball function defined as

$$f(x;\mu,\sigma,\alpha,n) = N \times \begin{cases} \exp\left(-\dfrac{(x-\mu)^2}{2\sigma^2}\right), & \text{for } \dfrac{x-\mu}{\sigma} > -\alpha, \\ A\left(B - \dfrac{x-\mu}{\sigma}\right)^{-n}, & \text{for } \dfrac{x-\mu}{\sigma} \leq -\alpha, \end{cases} \tag{8.1}$$

where variables A and B are defined as

$$A = \left(\frac{n}{|\alpha|}\right)^n \exp\left(-\frac{\alpha^2}{2}\right), \qquad B = \frac{n}{|\alpha|} - |\alpha|,$$

and the normalization constant N is

$$N = \frac{1}{\sigma\left[\dfrac{A}{n-1} + \sqrt{\dfrac{\pi}{2}}\left(1 + \mathrm{Erf}\left(\frac{\alpha}{\sqrt{2}}\right)\right)\right]}.$$

Crystal ball function is a standard function used in particle physics experiments. It is obtained from Gaussian and power law functions to make the left side decrease

less radical. Very often particles traversing the material of the experiment setup lose energy, which is why their reconstructed momentum is smaller than the actual value. In our case, it is mainly due to the fact, that either the J/ψ decays to $e^+e^-\gamma$, or one of the electrons undergoes bremsstrahlung. The left side of the crystal ball is parametrized with α and n parameters. Instead of letting these parameters loose in the fit (which could present some uncertainties), we fixed them to α and n results from the fit to embedding.
The other function part of the fit model is the continuum function described previously in Sec. 6.6.3, specifically Eq. 6.12. It is a standard function that describes the physical process $\gamma + \gamma \rightarrow e^+ + e^-$. Physically, the shape of such process is an exponential decrease in the invariant mass distribution. In our case, it is a steep rise from 0.5 GeV caused by the detector inefficiency to reconstruct such low-momentum particles (mainly BEMC). Then, from around 1.5 GeV we can see the exponential decrease. The process itself is background to J/ψ photoproduction seen also in other analyses [51], [52].
The invariant mass distribution for the analysis of J/ψ photoproduction without the use of the Roman Pot system of detectors (only information from the central system) can be seen in Fig. 8.1. The background (e^+e^+, e^-e^-) is subtracted from the signal (e^+e^-). As was previously mentioned, the $\gamma+\gamma \rightarrow e^+ + e^-$ spectrum starts at about 0.5 GeV and then continues well beyond the mass of the J/ψ. The J/ψ peak can be very clearly seen at around 3.1 GeV. The results of the fit parameters can be found in Tab. 8.1, in the first column.

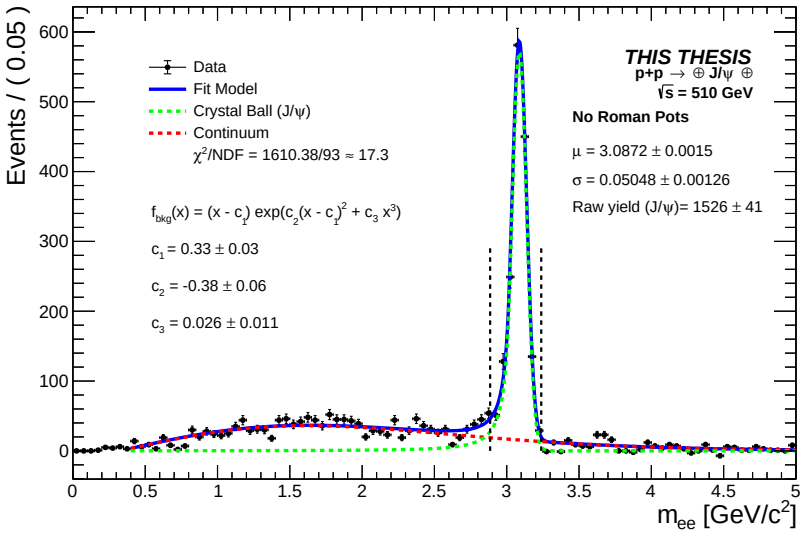


Figure 8.1: Invariant mass spectrum of e^+e^- pairs for analysis without the use of the Roman Pots. Distribution is fitted with a combination of a crystal ball function for the peak and a continuum function.

For the case with the Roman Pots, meaning we measure one of the forward scattered protons, the invariant mass spectrum can be seen in Fig. 8.2. Again, the background is subtracted. The statistics in this case are limited; it is possible to fit the distribution without any function for the background. Only the signal is fitted with a crystal ball function.

The comparison of the fit parameters for the crystal ball function can be seen in Tab. 8.1. In both cases, the parameters α and n were fixed to constants from the embedding. The extracted raw yields have been obtained from the integration of the crystal ball function in range $(\mu - 4\sigma,\ \mu + 3\sigma)$ and are also in the same table in the column denoted as Y.

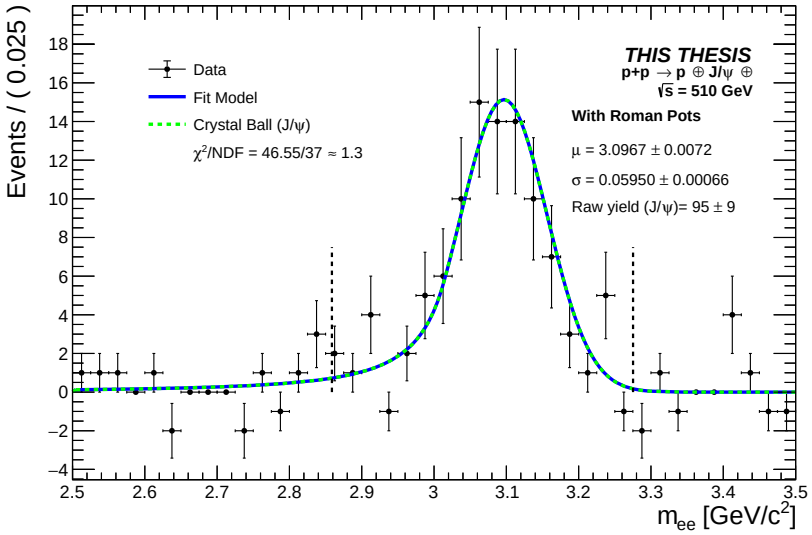


Figure 8.2: Invariant mass spectrum of e^+e^- pairs for analysis with the use of the Roman Pots. Distribution is fitted with a crystal ball function defined in Eq. 8.1.

	μ [GeV/c^2]	σ [GeV/c^2]	χ^2/NDF	Y
with RP	3.097 $\pm$ 0.007	0.0595 $\pm$ 0.0007	1.26/37 $\approx$ 0.03	95 $\pm$ 9
without RP	3.0872 $\pm$ 0.0015	0.050 $\pm$ 0.013	17.3/93 $\approx$ 0.2	1530 $\pm$ 40

Table 8.1: Table of crystal ball fit parameters for both J/ψ fits.

8.2 Cross Section

The cross section for J/ψ photoproduction in both cases (with and without Roman Pots) is obtained from extracted yields, corrections, and the integrated luminosity of the process. The exact relation is

$$\sigma = \frac{Y_{corr}}{\epsilon\ L_{int}^{eff}}, \tag{8.2}$$

where Y_c is the extracted yield corrected according to reconstruction efficiency (described in Sec. 6.5.3) w.r.t. the transverse momentum of J/ψ (shown in Fig. 6.17), ϵ is the combined efficiency, and L_{int}^{eff} is the effective luminosity. One very often shows the cross section per unit rapidity [53], [54]

$$\frac{\mathrm{d}\sigma}{\mathrm{d}y} = \frac{Y_{corr}}{\epsilon\ L_{int}^{eff}\ \Delta y}, \tag{8.3}$$

where factor Δy is the rapidity range of J/ψ , in our case it is 1.8. The combined efficiency is the product of all global efficiencies listed in Tab. 6.1. In the analysis without the RP, the Roman Pot efficiency is obviously not used.

	Y	Y_c	Y_g	ϵ [%]	L_{int} [pb^{-1}]	L_{int}^{eff} [pb^{-1}]
without RP	1509	15899	13258	32.3	315.0	76.2
with RP	93	1051	805	28.7	194.0	47.8

Table 8.2: Table of values used for the cross section calculation. Variables Y, Y_c, and Y_g represent extracted yield, corrected yield by the J/ψ transverse momentum spectrum, and the yield corrected with one overall reconstruction efficiency respectively. Variable ϵ is a product of all efficiencies summarized in Tab. 6.1, and L_{int} and L_{int}^{eff} are the integrated and the integrated effective luminosity.

The integrated luminosity without pile-up and veto correction is 315 pb for the analysis without the Roman Pots, and 194 pb for the case with the Roman Pots. The value differs in the 2 cases because for the case with the Roman Pots, we excluded runs with inactive RPs. After the veto and pile-up corrections are applied, the integrated effective luminosities are in Tab. 8.2, column denoted as L_{int}^{eff}. The table summarizes different results, such as Y_g, which is the extracted yield corrected with one overall value of the reconstruction efficiency. The integrated cross section values for the 2 separate cases in the dielectron decay channel are

$$\begin{aligned} \sigma^{p+p\to\oplus J/\psi(e^+e^-)\oplus}(|\eta_e|<0.9) &= 650 \ \pm \ 40(stat)\ ^{+70}_{-50}(sys) \text{ pb}, \\ \sigma^{p+p\to p\oplus J/\psi(e^+e^-)\oplus}(|\eta_e|<0.9) &= 77 \ \pm \ 5(stat)\ ^{+9}_{-7}(sys) \text{ pb}. \end{aligned} \tag{8.4}$$

The first value corresponds to the cross section without the Roman Pots, because we measure only the J/ψ decay products and ensure rapidity gaps ($\oplus$). The second value corresponds to the case when we measure one of the forward protons with the rapidity gaps between the particles in the final state. To get the cross section in all decay channels, one divides the values by the branching ratio BR($J/\psi\to e^+e^-$) = (5.971 $\pm$ 0.032) % [79]

$$\begin{aligned} \sigma^{p\ +\ p\ \to\ \oplus\ J/\psi\ \oplus}(|\eta|<0.9) &= 6.1 \ \pm \ 0.4(stat)\ ^{+0.7}_{-0.5}(sys) \text{ nb}, \\ \sigma^{p\ +\ p\ \to\ p\ \oplus\ J/\psi\ \oplus}(|\eta|<0.9) &= 720 \ \pm \ 60(stat)\ ^{+80}_{-70}(sys) \text{ pb}. \end{aligned} \tag{8.5}$$

The results of the cross sections in the dielectron channel per unit rapidity of the J/ψ are below. Similarly as in the integrated cross section, to obtain the J/ψ photoproduction cross sections in all channels, one would need to divide it by the branching ratio of J/ψ decay to dielectron channel.

$$\begin{aligned} \frac{\mathrm{d}\sigma^{p+p\to\oplus J/\psi(e^+e^-)\oplus}}{\mathrm{dy}}(|\eta_e|<0.9) &= 360 \ \pm \ 20(stat)\ ^{+40}_{-30}(sys) \text{ pb}, \\ \frac{\mathrm{d}\sigma^{p+p\to p\oplus J/\psi(e^+e^-)\oplus}}{\mathrm{dy}}(|\eta_e|<0.9) &= 43 \ \pm \ 3(stat)\ ^{+5}_{-4}(sys) \text{ pb}. \end{aligned} \tag{8.6}$$

8.3 Discussion

We studied 2 specific processes, the analysis with and without the Roman Pots, with which we are able to measure one of the forward protons. The analysis with the Roman Pots is not directly comparable to an existing measurement, because a similar measurement has not been done yet. On the other hand, the analysis without the Roman Pots, when we look at the daughter electrons in the central system only, is directly comparable to both experimental measurements and theoretical predictions. It is important to note that the 2 cross sections are not directly comparable: in the analysis with the Roman Pots, we look at protons in small windows of 5 cm by 8 cm above and below the beamline with certain specific transverse momenta (F2). We do not attempt to extrapolate the cross section outside this fiducial phase space, therefore, the cross section is bound to be smaller than the case without the Roman Pots.

By comparing the result to STARlight, we are comparing it to results from the HERA collider, because parameters in STARlight were restrained by fits to HERA data. The comparable cross section between STARlight and this measurement is about a factor of 16, which is a large discrepancy. The STARlight result for central exclusive J/ψ photoproduction, discussed at the end of Sec. 5.1, is $\sigma^{|\eta_e|<0.9}_{STARlight}(p+p \rightarrow J/\psi(e^+e^-)) = 40$ pb.

One of the possible sources of the difference could be background events that we have not corrected for. We are interested in the so-called central exclusive J/ψ photoproduction[1], and although we consider events in the offline selection that have exactly 2 BEMC tracks (C4.6), there still could be some contamination from the central inclusive production (J/ψ in the central system produced with other particles). The inclusive production could simply mean production of J/ψ along with some low-momenta or neutral particles, which the STAR experiment is not able to measure. This is not photoproduction anymore.

The exclusive and inclusive productions are difficult to differentiate from the experimental perspective, only their transverse momentum distributions are different. The transverse momentum distribution for the exclusive production is characterized by low momentum (below 1 GeV/c). The inclusive production tends to produce J/ψ particles with higher transverse momentum (up to several GeV/c), but also has some overlap in the low momentum region [83]. This process can happen via the Double IPomeron Exchange, which is a source of background to photoproduction. The trigger used in this analysis (*HTTP JPsi*) limits the contamination from J/ψ candidates with higher p_T with the back to back condition in BEMC (T1).

The plot shown in Fig. 8.3 supports the claim for significant contamination. The plot shows the transverse momentum of J/ψ candidates in blue (without the RP) compared to the STARlight sample in red. STARlight produces central exclusive events only. If one neglects the fact that the STARlight generator works as a Vector Dominance model and therefore does not perfectly describe the p_T distribution of exclusively produced J/ψ, one shall recognize some level of contamination. The

[1]The central exclusive production in this case means that there are no other particles produced in the central system, which we would measure, not in the case that we would measure all final state particles produced in the interaction.

same could be said for proton dissociation J/ψ production, when one of the protons breaks down (this contamination is less pronounced due to the fact that the J/ψ candidates tend to have larger p_T than inclusive and exclusive). One could argue that the discrepancy can be explained by detector smearing effects. This can be refuted with Fig. 6.17, which shows the comparison between p_T of measured J/ψ candidates and J/ψ candidates from STARlight which have passed through STARsim and reconstruction. The comparison in rapidity of J/ψ between the data and STARlight is shown in Fig. 8.4 which shows almost no discrepancy.

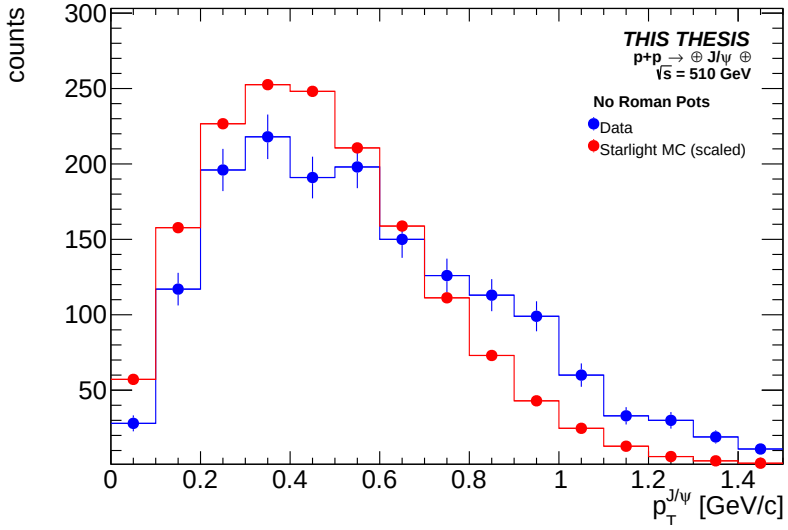


Figure 8.3: The comparison of transverse momentum distribution between the measurement and the STARlight sample.

Another important correction, which is missing at this point is the feeddown correction. Simultaneously to our process, there are particles such as $\psi(2S)$ and χ_c produced, which then usually decay to a photon ($\tilde{3}00$ MeV) and J/ψ. Again, these are background DIPE processes that increase the final cross section of our process. The cross section for these background processes is fairly small though. The contamination from these processes is expected to be around several percent, but further study is necessary.

Another possible source of discrepancy that could be improved is the veto efficiency, which enlarges the cross section by about a factor of 2.5. This value is significantly higher than, for example, J/ψ photoproduction in Au+Au at $\sqrt{s} = 200$ GeV [57]. A STAR measurement which used a very similar trigger including the veto in BBC, and estimated the veto efficiency at 3.9 %. Studying veto efficiency is very difficult, because by definition, we have no access to the excluded events. STAR experiment and the collaboration would certainly benefit from having a some sort of trigger simulation software. In addition, there are several runs in Fig. 6.1 which are well below all the rest of the values, yet their error bars are fairly small. These could be hints of some effects that could be significant and should be examined going forward. The last comment on the difficulty of the measurement is that photoproduction of J/ψ mesons has not been studied in proton-proton collisions at experiment STAR before. In fact, experiment STAR is not an ideal experiment for this type of study.

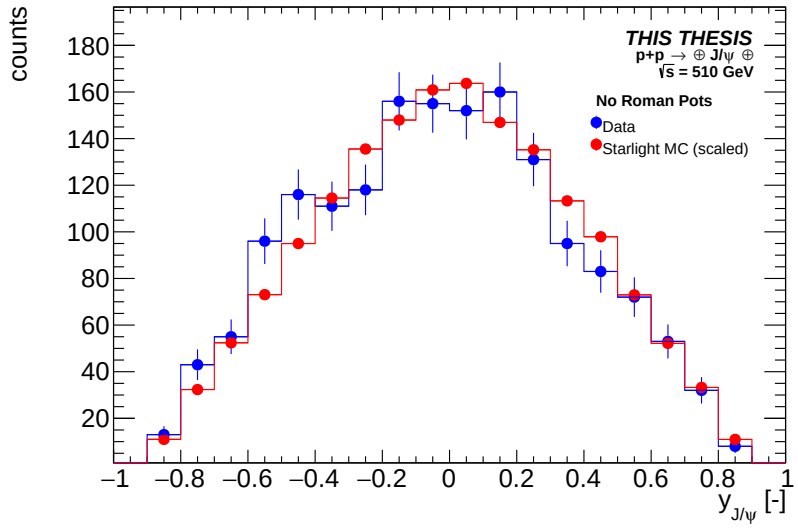


Figure 8.4: The comparison of J/ψ rapidity between the data (blue) and the STARlight sample (red).

Originally, experiment STAR was created for high multiplicity heavy ion collisions. Our 2 electrons with transverse momentum between 1 and 2 GeV are right at the limiting range of the Barrel Electromagnetic Calorimeter. The BEMC was created to measure jet-like events with transverse momentum from several GeV up to tens of GeV. Even the simulation of the detector output had to be corrected, because the calorimeter was simply not created for these particles. Yet, we have been able to observe a clean signal of J/ψ with rapidity gaps between the produced particles.

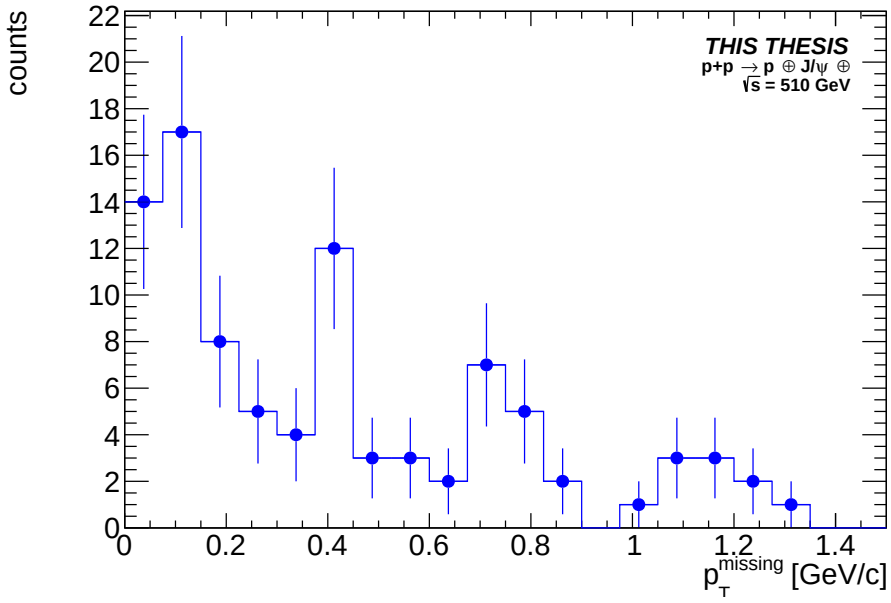


Figure 8.5: Distributions of missing transverse momenta in the analysis with the Roman Pots after background subtraction.

In the future, this study would certainly benefit from comparison to other Monte Carlo generators and more complex theoretical models describing the quarkonia or vector meson production. Some have been mentioned in Sec. 1.3.2; another example would be SuperChic 3 [84]. These more complex models are able to account for more effects stemming from the Quantum Chromodynamics.
The analysis with the Roman Pots shows potential for studying the forward protons. We were able to see a clear signal, although with lower statistics. The reason for measuring only 1 of the forward protons is based on the assumption that the proton coming from the photon vertex (see Fig. 1.4) does not have enough transverse momentum to reach the fiducial region of the Roman Pots (F2). In the future, it would be beneficial to calculate how factual the assumption is that the proton we measure comes from the photon interaction instead of the Pomeron. Based on the answer to this question, we could potentially make claims regarding the missing transverse momentum which equals to the WWA photon[2]. The transverse momentum distribution after the background subtraction is shown in Fig. 8.5.
To further improve the study, one should try to compare the analysis with the Roman Pots to a theoretical model. A study such as that would demand alteration of the codes of some of the MC generators, but in the end, one could gain further insight into the large difference between our measurement and STARlight. More insight could also be gained from studying the cross section of process $\gamma+\gamma \to e^+ + e^-$. This could, in principle, confirm or refute the ambiguity in the veto efficiency and explain the difference between our result and STARlight or HERA and LHCb experiment results.

[2]We measure the J/ψ daughter particles, electrons, and the other forward proton. After summation of their transverse momenta, we would get the transverse momentum of the emitted photon.

Summary

Studying J/ψ photoproduction provides a great way to study Quantum Chromodynamics, because it is possible to directly compare experimental results to existing models and theories. In fact, this ability is quite rare in particle physics. Photoproduction provides a way to probe the inner structure of protons, especially the gluon component. This thesis began with an introduction to particle physics variables and observables, then discussed various models and theoretical descriptions of J/ψ photoproduction, along with recent similar measurements, but the main goal was to analyze data and extract results.
The measurement itself was done at the experiment STAR located at the site of Brookhaven National Laboratory, in the U.S. The STAR experiment is a versatile detector capable of measuring a number of different processes. An overview of the STAR detector was given in chapter 2, focusing on the detectors used in the online and offline data analysis. The majority of the work done in this analysis is discussed in detail in chapter 3, chapter 4, chapter 6, and chapter 7, ranging from the introduction to the data format, discussion of the event selection, the applied corrections, and the estimation of the systematic uncertainties. The calculated cross section is shown in chapter 8 and compared to the STARlight Monte Carlo event generator, which was described in chapter 5. The discussion of the results, the possible sources of ambiguity, and the outlook towards the future improvements are also included.
In summary, exclusive J/ψ photoproduction in proton-proton collisions at an energy of collision $\sqrt{s} = 510$ GeV at the experiment STAR has been studied. Two separate analyses in the dielectron channel are studied, one with the use of the Roman Pots, the forward proton detectors, and one without. From the extracted yields, calculated efficiencies, corrections, and luminosity, 2 values of cross section have been extracted. In the case without the Roman Pots, the cross section in the dielectron channel in central pseudorapidity region was measured at $\sigma(p+p \to \oplus J/\psi \oplus) = 650 \pm 40(stat) \,^{+70}_{-50}(sys)$ pb and, in the case when we measure one of the forward protons, the result is $\sigma(p+p \to p \oplus J/\psi \oplus) = 77 \pm 5(stat) \,^{+9}_{-7}(sys)$ pb. In the comparable measurement, a significant discrepancy with existing previous measurements and theoretical predictions is observed. The origin of this discrepancy has not yet been identified.

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